

# MATH 8803: Representation Theory II

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# Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Jan. 12 — Introduction and Review</b>                        | <b>4</b>  |
| 1.1      | Review and Overview . . . . .                                   | 4         |
| 1.2      | Representations of Semisimple Lie Algebras . . . . .            | 5         |
| 1.3      | Representations of $SL_n$ and $GL_n$ . . . . .                  | 5         |
| <b>2</b> | <b>Jan. 14 — Applications of Schur-Weyl Duality</b>             | <b>7</b>  |
| 2.1      | The Schur Functor . . . . .                                     | 7         |
| 2.2      | Invariant Theory . . . . .                                      | 8         |
| 2.3      | Weyl Character Formula for $GL_n$ . . . . .                     | 9         |
| 2.4      | Howe Duality . . . . .                                          | 10        |
| <b>3</b> | <b>Jan. 21 — Minuscule Weights</b>                              | <b>11</b> |
| 3.1      | Minuscule Weights . . . . .                                     | 11        |
| 3.2      | Applications of Minuscule Weights . . . . .                     | 12        |
| <b>4</b> | <b>Jan. 26 — Other Classical Lie Algebras</b>                   | <b>14</b> |
| 4.1      | Applications of Minuscule Weights, Continued . . . . .          | 14        |
| 4.2      | Other Classical Lie Algebras . . . . .                          | 15        |
| 4.3      | Representations of Symplectic Lie Algebras . . . . .            | 15        |
| 4.4      | Representations of Orthogonal Lie Algebras . . . . .            | 16        |
| <b>5</b> | <b>Jan. 28 — Other Classical Lie Algebras, Part 2</b>           | <b>17</b> |
| 5.1      | More on Orthogonal Lie Algebras . . . . .                       | 17        |
| 5.2      | Clifford Algebras . . . . .                                     | 18        |
| <b>6</b> | <b>Feb. 2 — Duals, Maximal Weights, Exponents</b>               | <b>20</b> |
| 6.1      | Dual Representations . . . . .                                  | 20        |
| 6.2      | Maximal Weights . . . . .                                       | 20        |
| 6.3      | Principal $\mathfrak{sl}_2$ -Subalgebra and Exponents . . . . . | 21        |
| 6.4      | Complex, Real, and Quaternionic Types . . . . .                 | 21        |
| 6.5      | Review of Compact Lie Groups . . . . .                          | 22        |
| <b>7</b> | <b>Feb. 4 — Compact Groups</b>                                  | <b>23</b> |
| 7.1      | More on Exponents . . . . .                                     | 23        |
| 7.2      | Matrix Coefficients . . . . .                                   | 23        |
| 7.3      | Peter-Weyl Theorem . . . . .                                    | 24        |
| 7.4      | Introduction to Quantum Mechanics . . . . .                     | 25        |
| <b>8</b> | <b>Feb. 9 — Hydrogen Atom</b>                                   | <b>26</b> |
| 8.1      | Bound States of the Hydrogen Atom . . . . .                     | 26        |

|           |                                                    |           |
|-----------|----------------------------------------------------|-----------|
| 8.2       | Spin . . . . .                                     | 28        |
| 8.3       | Pauli Exclusion Principle . . . . .                | 28        |
| <b>9</b>  | <b>Feb. 11 — Real Forms</b>                        | <b>30</b> |
| 9.1       | Automorphisms of Semisimple Lie Algebras . . . . . | 30        |
| 9.2       | Real Forms of Semisimple Lie Algebras . . . . .    | 30        |
| <b>10</b> | <b>Feb. 16 — Real Forms, Part 2</b>                | <b>33</b> |
| 10.1      | Compact Real Forms . . . . .                       | 33        |
| 10.2      | Classification of Real Forms . . . . .             | 34        |
| <b>11</b> | <b>Feb. 18 — Real Forms, Part 3</b>                | <b>36</b> |
| 11.1      | Classification of Real Forms, Continued . . . . .  | 36        |
| 11.2      | Real Forms of Classical Lie Algebras . . . . .     | 37        |
| 11.3      | More on Compact Groups . . . . .                   | 38        |
| <b>12</b> | <b>Feb. 23 — Classifications of Lie Groups</b>     | <b>39</b> |
| 12.1      | Classification of Compact Lie Groups . . . . .     | 39        |
| 12.2      | Polar Decomposition . . . . .                      | 40        |
| <b>13</b> | <b>Feb. 25 — Maximal Tori</b>                      | <b>42</b> |
| 13.1      | Classification of Reductive Lie Groups . . . . .   | 42        |
| 13.2      | Maximal Tori . . . . .                             | 42        |
| 13.3      | Semisimple and Unipotent Elements . . . . .        | 43        |
| 13.4      | Cartan Decomposition . . . . .                     | 44        |
| <b>14</b> | <b>Mar. 2 — Verma Modules</b>                      | <b>45</b> |
| 14.1      | Integral Form of Weyl Character Formula . . . . .  | 45        |
| 14.2      | More on Verma Modules . . . . .                    | 46        |
| 14.3      | Maximal Roots and the Casimir Operator . . . . .   | 47        |
| 14.4      | Affine Lie Algebras . . . . .                      | 48        |
| <b>15</b> | <b>Mar. 4 — Affine Lie Algebras</b>                | <b>49</b> |
| 15.1      | Affine Lie Algebras, Continued . . . . .           | 49        |
| 15.2      | Generalized Cartan Matrices . . . . .              | 50        |
| 15.3      | Verma Modules and Weyl Modules . . . . .           | 50        |
| 15.4      | Integrable Representations . . . . .               | 52        |
| <b>16</b> | <b>Mar. 9 —</b>                                    | <b>53</b> |
| <b>17</b> | <b>Mar. 11 —</b>                                   | <b>54</b> |
| <b>18</b> | <b>Mar. 16 — Correlation Functions</b>             | <b>55</b> |
| 18.1      | Correlation Functions . . . . .                    | 55        |
| 18.2      | Analyticity of Correlation Functions . . . . .     | 57        |
| <b>19</b> | <b>Mar. 18 — KZ Equations, Part 2</b>              | <b>59</b> |
| 19.1      | Correlation Functions, Continued . . . . .         | 59        |
| 19.2      | Trigonometric KZ Equations . . . . .               | 60        |

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| 19.3 Factorizable Systems . . . . .                               | 61        |
| <b>20 Mar. 30 — KZ Equations for <math>\mathfrak{sl}_2</math></b> | <b>63</b> |
| 20.1 KZ Equations for $\mathfrak{sl}_2$ . . . . .                 | 63        |
| 20.2 Integral Formulas for Hypergeometric Equation . . . . .      | 65        |
| 20.3 Solutions in Higher Levels . . . . .                         | 66        |

# Lecture 1

## Jan. 12 — Introduction and Review

### 1.1 Review and Overview

**Remark.** Recall that we are interested in representations of Lie groups  $G$ , which is closely related to representations of Lie algebras  $\mathfrak{g}$ .

We are primarily interested in semisimple Lie algebras. In this case, we fix a *Cartan subalgebra*  $\mathfrak{h} \subseteq \mathfrak{g}$ , where  $r = \dim \mathfrak{h}$  is called the *rank*. We have the Serre generators  $\{h_i, e_i, f_i\}_{i=1}^r$  and relations

$$[h_i, e_j] = a_{i,j} e_j, \quad [h_i, f_j] = a_{i,j} f_j, \quad \text{ad}_{e_i}^{1-a_{i,j}} e_j = 0, \quad \text{ad}_{f_i}^{1-a_{i,j}} f_j = 0,$$

where  $a_{i,j} = \langle \alpha_i^\vee, \alpha_j \rangle$  for  $\alpha_i^\vee = 2\alpha_i / (\alpha_i, \alpha_i)$ . Here  $\{\alpha_i\} \subseteq \mathfrak{h}^*$  and we identify  $\alpha_i^\vee \leftrightarrow h_i \in \mathfrak{h}$ . Then

$$\mathfrak{g} = \mathfrak{n}_+ \oplus \mathfrak{h} \oplus \mathfrak{n}_-,$$

where  $\mathfrak{n}_+$  is generated by  $\{e_i\}$  and  $\mathfrak{n}_-$  is generated by  $\{f_i\}$ . We also have

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha,$$

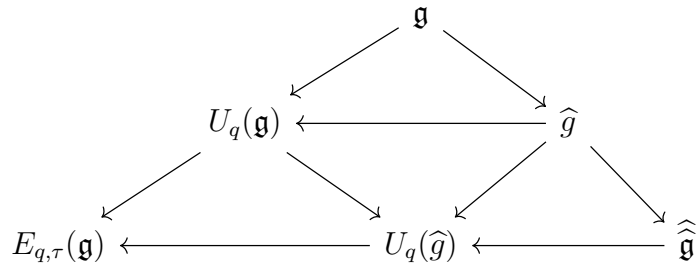
where  $R = R_+ \sqcup R_-$ . We have  $R_+ \subseteq Q_+$  and  $R_- \subseteq Q_-$ , where  $Q_+ = \{\sum_{i=1}^r n_i \alpha_i : n_i \geq 0\}$ . If the  $a_{i,j}$  are degenerate, then we can define  $\widehat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}c \oplus \mathbb{C}d$ , where  $\mathbb{C}c$  is called the *central extension* and  $d = t \frac{d}{dt}$ . We can think of these as maps  $S^1 \rightarrow \mathfrak{g}$ .

We can also consider the universal enveloping algebra  $U(\mathfrak{g})$ , and the related object.  $U_q(\mathfrak{g})$  We have an  $R$ -matrix  $R_{V,W}$  for the representations  $V \otimes W$  and  $W \otimes V$ , and we have the relation

$$R_{1,2} R_{1,3} R_{2,3} = R_{2,3} R_{1,3} R_{1,2}$$

in  $V_1 \otimes V_2 \otimes V_3$ . A main goal later in the course will be to relate the representations of  $U_q(\mathfrak{g})$  and  $\widehat{\mathfrak{g}}$ .

In this case, we have the diagram:



The object  $U_q(\widehat{\mathfrak{g}})$  is related to quantum integrable models of spin chain type (XXX and XXZ), and  $E_{q,\tau}(\mathfrak{g})$  is the *elliptic quantum group* (XYZ).

## 1.2 Representations of Semisimple Lie Algebras

**Remark.** Recall the *Weyl group*  $W = \{s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha\}$ . The *weight lattice* is

$$P = \{\lambda \in E : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}, \alpha \in R\} = \bigoplus_i \mathbb{Z}\omega_i,$$

where  $\omega_i$  are the fundamental weights satisfying  $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{i,j}$ .

We can consider the *highest weight representation*. The *Verma module* is  $M_\lambda = U(\mathfrak{g}) \otimes_{U(\mathfrak{h} \oplus \mathfrak{n}_+)} \mathbb{C}_\lambda$ , where  $\mathbb{C}_\lambda$  is the 1-dimensional representation of  $U(\mathfrak{h} \oplus \mathfrak{n}_+)$  on which  $\mathfrak{h}$  acts by  $\lambda(h)$ . Then

$$P(M_\lambda) = \lambda - \mathbb{Q}_+,$$

and for each  $\lambda \in \mathfrak{h}^*$ ,  $M_\lambda$  has a unique irreducible quotient  $L_\lambda$ . The *dominant integral weights*  $\lambda$  satisfy

$$\langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_+, \quad 1 \leq i \leq r,$$

where  $\lambda = \sum_{i=1}^r n_i \omega_i$  with  $n_i \in \mathbb{Z}_+$ .

**Theorem 1.1.** *The finite-dimensional irreps of  $\mathfrak{g}$  are classified up to isomorphism by  $\lambda \in P_+$ . Moreover,  $P(V)$  is Weyl invariant, and for any  $\mu \in P(V)$ ,  $w \in W$ ,*

$$\dim L_\lambda[\mu] = \dim L_\lambda[w\mu].$$

**Example 1.0.1.** For  $\mathfrak{g} = \mathfrak{sl}_2$ , the dominant integral weights are  $n \in \mathbb{Z}_{\geq 0}$ ,  $L_n = V_n$ , and the Weyl group  $W$  acts by reflection.

**Remark** (Weyl character formula). Let  $\chi_V(g) = \text{tr}_V(g)$ . We can represent  $g \sim e^h$ , where  $h \in \mathfrak{h}$ . Then

$$\chi_V(e^h) = \sum_{\mu \in P} (\dim V(\mu)) e^{\mu(h)}.$$

We can then formally define  $\chi_V = \sum_{\mu \in P} (\dim V(\mu)) e^\mu$ . The *Weyl character formula* is

$$\chi_{L_\lambda} = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda + \rho)}}{\Delta},$$

where  $\Delta = \prod_{\alpha \in R_+} (e^{\alpha/2} - e^{-\alpha/2}) = \prod_{w \in W} (-1)^{\ell(w)} e^{w\rho}$  is the *Weyl denominator*. Here  $\rho = \frac{1}{2} \sum_{\alpha \in R_+} \alpha = \sum_{i=1}^r \omega_i$ . The *Weyl dimension formula* is then

$$\dim L_\lambda = \frac{\prod_{\alpha \in R_+} (\alpha, \lambda + \rho)}{\prod_{\alpha \in R_+} (\alpha, \rho)}.$$

Recall the *Casimir operator*  $\sum_{i=1}^{\dim \mathfrak{g}} x_i x^i \in U(\mathfrak{g})$ , which acts by the scalar  $(\lambda, \lambda + 2\rho)$ .

## 1.3 Representations of $\text{SL}_n$ and $\text{GL}_n$

**Proposition 1.1.** *For general simple  $\mathfrak{g}$ , let  $\lambda = \sum_{i=1}^r m_i \omega_i$  be a dominant integral weight. Let  $T_\lambda = \bigotimes_i L_{\omega_i}^{\otimes m_i}$  and  $v = \bigotimes_i v_{\omega_i}^{\otimes m_i}$ . Let  $V$  be the subrepresentation of  $T_\lambda$  generated by  $v$ . Then  $V \cong L_\lambda$ .*

**Remark.** For  $\mathfrak{sl}_n$ , we have  $\lambda = \sum_{i=1}^{n-1} m_i \omega_i$ . The Cartan subalgebra is

$$\mathfrak{h} = \mathbb{C}_0^n = \{(x_1, \dots, x_n) \in \mathbb{C}^n : x_1 + \dots + x_n = 0\}.$$

We have  $\alpha_i^\vee = e_i - e_{i-1}$  and  $\delta_{i,j} = (\omega_i, \alpha_j^\vee) = (\omega_i, e_j - e_{j+1})$ , where  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  with  $i$  ones. We can associate  $\lambda$  with the partition

$$\lambda = (m_1 + \dots + m_{n-1}, m_2 + \dots + m_{n-1}, \dots, m_{n-1}, 0) = (\lambda_1, \lambda_2, \dots, \lambda_{n-1}, 0),$$

and  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{n-1}$ . Note that  $L_{\omega_1}$  is the defining representation, where  $v_{\omega_1} = (1, 0, \dots, 0)^T = v_1$ , where  $\{v_1, \dots, v_n\}$  is a basis of the defining representation. Then we have that  $L_{\omega_m} = \wedge^m V$  with highest weight  $v_1 \wedge \dots \wedge v_m$ . Here  $e_i = E_{i,i+1}$ . Then we see that  $L_\lambda \subseteq \bigotimes_{i=1}^{n-1} (\wedge^i V)^{\otimes m_i}$ .

**Remark.** To move to  $\mathrm{GL}_n$ , we can write

$$\mathrm{GL}_n(\mathbb{C}) = (\mathbb{C}^\times \times \mathrm{SL}_n(\mathbb{C})) / \mu_n,$$

where  $\mu_n$  are the roots of unity embedded by  $z \mapsto (z^{-1}, zI)$ . We have a covering homomorphism

$$\begin{aligned} \mathbb{C}^\times \times \mathrm{SL}_n(\mathbb{C}) &\longrightarrow \mathrm{GL}_n(\mathbb{C}) \\ (z, A) &\longmapsto zA. \end{aligned}$$

We need to determine the holomorphic representations of  $\mathbb{C}^\times$ . Its Lie algebra is spanned by  $h$  such that  $e^{2\pi i h} = 1$ . Within a representation,  $h$  acts by an operator  $H$  such that  $e^{2\pi i H} = 1$ . Thus all irreducible representations of  $\mathbb{C}^\times$  are of the form  $\chi_N(z) = z^N$ . So for  $\mathbb{C}^\times \times \mathrm{SL}_n(\mathbb{C})$ , we have  $L_{\lambda,N} = \chi_N \otimes L_\lambda$ .

**Exercise 1.1.** Show that if  $L_{\lambda,N} = \chi_N \otimes L_\lambda$ , then  $N = nr + \sum_{i=1}^{n-1} \lambda_i$  for some integer  $r$ .

**Remark.** Letting  $m_n = r \geq 0$  in the above exercise, the representation  $L_{\lambda, nm_n + \sum_{i=1}^{n-1} \lambda_i}$  for  $\mathfrak{gl}_n$  corresponds to the partition  $(m_1 + \dots + m_n, \dots, m_{n-1} + m_n, m_n)$ .

**Remark.** For  $\mathrm{SL}_n$ , the representation  $\wedge^n V$  is trivial, but it is the determinant for  $\mathrm{GL}_n$ . For  $\mathrm{GL}_n$ , we also have  $\chi^k$  and  $(\chi^*)^k = \chi^{-k}$ , these are called the *polynomial representations*.

**Remark.** Let  $\lambda = (\lambda_1, \dots, \lambda_n)$  with  $\lambda_i \geq \dots \geq \lambda_n$  be a partition with at most  $n$  parts. Then  $|\lambda| = \sum_i \lambda_i$  is an eigenvalue of  $1_n = \sum_{i=1}^n e_{i,i} \in \mathfrak{gl}_n$ . We can realize  $\lambda$  as a Young diagram. Note that  $L_\lambda$  occurs in  $V^{\otimes N}$ , where  $V$  is the defining representation. We can decompose

$$V^{\otimes N} = \bigoplus_{\lambda: |\lambda|=N} L_\lambda \otimes \pi_\lambda,$$

where  $\pi_\lambda = \mathrm{Hom}_{\mathrm{GL}_n(\mathbb{C})}(L_\lambda, V^{\otimes N})$ . There is a natural action of  $S_N$  on  $V^{\otimes N}$ .

**Theorem 1.2** (Schur-Weyl duality). *Let  $A$  be the image of  $U(\mathfrak{gl}_n)$  in  $\mathrm{End}(V^{\otimes N})$  and  $B$  be the image of  $\mathbb{C}S_N$  in  $\mathrm{End}(V^{\otimes N})$ . Then*

1. *the centralizer of  $A$  is  $B$  and vice versa;*
2. *if  $\lambda$  has at most  $n$  parts, then the representation  $\pi_\lambda$  of  $B$  (and hence of  $S_N$ ) is irreducible, and such representations are pairwise non-isomorphic;*
3. *if  $\dim V \geq N$ , then the  $\pi_\lambda$  exhaust all irreducible representations of  $S_N$ .*

# Lecture 2

## Jan. 14 — Applications of Schur-Weyl Duality

### 2.1 The Schur Functor

**Remark.** Let  $V$  be the defining representation for  $\mathrm{GL}_n$ . Then

$$V^{\otimes N} = \bigoplus_{\lambda: |\lambda|=N} L_\lambda \otimes \pi_\lambda.$$

Recall that if  $\lambda = (\lambda_1, \dots, \lambda_n)$ , then we have

$$\lambda_1 = m_1 + \dots + m_n, \quad \lambda_2 = m_2 + \dots + m_n, \quad \dots, \quad \lambda_n = m_n.$$

**Definition 2.1.** Suppose we are given the partition  $\lambda$  of  $N$ . The *Schur functor*  $S^\lambda$  is given by

$$S^\lambda V = \mathrm{Hom}_{S_N}(\pi_\lambda, V^{\otimes N})$$

for a vector space  $V$ . Note that this language, we have  $V^{\otimes N} = \bigoplus_\lambda S^\lambda V \otimes \pi_\lambda$ .

**Example 2.1.1.** Consider the following:

1.  $S^{(n)}V = S^n V$ , where  $(n)$  is the partition of  $n$  with a single part.
2.  $S^{(1^n)}V = \wedge^n V$ , where  $(1^n)$  is the partition of  $n$  with  $n$  parts equal to 1.
3.  $V \otimes V = S^{(2)}V \otimes \mathbb{C}_+ \oplus S^{(1,1)}V \otimes \mathbb{C}_-$ , where  $\mathbb{C}_2$  acts trivially on  $\mathbb{C}_+$  and by the sign on  $\mathbb{C}_-$ .
4.  $V \otimes V \otimes V = S^{(3)}V \otimes \mathbb{C}_+ \oplus S^{(2,1)}V \otimes \mathbb{C}^2 \oplus S^{(1,1,1)}V \otimes \mathbb{C}_-$ , where  $S_3$  acts trivially on  $\mathbb{C}_+$  and by sign on  $\mathbb{C}_-$  as before, and  $\mathbb{C}^2 = \{(x, y, z) : x + y + z = 0\}$ .

Note that  $V \otimes V = S^2 V \oplus \wedge^2 V$ , so  $S^2 V \otimes V = S^3 V \oplus S^{(2,1)}V$  and  $\wedge^2 V \otimes V = \wedge^3 V \oplus S^{(2,1)}V$ .

**Remark.** Let  $\dim V = N$  and  $\lambda$  have  $k$  parts. Recall that by the Weyl dimension formula,

$$\dim L_\lambda = \frac{\prod_{\alpha \in R_+} (\alpha, \lambda + \rho)}{\prod_{\alpha \in R_+} (\alpha, \rho)}.$$

We have  $R_+ = \{\alpha_{i,j} = e_i - e_j : i < j\}$  and  $\rho = \sum_{i=1}^{N-1} \omega_i = (N-1, N-2, \dots, 1, 0)$  (recall that  $\omega_i$  is  $i$  ones followed by zeros). Thus we see that

$$\dim S^\lambda V = \prod_{1 \leq i < j \leq N} \frac{\lambda_i - \lambda_j + j - i}{j - i} = \prod_{1 \leq i < j \leq k} \frac{\lambda_i - \lambda_j + j - i}{j - i} \prod_{1 \leq i < k < j \leq N} \frac{\lambda_i + j - i}{j - i}.$$



We can rewrite the second product as

$$\prod_{1 \leq k < j \leq N} \frac{\lambda_i + j - i}{j - i} = \prod_{i=1}^k \frac{(N+1-i) \cdots (N+\lambda_i-i)}{(k+1-i) \cdots (k+\lambda_i-i)}.$$

**Proposition 2.1.** *We have  $\dim S^\lambda V = P_\lambda(N)$ , where  $P_\lambda$  is a polynomial of degree  $|\lambda|$  with rational coefficients and integer roots. The roots of  $P_\lambda$  are all integers from the interval  $[1 - \lambda_1, k - 1]$  (occurring with multiplicities).*

**Example 2.1.2.** Let  $P_n(N)$  correspond to  $S^n V$ . Then  $\lambda_1 = n$  and  $k = 1$ , and

$$P_n(N) = \dim S^n V = \binom{N+n-1}{n}.$$

Similarly, one can see that

$$P_{1^n}(N) = \dim \wedge^n V = \binom{N}{n}.$$

One can also consider  $P_{(a,b)}(N)$  corresponding to partitions with two parts. The values  $P_{(a,n)}(N)$  are called the Narayana numbers, which are of use in combinatorics.

## 2.2 Invariant Theory

**Remark.** Let  $V$  be a finite-dimensional vector space and  $\{T_i\} \in (V^*)^{\otimes m_i} \otimes V^{\otimes n_i}$  for  $i = 1, \dots, k$ . One would like to characterize *invariants* of such collections, i.e. polynomial functions  $F(T_1, \dots, T_k)$  which are invariant under the action of  $\mathrm{GL}(V)$ .

One can think of such a tensor in  $(V^*)^{\otimes m_i} \otimes V^{\otimes n_i}$  as a vertex with  $m_i$  incoming edges and  $n_i$  outgoing edges. Then constructing invariants  $\{T_i\}$  reduces to studying graphs where  $T_i$  corresponds to a vertex  $v_i$  of the graph  $\Gamma$ . This allows us to assign to a given graph an invariant function  $F_\Gamma$ .

**Theorem 2.1.** *The functions  $F_\Gamma$  for various  $\Gamma$  span the space of invariant functions.*

*Proof.* We can view an invariant as an invariant element of the space  $\bigotimes_{i=1}^k ((V^*)^{\otimes m_i} \otimes V^{\otimes n_i})$ , which we can view as  $\mathrm{End}_{\mathrm{GL}(n)}(V^{\otimes M}, V^{\otimes N})$ , where  $M = \sum d_i m_i$  (the number of incoming edges) and  $N = \sum d_i n_i$  (the number of outgoing edges). Note that this space is empty when  $M \neq N$ , and the statement follows by Schur-Weyl duality when  $M = N$ .  $\square$

**Example 2.1.3.** Let  $m_i = n_i = 1$ . Then  $T_1, \dots, T_k$  are matrices. Then the graph  $\Gamma$  must look like a cycle, hence the invariants are all of the form

$$F_{j_1, \dots, j_r}(T_1, \dots, T_k) = \mathrm{tr}(T_{j_1} \cdots T_{j_r}).$$

Note that these invariants are asymptotically algebraically independent (when  $V$  is large enough). In particular, if  $P(T_1, \dots, T_k) = 0$  in all dimensions, then  $\mathrm{tr}(P(T_1, \dots, T_k)T_{k+1}) = 0$ , which cannot be true as the trace decomposes in terms of the  $F_{j_1, \dots, j_r}$ . (However, note that  $[X, Y] = 0$  for  $1 \times 1$  matrices and  $[Z, [X, Y]^2] = 0$  for  $2 \times 2$  matrices.) This also implies the uniqueness of the  $\mu_n$  in the BCH formula:

$$\log(\exp(x) \exp(y)) = \sum_{n \geq 1} \frac{\mu_n(x, y)}{n!}.$$

## 2.3 Weyl Character Formula for $GL_n$

**Remark** (Weyl character formula for  $GL_n$ ). Recall that Weyl's character formula gives

$$\chi_\lambda = \frac{\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda + \rho)}}{\sum_{\alpha \in R_+} (e^{\alpha/2} - e^{-\alpha/2})}, \quad (*)$$

where the denominator is  $\Delta = \prod_{\alpha \in R_+} (e^{\alpha/2} - e^{-\alpha/2}) = \prod_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}$ . Letting  $\rho = \frac{1}{2} \sum_{\alpha \in R_+} \alpha$ ,

$$\Delta = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = x_1^{n-1} x_2^{n-2} \cdots x_n^0 \prod_{i < j} (1 - x_j/x_i),$$

where  $\rho = (n-1, n-2, \dots, 1, 0)$  and  $x_i = e^{e_i}$  (e.g.  $x_1 = e^{(1,0,\dots,0)}$ ). After multiplying we get that

$$\Delta = \prod_{i < j} (x_i - x_j).$$

On the other hand, using  $\Delta = \prod_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}$ , we have

$$\Delta = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)} = \sum_{s \in S_n} \text{sign}(s) x_{s(1)}^{n-1} \cdots x_{s(n)}^0.$$

Comparing these two formulas, we recover the formula for the Vandermonde determinant:

$$\det(\{x_j^{n-i}\}_{1 \leq i, j \leq n}) = \sum_{s \in S_n} \text{sign}(s) x_{s(1)}^{n-1} \cdots x_{s(n)}^0 = \prod_{i < j} (x_i - x_j).$$

Now applying this to the numerator of (\*), we have

$$\sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda + \rho)} = \sum_{s \in S_n} \text{sign}(s) x_{s(1)}^{\lambda_1 + n - 1} \cdots x_{s(n)}^{\lambda_n + 0}.$$

Thus in total, the character  $\chi_\lambda$  is given by

$$\chi_\lambda = \frac{\sum_{s \in S_n} \text{sign}(s) x_{s(1)}^{\lambda_1 + n - 1} \cdots x_{s(n)}^{\lambda_n + 0}}{\prod_{i < j} (x_i - x_j)} = \frac{\det(\{x_i^{\lambda_j + n - i}\})}{\prod_{i < j} (x_i - x_j)}.$$

These functions are known as the *Schur polynomials*  $s_\lambda(x_1, \dots, x_n)$ .

**Example 2.1.4** (Character of  $S^m V$ ). Using the above formula, we get the identity

$$s_{(m)}(x_1, \dots, x_n) = \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} x_{j_1} \cdots x_{j_m} = h_m(x_1, \dots, x_n),$$

the  $m$ th complete symmetric function.

**Example 2.1.5** (Character of  $\wedge^m V$ ). Similarly, one gets the identity

$$s_{(1^m)}(x_1, \dots, x_n) = \sum_{1 \leq j_1 < \dots < j_m \leq n} x_{j_1} \cdots x_{j_m} = e_m(x_1, \dots, x_n),$$

the  $m$ th elementary symmetric function.

**Example 2.1.6** (Trace in  $V^{\otimes N}$ ). Consider  $x \otimes \sigma$ , where  $x = \text{diag}(x_1, \dots, x_n)$  and  $\sigma$  has  $m_i$  cycles of length  $i$ . Then we have

$$\text{tr}_{V^{\otimes N}}(x \otimes \sigma) = \prod_i (x_1^i + \dots + x_n^i)^{m_i}.$$

By Schur-Weyl duality, we have that

$$\text{tr}_{V^{\otimes N}}(x \otimes \sigma) = \sum_{\lambda} \chi_{\lambda}(\sigma) s_{\lambda}(x) = \prod_i (x_1^i + \dots + x_n^i)^{m_i}.$$

Using the formula for the Schur polynomial, we get the identity

$$\sum_{\lambda} \chi_{\lambda}(\sigma) \det(\{x_i^{\lambda_j + N - j}\}) = \prod_{i < j} (x_i - x_j) \prod_i (x_1^i + \dots + x_n^i)^{m_i}.$$

**Theorem 2.2** (Frobenius character formula).  $\chi_{\lambda}(\sigma)$  is the coefficient of  $x_1^{\lambda_1 + N - 1} \dots x_N^{\lambda_N}$  in the polynomial

$$\prod_{i < j} (x_i - x_j) \prod_i (x_1^i + \dots + x_n^i)^{m_i}.$$

## 2.4 Howe Duality

**Remark.** Fix  $V, W$  and consider  $S^n(V \otimes W)$ , which is a representation of  $\text{GL}(V) \otimes \text{GL}(W)$ .

**Theorem 2.3** (Howe duality). *We have a decomposition*

$$S^n(V \otimes W) = \bigoplus_{\lambda: |\lambda|=n} S^{\lambda}V \otimes S^{\lambda}W.$$

*Proof.* We can write

$$S^n(V \otimes W) = ((V \otimes W)^{\otimes n})^{S_n} = (V^{\otimes n} \otimes W^{\otimes n})^{S_n}.$$

Using Schur-Weyl duality for each part, we get that

$$\begin{aligned} S^n(V \otimes W) &= \left( \left( \bigoplus_{\lambda: |\lambda|=n} S^{\lambda}V \otimes \pi_{\lambda} \right) \otimes \left( \bigoplus_{\mu: |\mu|=n} S^{\mu}W \otimes \pi_{\mu} \right) \right)^{S_n} \\ &= \bigoplus_{\lambda, \mu: |\lambda|=|\mu|=n} S^{\lambda}V \otimes S^{\mu}W \otimes (\pi_{\lambda} \otimes \pi_{\mu})^{S_n}. \end{aligned}$$

Since  $\pi_{\lambda} = \pi_{\lambda}^*$ , by Schur's lemma we have  $(\pi_{\lambda} \otimes \pi_{\mu})^{S_n} = \mathbb{C}^{\delta_{\lambda, \mu}}$ . □

**Corollary 2.3.1** (Cauchy identity). *Let  $x = (x_1, \dots, x_r)$  and  $y = (y_1, \dots, y_s)$ . Then*

$$\sum_{\lambda} s_{\lambda}(x) s_{\lambda}(y) z^{|\lambda|} = \prod_{i=1}^r \prod_{j=1}^s \frac{1}{1 - z x_i y_j}.$$

# Lecture 3

## Jan. 21 — Minuscale Weights

### 3.1 Minuscale Weights

**Remark.** Let  $\mathfrak{g}$  be a simple complex Lie algebra.

**Definition 3.1.** A dominant integral weight  $\omega$  for  $\mathfrak{g}$  is called *minuscale* if  $\langle \omega, \beta \rangle \leq 1$  for every positive coroot  $\beta$  (equivalently, if  $|\langle \omega, \alpha \rangle| \leq 1$  for any coroot  $\beta$ ).

**Example 3.1.1.** Clearly  $\omega = 0$  is minuscale.

**Example 3.1.2.** Let  $\mathfrak{g} = \mathfrak{sl}_n$  with fundamental weights  $\{\omega_i\}_{i=1}^{n-1}$ ,<sup>1</sup> where

$$\omega_i = (\underbrace{1, \dots, 1}_i, 0, \dots, 0)$$

Let  $\alpha_{i,j} = \alpha_{i,j}^\vee = e_i - e_j$ . Note that  $\langle \omega_i, e_j - e_k \rangle = 0$  when  $j, k \leq i$  or  $j, k > i$ , and  $\langle \omega_i, e_j - e_k \rangle = 1$  when  $j \leq i < k$ . So all of the  $\omega_i$  are minuscale in this case.

**Lemma 3.1.** *Every nonzero minuscale weight is fundamental.*

*Proof.* Suppose  $\omega$  is minuscale. Then there exists  $i$  with  $\langle \omega, \alpha_i^\vee \rangle = 1$ . Moreover, there can only be one such  $i$ , since if there were many, then  $\langle \omega, \theta^\vee \rangle \geq 2$ , where  $\theta^\vee$  is the longest coroot (i.e. if  $\theta = \sum_{m_i > 0} m_i \alpha_i$  is the longest root, then  $\theta^\vee = \sum_{m_i > 0} m_i \alpha_i^\vee$ ). So  $\omega$  is necessarily fundamental.  $\square$

**Example 3.1.3.** For  $G_2$ ,  $F_4$ , and  $F_8$ , none of the fundamental weights are minuscale.

**Lemma 3.2.** *A fundamental weight  $\omega_i$  is minuscale if and only if  $m_i = 1$  where  $\theta^\vee = \sum_j m_j \alpha_j^\vee$ .*

*Proof.* By the minuscale condition, we know  $m_i \leq 1$ . If  $m_i = 1$ , then for any positive coroot  $\beta = \sum n_j \alpha_j^\vee$  we have  $n_j \leq m_j$ , so  $n_i \leq 1$ . Thus  $\langle \omega_i, \beta \rangle = n_i \leq 1$ , so  $\omega_i$  is minuscale.  $\square$

**Lemma 3.3.** *If  $\omega \in Q$  with  $|\langle \omega, \beta \rangle| \leq 1$  for all coroots  $\beta$ , then  $\omega = 0$ .*

*Proof.* Assume to the contrary that  $\omega = \sum_i \alpha_i \neq 0$ . We may assume that  $\sum_i |m_i|$  is smallest possible. Then  $0 < (\omega, \omega) = \sum_i m_i (\omega, \alpha_i)$ , since the form is positive definite. Thus there exists  $j$  such that  $m_j$  and  $\langle \omega, \alpha_j^\vee \rangle$  have the same sign. By replacing  $\omega$  with  $-\omega$  if necessary, we may assume both are positive. Then  $\langle \omega, \alpha_j^\vee \rangle = 1$ . Consider the reflection  $s_j(\omega) = \omega - \alpha_j = \sum_i m'_i \alpha_i$ . So  $m'_i = m_j - 1$  and  $m'_i = m_i$ . But then  $\sum_i |m'_i| = \sum_i |m_i| - 1 < \sum_i |m_i|$ , contradicting the minimality of  $\omega$ .  $\square$

<sup>1</sup>Recall a *fundamental weight* is a weight  $\omega_i$  such that  $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{i,j}$  for all simple coroots  $\alpha_j^\vee$ .

**Proposition 3.1.** *The following conditions are equivalent:*

1.  $\omega$  is minuscule;
2. all weights of  $L_\omega$  belong to the Weyl orbit  $W\omega$ ;
3. if  $\lambda$  is a dominant integral weight such that  $\omega - \lambda \in Q_+$ , then  $\lambda = \omega$ .

*Proof.* (1  $\Rightarrow$  3) If  $\omega = 0$ , then  $-\lambda \in Q_+$ , so  $(\lambda, \rho) \leq 0$  where  $\rho = \sum_{i=1}^r \omega_i$ , so  $\lambda = 0$ . Now let  $\omega = \omega_i$  be minuscule. Then  $\omega_i - \lambda = \sum_k m_k \alpha_k$  with  $m_k \geq 0$ . If  $m_k = 0$  for  $k \neq i$ , then the problem reduces to a lower rank Dynkin diagram. So we can assume  $m_k > 0$  for every  $k \neq i$ . Let  $\beta$  be a positive coroot, then

$$\langle \omega_i - \lambda, \beta \rangle = \langle \omega_i, \beta \rangle - \langle \lambda, \beta \rangle \leq \langle \omega_i, \beta \rangle \leq 1.$$

If  $\alpha_j^\vee$  does not occur in  $\beta$ , then the above is  $\leq 0$ . In particular, we have  $\langle \omega_i - \lambda, \alpha_j^\vee \rangle \leq 0$  for  $j \neq i$ . If we also have  $\langle \omega_i - \lambda, \alpha_i^\vee \rangle \leq 0$ , then  $(\omega_i - \lambda, \omega_i - \lambda) \leq 0$ , so  $\omega_i = \lambda$ . Otherwise,  $\langle \omega_i - \lambda, \alpha_i^\vee \rangle = 1$ . Then  $m_j > 0$  for every  $j$ , so  $\langle \omega_i - \lambda, \theta^\vee \rangle \geq 1$ , since  $\theta^\vee$  is a dominant coweight. Then  $\langle \lambda, \theta^\vee \rangle \leq 0$ , so we must have  $\lambda = 0$  since  $\theta^\vee$  contains all  $\alpha_j^\vee$  with positive coefficients. But then  $\omega_i \in Q$ , which is impossible by Lemma 3.3.

(3  $\Rightarrow$  2) If  $\mu$  is any weight of  $L_\omega$ , then there exists  $w \in W$  such that  $\lambda = w\mu$  is dominant (since every orbit of  $W$  intersects the dominant chamber at exactly 1 point). Then  $\omega - \lambda \in Q_+$ , so  $\lambda = \omega$ , hence  $\mu = w^{-1}\omega \in W\omega$ .

(2  $\Rightarrow$  1) Suppose otherwise  $\omega$  is not minuscule. Then  $\langle \omega, \alpha^\vee \rangle > 1$  for some positive coroot  $\alpha^\vee$ . Then

$$2(\omega, \alpha) > (\alpha, \alpha).$$

Note that  $\omega - \alpha$  is a weight of  $L_\omega$  (weight of  $f_\alpha v_\omega$ , where  $v_\omega$  is a highest weight vector and  $\{e_\alpha, f_\alpha, \alpha^\vee\}$  is an  $\mathfrak{sl}_2$ -triple). But  $\omega - \alpha$  is not  $W$ -conjugate to  $\omega$ , since

$$(\omega - \alpha, \omega - \alpha) = (\omega, \omega) - 2(\omega, \alpha) + (\alpha, \alpha) < (\omega, \omega)$$

but the pairing is  $W$ -invariant. Contradiction.  $\square$

**Corollary 3.0.1.** *If  $\omega$  is minuscule, then  $\chi_\omega = \sum_{\gamma \in W\omega} e^\gamma$ .*

## 3.2 Applications of Minuscule Weights

**Proposition 3.2.**  *$\omega \in P_+$  is minuscule if and only if the restriction of  $L_\omega$  to any root  $\mathfrak{sl}_2$ -subalgebra of  $\mathfrak{g}$  is the direct sum of 1-dimensional and 2-dimensional representations.*

*Proof.* ( $\Rightarrow$ ) Let  $\omega$  be minuscule and  $v \in L_\omega$  the highest weight vector (of weight  $w\omega$ ) for  $(\mathfrak{sl}_2)_\alpha$ . Then

$$h_\alpha v = \langle w\omega, \alpha^\vee \rangle v = \langle \omega, w^{-1}\alpha^\vee \rangle v.$$

Then  $h_\alpha v = 0$  or  $h_\alpha v = v$ , so the representation is 1-dimensional or 2-dimensional.

( $\Leftarrow$ ) Suppose  $\omega$  is not minuscule. Then there exists  $\alpha \in Q_+$  with  $\langle \omega, \alpha^\vee \rangle = m > 1$ . Let  $v_\omega$  be a highest weight vector, then  $h_\alpha v_\omega = \langle \omega, \alpha^\vee \rangle v_\omega$ , which leads to a higher-dimensional  $\mathfrak{sl}_2$ -representation.  $\square$

**Corollary 3.0.2.** *If  $\omega$  is minuscule, then for every dominant integral weight  $\lambda$  of  $\mathfrak{g}$ , we have*

$$L_\omega \otimes L_\lambda = \bigoplus_{\gamma \in W\omega} L_{\lambda+\gamma}.$$

(It is assumed that if  $\lambda + \gamma$  is not dominant, then  $L_{\lambda+\gamma} = 0$ .)

*Proof.* We know  $\chi_\omega = \sum_{\mu \in W\omega} e^\mu$ . Then we have

$$\chi_{L_\omega \otimes L_\lambda} = \frac{\sum_{\mu \in W\omega} \sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\rho)+\mu}}{\Delta} = \frac{\sum_{\gamma \in W\omega} \sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\gamma+\rho)}}{\Delta}$$

where  $\Delta$  is the Weyl denominator. If  $\lambda + \gamma \notin P_+$ , then for some  $\alpha_i^\vee$ , we get  $\langle \lambda + \gamma, \alpha_i^\vee \rangle < 0$ . But we know  $\langle \gamma, \alpha_i^\vee \rangle \geq -1$ , so  $\langle \lambda + \gamma, \alpha_i^\vee \rangle = -1$ . Thus  $\langle \lambda + \gamma + \rho, \alpha_i^\vee \rangle = 0$ , so for any  $w\gamma$ , the term  $ws_i\gamma$  comes with the opposite sign. So we get that

$$\chi_{L_\omega \otimes L_\lambda} = \frac{\sum_{\gamma \in W\omega : \lambda+\gamma \in P_+} \sum_{w \in W} (-1)^{\ell(w)} e^{w(\lambda+\gamma+\rho)}}{\Delta} = \sum_{\gamma \in W\omega : \lambda+\gamma \in P_+} \chi_{\lambda+\gamma},$$

which proves the desired result.  $\square$

**Example 3.1.4.** For  $\mathfrak{sl}_2$ , we have  $L_1 \otimes L_m = L_{m+1} \oplus L_{m-1}$ , which leads to the formula

$$L_m \otimes L_n = \bigoplus_{k=|m-n|}^{m+n} L_k$$

**Example 3.1.5.** Let  $V = V_{\omega_1}$  be the defining representation for  $\mathrm{GL}_n$ . Then

$$L_{\omega_1} \otimes L_\lambda = \bigoplus_{\mu \in \lambda + \square} L_\mu,$$

where  $\lambda$  is a partition and  $\lambda + \square$  denotes the set of partitions obtained by adding a single box to  $\lambda$ . For example, for  $\lambda = (3, 3, 2, 1)$  we have

$$L_{\omega_1} \otimes S^{(3,3,2,1)}V = S^{(4,3,2,1)}V \oplus S^{(3,3,3,1)}V \oplus S^{(3,3,2,2)}V \oplus S^{(3,3,2,1,1)}V.$$

Similarly, for  $\wedge^m V = L_{\omega_m}$  (where  $\omega_m = (1, \dots, 1, 0, \dots, 0)$  with  $m$  ones), we have

$$L_{\omega_m} \otimes L_\lambda = \bigoplus_{\mu \in \lambda + m\square} L_\mu,$$

where we are allowed to add  $m$  boxes to  $\lambda$  in  $\lambda + m\square$ . For example,

$$\wedge^2 V \otimes S^{(3,1)}V = S^{(4,2)}V \oplus S^{(4,1,1)}V \oplus S^{(3,2,1)}V \oplus S^{(3,1,1,1)}V.$$

# Lecture 4

## Jan. 26 — Other Classical Lie Algebras

### 4.1 Applications of Minuscule Weights, Continued

**Proposition 4.1.** *We have the following:*

1. Let  $\lambda$  be a partition of  $N$ . Then  $\mathbb{C}S_{N+1} \otimes_{S_N} \pi_\lambda = \bigoplus_{\mu \in \lambda + \square} \pi_\mu$ .
2. Let  $\mu$  be a partition of  $N + 1$ . Then  $\pi_\mu|_{S_N} = \bigoplus_{\lambda \in \mu - \square} \pi_\lambda$ .

*Proof.* (1) Let  $V$  be a vector space of sufficiently large dimension. By Frobenius reciprocity,

$$\mathrm{Hom}_{S_{N+1}}(\mathbb{C}S_{N+1} \otimes_{S_N} \pi_\lambda, V^{\otimes(N+1)}) \cong \mathrm{Hom}_{S_N}(\pi_\lambda, V^{\otimes N} \otimes V) = V \otimes S^\lambda V.$$

Now by Schur-Weyl duality, we have

$$\mathrm{Hom}_{S_{N+1}}\left(\bigoplus_{\mu \in \lambda + \square} \pi_\mu, V^{\otimes(N+1)}\right) = \bigoplus_{\mu \in \lambda + \square} S^\mu V.$$

Since  $V \otimes S^\lambda V = \bigoplus_{\mu \in \lambda + \square} S^\mu V$ , we conclude that  $\mathbb{C}S_{N+1} \otimes_{S_N} \pi_\lambda = \bigoplus_{\mu \in \lambda + \square} \pi_\mu$ .

(2) This is left as an exercise. Use a different version of Frobenius reciprocity.  $\square$

**Definition 4.1.** Let  $\lambda$  be a partition, and  $\lambda^\dagger$  be the *conjugate partition* (the one corresponding to the transposed diagram). For example,  $(3, 3, 2, 1)^\dagger = (4, 3, 2)$ .

**Corollary 4.0.1.** Let  $\mathbb{C}_-$  be the sign representation of  $S_N$ . Then  $\pi_\lambda \otimes \mathbb{C}_- \cong \pi_{\lambda^\dagger}$ .

*Proof.* This is left as an exercise. The proof is by induction on  $N = |\lambda|$ . Let  $C = \sum_{i < j} (i \ j)$ , and note that its eigenvalues are the same as the Casimir operator of  $\mathrm{SL}_N$ .  $\square$

**Proposition 4.2** (Skew Howe duality). *We have a decomposition  $\wedge^n(V \otimes W) = \bigoplus_\lambda S^\lambda V \otimes S^{\lambda^\dagger} W$  (as  $\mathrm{GL}(V) \otimes \mathrm{GL}(W)$ -modules).*

**Proposition 4.3.** *Every coset in  $P/Q$  contains a unique minuscule weight. This gives a bijection between  $P/Q$  and minuscule weights, so the number of minuscule weights is equal to  $\det A$ , where  $A$  is the Cartan matrix.*

*Proof.* Let  $C = a + Q \in P/Q$  be a coset. Let  $\omega \in C \cap P_+$  be the element which minimizes  $\langle \omega, \rho^\vee \rangle$ . If  $\lambda$  is the dominant weight for  $L_\omega$ , then  $\lambda \in C \cap P_+$  implies that

$$(\lambda, \rho^\vee) \geq (\omega, \rho^\vee).$$

Thus  $(\omega - \lambda, \rho^\vee) \leq 0$ , so  $\omega - \lambda \in Q_+$ . Thus  $\lambda = \omega$ , so  $\omega$  is minuscule. Now suppose  $\omega_1, \omega_2 \in C$  are minuscule and  $\omega_1 \neq \omega_2$  with  $\omega_1 - \omega_2 \in Q$ . By Lemma 3.3, we must have  $\langle \omega_1 - \omega_2, \beta \rangle \geq 2$  for all coroots  $\beta$ . But then  $\langle \omega_1, \beta \rangle = 1$  (which implies  $\beta > 0$ ) and  $\langle \omega_2, \beta \rangle = -1$  (which implies  $\beta < 0$ ), a contradiction.  $\square$

**Remark.** Let  $A$  be the Cartan matrix. For every root, we can write

$$\alpha_i = \sum_{j=1}^r A_{i,j} \omega_j.$$

We have a covering map  $\mathbb{R}^r / \Lambda_2 \rightarrow \mathbb{R}^r / \Lambda_1$ , where  $\Lambda_2 = P$  and  $\Lambda_1 = Q$ . Then  $\det A$  is precisely the degree of this covering, which counts the number of cosets.

## 4.2 Other Classical Lie Algebras

**Example 4.1.1.** Recall that  $\mathfrak{g} = \mathfrak{sp}_{2n}$  corresponds to the Dynkin diagram  $C_n$  ( $\bullet \cdots \bullet \rightleftarrows \bullet$ ), where the arrow points from longer roots to shorter roots. We have  $R_+ = e_i \pm e_j, 2e_j$ . The simple roots are

$$\alpha_1 = e_1 - e_2, \quad \alpha_2 = e_2 - e_3, \quad \dots, \quad \alpha_{n-1} = e_{n-1} - e_n, \quad \alpha_n = 2e_n.$$

We have  $\alpha_i^\vee = \alpha_i$  for  $i \neq n$  and  $\alpha_n^\vee = e_n$ , and  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  (with  $i$  ones) for  $1 \leq i \leq n$ .

**Example 4.1.2.** The Dynkin diagram  $B_n$  ( $\bullet \cdots \bullet \rightleftarrows \bullet$ ) corresponds to  $\mathfrak{g} = \mathfrak{so}_{2n+1}$ . Most things are the same as above, but we will have  $\alpha_n = e_n$  and  $\alpha_n^\vee = 2e_n$ . We have the same  $\omega_i$  for  $i < n$ , but we get  $\omega_n = (1/2, \dots, 1/2)$ . We have  $R_+ = e_i \pm e_j, e_i$ .

**Example 4.1.3.** The Dynkin diagram  $D_n$  ( $\bullet \cdots \bullet \begin{smallmatrix} \nearrow \\ \searrow \end{smallmatrix}$ ) corresponds to  $\mathfrak{g} = \mathfrak{so}_{2n}$ . In this case we have  $R_+ = e_i \pm e_j$ , and simple roots given by

$$\alpha_1 = e_1 - e_2, \quad \dots, \quad \alpha_{n-2} = e_{n-1}, \quad \alpha_{n-1} = e_{n-1} - e_n, \quad \alpha_n = e_{n-1} + e_n.$$

We have  $\omega_i = (1, \dots, 1, 0, \dots, 0)$  (with  $i$  ones) for  $i = 1, \dots, n-2$ , but we get  $\omega_{n-1} = (1/2, \dots, 1/2, 1/2)$  and  $\omega_n = (1/2, \dots, 1/2, -1/2)$ .

**Remark.** We have the following:

- For  $G_2, F_4, F_8$ , we have  $\det A = 1$  (here  $A$  is the Cartan matrix), so the only minuscule weight is 0.
- For  $B_n$ , we have  $\det A = 2$  (the nontrivial minuscule weight is  $(1/2, \dots, 1/2)$ , and the representation has weights  $(\pm 1/2, \dots, \pm 1/2)$  with all possible combinations of  $\pm$  and dimension  $2^n$ ).
- For  $D_n$ , we have  $\det A = 4$ . The minuscule weights are  $\omega_1, \omega_{n-1}, \omega_n$ . Here  $\omega_1$  is the  $2n$ -dimensional defining representation. The other two are spin representations of dimension  $2^{n-1}$ , with weights  $(\pm 1/2, \dots, \pm 1/2)$ , taking even or odd numbers of  $-$  signs.

## 4.3 Representations of Symplectic Lie Algebras

**Remark.** For  $\mathfrak{g} = \mathfrak{sp}_{2n}$ , we have the Dynkin diagram  $C_n$  and

$$\omega_i = (\underbrace{1, \dots, 1}_{i \text{ ones}}, 0, \dots, 0).$$



The elements of the Cartan subalgebra are given by  $\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n)$ . So  $L_{\omega_1} = V$  (the defining representation) with highest weight  $e_1$ . Note that  $\wedge^2 V$  is not irreducible:

$$\wedge^2 V = \wedge_0^2 V \oplus \mathbb{C},$$

where  $\mathbb{C}$  is the trivial representation spanned by  $B^{-1} = \sum_i e_{i+n} \wedge e_i$  (note that  $B^{-1}$  is invariant under  $\mathfrak{sp}_{2n}$ ). However, one can check that  $\wedge_0^2 V$  is irreducible.

Now let us consider  $L_{\omega_j}$  for  $j \geq 2$ . Let  $B = \sum_i e_i^* \wedge e_{i+n}^*$ . We have an operator

$$i_B : \wedge^{i+1} V \longrightarrow \wedge^{i-1} V,$$

and we can denote  $\wedge_0^i V = \ker(i_B|_{\wedge^i V})$  (note that  $i_B|_{\wedge^i V}$  is injective when  $i \geq n$ ). The  $\wedge_0^i V$  are irreducible for  $i \leq n$ , and one can check that these form all of the irreducible representations of  $\mathfrak{sp}_{2n}$  (compute their dimensions and compare them to the highest weight representations).

We can also define an operator

$$\begin{aligned} m_B : \wedge^{i-1} V &\longrightarrow \wedge^{i+1} V \\ u &\mapsto B^{-1} \wedge u. \end{aligned}$$

One can check that  $m_B$  and  $i_B$  together with  $h$  (acting as  $i - n$  on  $\wedge^i V$ ) form an  $\mathfrak{sl}_2$ -triple. Then

$$\wedge V = \bigoplus_{i=0}^n L_{\omega_i} \otimes L_{n-j}$$

(where  $\omega_0 = 0$  and  $L_{n-j}$  is the representation of  $\mathfrak{sl}_2$  of weight  $n - j$ ) as representations of  $\mathfrak{sp}_{2n} \oplus \mathfrak{sl}_2$ .

## 4.4 Representations of Orthogonal Lie Algebras

**Remark.** First consider  $B_n$ , which corresponds to  $\mathfrak{g} = \mathfrak{so}_{2n+1}$ . Let  $Q = \sum_{i=1}^n x_i x_{i+n} + x_{2n+1}^2$ . In this case, the Cartan subalgebra is given by elements of the form  $\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n, 0)$ . Let  $V$  be the  $(2n+1)$ -dimensional defining representation. Then for  $1 \leq i \leq n-1$ , the representation  $\wedge^i V$  is irreducible (one can check this using the dimension formula) with highest weight

$$\omega_i = (\underbrace{1, \dots, 1}_i, 0, \dots, 0).$$

On the other hand,  $\wedge^n V$  is irreducible but not fundamental, with highest weight  $(1, \dots, 1) = 2\omega_n$ .

Now we consider the spin representation  $S$  (whose elements are called *spinors*). It has weights

$$(\pm 1/2, \pm 1/2, \dots, \pm 1/2)$$

(all possible combinations of  $\pm$ ). The character of  $S$  is given by

$$\chi_S(x_1, \dots, x_n) = (x_1^{1/2} + x_1^{-1/2}) \cdots (x_n^{1/2} + x_n^{-1/2}).$$

**Remark.** We will want to look at the Lie group  $\text{Spin}_{2n+1}(\mathbb{C})$ , the universal cover of  $\text{SO}_{2n+1}(\mathbb{C})$ . For  $n = 1$ , we have  $\mathfrak{g} = \mathfrak{so}_3(\mathbb{C}) = \mathfrak{sl}_2(\mathbb{C})$ . We will see that  $S$  is 2-dimensional, and  $\pi_1(\text{SO}_3(\mathbb{C})) = \mathbb{Z}/2\mathbb{Z}$ .

# Lecture 5

## Jan. 28 — Other Classical Lie Algebras, Part 2

### 5.1 More on Orthogonal Lie Algebras

**Proposition 5.1.** *For  $n \geq 3$ , we have  $\pi_1(\mathrm{SO}_n(\mathbb{C})) = \mathbb{Z}/2\mathbb{Z}$ .*

*Proof.* There is a deformation retract from the surface  $X_n$  defined by  $z_1^2 + \cdots + z_n^2 = 1$  in  $\mathbb{C}^n$  to the sphere  $X_n^{\mathbb{R}} = X_n \cap \mathbb{R}^n$  defined by  $x_1^2 + \cdots + x_n^2 = 1$  in  $\mathbb{R}^n$ : Let  $\vec{z} = \vec{x} + i\vec{y} \in X_n$  for  $\vec{x}, \vec{y} \in \mathbb{R}^n$ , and note that  $|\vec{z}|^2 = 1$  if and only if  $|\vec{x}|^2 - |\vec{y}|^2 = 1$  and  $\vec{x} \cdot \vec{y} = 0$ . We also have

$$(\vec{x} + ti\vec{y})^2 = |\vec{x}|^2 - t^2|\vec{y}|^2 = 1 + (1 - t^2)|\vec{y}|^2 \geq 1.$$

So we can define a homotopy  $f_t : X_n \rightarrow X_n$  by

$$f_t(\vec{z}) = \frac{\vec{x} + ti\vec{y}}{\sqrt{|\vec{x}|^2 - t^2|\vec{y}|^2}},$$

which satisfies  $|f_t(z)|^2 = 1$ ,  $f_1(z) = z$ , and  $f_0(z) \in X_n^{\mathbb{R}}$ . Now observe that  $\mathrm{SO}_n$  acts on  $X_n$  with fibers isomorphic to  $\mathrm{SO}_{n-1}$ , so we have a long exact sequence

$$\pi_2(X_n) \longrightarrow \pi_1(\mathrm{SO}_{n-1}(\mathbb{C})) \longrightarrow \pi_1(\mathrm{SO}_n(\mathbb{C})) \longrightarrow \pi_1(X_n).$$

The first and last groups are trivial for  $n \geq 4$ , so we have that  $\pi_1(\mathrm{SO}_{n-1}(\mathbb{C})) \cong \pi_1(\mathrm{SO}_n(\mathbb{C}))$ . Thus the result follows once one checks that  $\pi_1(\mathrm{SO}_3(\mathbb{C})) \cong \mathbb{Z}/2\mathbb{Z}$  (left as an exercise).  $\square$

**Remark.** Now consider  $D_n$ , which corresponds to  $\mathfrak{g} = \mathfrak{so}_{2n}$ . Let  $Q = \sum_{i=1}^n x_i x_{i+n}$ . The elements of the Cartan subalgebra are given by  $\mathrm{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n)$ . Let  $V$  be the  $2n$ -dimensional defining representation, and consider  $\wedge^i V$  for  $1 \leq i \leq n$ . We have  $\wedge^i V$  is irreducible for  $0 \leq i \leq n-1$ , and  $L_{\omega_i} = \wedge^i V$  for  $1 \leq i \leq n-2$ . Note that  $L_{(1, \dots, 1, 0)}$  is irreducible but not fundamental. Letting

$$\omega_{n-1} = (1/2, \dots, 1/2, 1/2) \quad \text{and} \quad (1/2, \dots, 1/2, -1/2),$$

the corresponding  $S_+ = L_{\omega_{n-1}}$  and  $S_- = L_{\omega_n}$  are the spin representations. The characters are

$$\chi_{S_{\pm}} = ((x_1^{1/2} + x_1^{-1/2}) \cdots (x_n^{1/2} + x_n^{-1/2}))_{\pm},$$

where the  $\pm$  denotes an even or odd number of  $-$  signs.

**Example 5.0.1.** We have  $\text{Spin}_4 = \text{SL}_2 \times \text{SL}_2$ , where factors correspond to  $S_+$  and  $S_-$ . We have  $\text{Spin}_5 = \text{Sp}_4$ , where  $S$  is the 4-dimensional defining representation, and  $\text{SO}_5 = \text{Sp}_4/\{\pm 1\}$ . We have  $\text{Spin}_6 = \text{SL}_4$ , where  $S_+, S_-$  are the 4-dimensional defining representation and its dual, and  $\text{SO}_6 = \text{SL}_4/\{\pm 1\}$ .

**Example 5.0.2.** Let  $V$  be a finite-dimensional vector space, and consider  $SV = \mathbb{C}[x_1, \dots, x_n]$ , where  $x_1, \dots, x_n$  is an orthonormal basis. Denote  $R^2 = \sum_{i=1}^n x_i^2 = S^2V$  and  $\Delta = \sum_{i=1}^n \partial^2/\partial x_i^2$ . Then:

1. Find a first-order differential operator making  $\{R^2, \Delta, \cdot\}$  an  $\mathfrak{sl}_2$ -triple. Make sure that it commutes with the  $\text{SO}(V)$  action.
2. Let  $H_m \subseteq S^m V$  be the subspace of harmonic polynomials. Then

$$SV = \bigoplus_{m=0}^{\infty} H_m \otimes W_m,$$

where  $H_m = L_{m\omega_1}$  is the irreducible representation of  $\text{SO}(V)$ , and  $W_m$  is the Verma module for  $\mathfrak{sl}_2$  of highest weight  $m$ .

## 5.2 Clifford Algebras

**Definition 5.1.** Let  $V$  be a finite-dimensional vector space (over  $\mathbb{C}$ ) and  $(\cdot, \cdot)$  a non-degenerate inner product on  $V$ . Give an associative algebra structure to  $V$  by

$$v^2 = \frac{1}{2}(v, v).$$

Such an algebra is called a *Clifford algebra*, and is denoted by  $\text{Cl}(V)$ .

**Corollary 5.0.1.**  $ab + ba = (a + b)^2 - a^2 - b^2 = (a, b).$

**Example 5.1.1.** The operators  ${}^i\partial/\partial x_i$  and  $dx_i \wedge \cdot$  define a Clifford algebra.

**Example 5.1.2.** Let  $e^i e^j + e^j e^i = \delta_{i,j}$ . Then  $D = \sum_{i=1}^n e^i \partial_i$  (the *Dirac operator*) satisfies  $D^2 = \Delta$ .

**Theorem 5.1.** The algebra  $\text{Cl}(V)$  is isomorphic to  $\text{Mat}_{2^n}(\mathbb{C})$  if  $\dim V = 2n$  and to  $\text{Mat}_{2^n}(\mathbb{C}) \oplus \text{Mat}_{2^n}(\mathbb{C})$  if  $\dim V = 2n + 1$ .

*Proof.* First consider the even case. Choose a basis  $a_1, \dots, a_n, b_1, \dots, b_n$  such that

$$(a_i, a_j) = (b_i, b_j) = 0, \quad (a_i, b_j) = \delta_{i,j}, \quad a_i a_j + a_j a_i = 0, \quad b_i b_j + b_j b_i = 0, \quad b_i a_i + a_i b_i = 1.$$

Consider  $\text{Cl}(V)$ -module  $M = \wedge(a_1, \dots, a_n)$  (note that  $\dim M = 2^n$ ) with action defined by

$$\rho(a_i)w = a_i w \quad \text{and} \quad \rho(b_i)w = \frac{\partial w}{\partial a_i}.$$

We have the relations

$$1 = \left[ a_i, \frac{\partial}{\partial a_i} \right] = a_i \frac{\partial}{\partial a_i} + \frac{\partial}{\partial a_i} a_i \quad \text{and} \quad a_j \frac{\partial}{\partial a_i} = -\frac{\partial}{\partial a_i} a_j$$

for  $i \neq j$ . Let  $c_{I,J} = a_{i_1} \cdots a_{i_k} b_{j_1} \cdots b_{j_m}$  for  $I = \{i_1, \dots, i_k\}$  and  $J = \{j_1, \dots, j_m\}$ . Check as an exercise that the  $c_{I,J}$  are linearly independent, then  $\rho : \text{Cl}(V) \rightarrow \text{End}(M)$  is an isomorphism.

If  $\dim V = 2n + 1$ , then we can pick an extra element  $z$  satisfying

$$(z, a_i) = (z, b_i) = 0 \quad \text{and} \quad (z, z) = 2,$$

with relations  $za_i + a_iz = zb_i + b_iz = 0$  and  $z^2 = 1$ . Then  $zw = \pm(-1)^{\deg w}wz$  for  $w \in M_{\pm}$ .  $\square$

**Remark.** There is an embedding  $\mathfrak{so}(V) \rightarrow \text{Cl}(V)$ . Define a map

$$\begin{aligned} \xi : \wedge^2 V = \mathfrak{so}(V) &\longrightarrow \text{Cl}(V) \\ a \wedge b &\longmapsto \frac{1}{2}(ab - ba) = ab - \frac{1}{2}(a, b). \end{aligned}$$

One can check that  $[\xi(a \wedge b), \xi(c \wedge d)] = \xi([a \wedge b, c \wedge d])$ , so  $\xi$  is a homomorphism of Lie algebras. We have  $\xi^*M$  for even dimensional  $V$  and  $\xi^*M_{\pm}$  for odd dimensional  $V$ , and

$$\rho_{\xi^*M}(a) = \rho_M(\xi(a))$$

gives  $\xi^*M$  the structure of an  $\mathfrak{so}(V)$ -representation (and similarly for  $\xi^*M_{\pm}$ . Notice that  $\chi^*M$  is reducible:

$$\xi^*M = (\xi^*M)_0 \oplus (\xi^*M)_1$$

as representations, where the first factor corresponds to even degree and the second to odd degree.

**Example 5.1.3.** We have the following:

1.  $(\xi^*M)_0 \cong S_+$  and  $(\xi^*M)_1 \cong S_-$  for even dimensional  $V$ .
2. If  $\dim V$  is odd, then  $\chi^*M_{\pm}$  are both isomorphic to  $S$ .

# Lecture 6

## Feb. 2 — Duals, Maximal Weights, Exponents

### 6.1 Dual Representations

**Remark.** Let  $L_\lambda$  be the irreducible representation of highest weight  $\lambda$ . What is the highest weight of the dual representation  $L_\lambda^*$ ? Let  $w_0$  be the maximal element in  $W$ .

**Proposition 6.1.** *We have  $L_\lambda^* = L_{-w_0(\lambda)}$ .*

*Proof.* Since  $\lambda$  is the highest weight in  $L_\lambda$ , for every weight  $\mu$  in  $L_\lambda$  we have  $\lambda - \mu \in Q_+$ . So

$$Q_- \ni w_0(\lambda - \mu) = w_0(\lambda) - w_0(\mu),$$

so  $w_0(\mu) - w_0(\lambda) \in Q_+$ . Thus  $w_0(\lambda) \leq w_0(\mu)$  for all  $\mu \in L_\lambda$ , so the length of  $w_0$  is  $|R_+|$ . Thus  $-w_0(\lambda)$  is the lowest weight of  $L_\lambda$ , which is the highest weight of  $L_\lambda^*$ .  $\square$

**Example 6.0.1.** Since the length of  $w_0$  is  $|R_+|$ ,  $w_0$  permutes the fundamental (co)weights and (co)roots, so  $w_0$  is an automorphism of Dynkin diagrams. Note that  $W$  acts on  $P/Q$ , and  $w_0$  acts as inversion.

- The Dynkin diagrams  $A_1, B_n, C_n, G_2, F_4, E_7, E_8$  have no automorphisms, so  $L_\lambda^* = L_\lambda$  for these.
- For  $A_n$  with  $n \geq 2$ , we have  $P/Q = \mathbb{Z}/n\mathbb{Z}$  (e.g. if  $V$  is the defining representation, then we have that  $L_{\omega_1}^* = V^* = \wedge^{n-1}V = L_{\omega_{n-1}}$ ).
- For  $E_6$ , we have  $P/Q = \mathbb{Z}/3\mathbb{Z}$ , where  $w_0$  exchanges the two minuscule weights.
- For  $D_{2n+1}$ , we have  $P/Q = \mathbb{Z}/4\mathbb{Z}$  and  $S_+^* = S_-$ . For  $D_{2n}$ ,  $P/Q = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ , and  $S_\pm^* = S_\pm$ .

### 6.2 Maximal Weights

**Definition 6.1.** Let *maximal weight* of  $\mathfrak{g}$ , denoted  $\theta$ , is the highest weight of the adjoint representation.

**Example 6.1.1.** If  $\mathfrak{g} = \mathfrak{sl}_n$ , then  $\theta$  is the highest weight for  $V^* \otimes V$  where  $V$  is the defining representation. Note that  $V^* = \wedge^{n-1}V$ , so the highest weight of  $V^* \otimes V$  is  $\theta = \omega_1 + \omega_{n-1}$ . It is not fundamental.

**Example 6.1.2.** For  $\mathfrak{g} = \mathfrak{sp}_{2n}$ , we have  $\mathfrak{g} = S^2V$  where  $V$  is the defining representation for  $\mathfrak{sp}_{2n}$ . Then  $\theta = 2\omega_1$ , which is also not fundamental.

**Proposition 6.2.** *For a simple Lie algebra with  $\mathfrak{g} \neq \mathfrak{sl}_n, \mathfrak{sp}_{2n}$ , the maximal weight  $\theta$  is fundamental.*

**Example 6.1.3.** For  $\mathfrak{so}_N$  with  $N \geq 7$  (type  $B$  or  $D$ ), we have  $\mathfrak{g} = \wedge^2V = L_{\omega_2}$ .

### 6.3 Principal $\mathfrak{sl}_2$ -Subalgebra and Exponents

**Definition 6.2.** Let  $\mathfrak{g}$  be a simple Lie algebra and  $\{e_i, f_i, h_i\}$  (where  $h_i = \alpha_i^\vee$ ) be Chevalley generators. Let  $e = \sum_{i=1}^r e_i$ , and  $h$  such that  $\alpha_i(h) = 2$  for all  $i$  (so  $h = 2\rho^\vee$ ). Note that we have  $[h, e] = 2e$  and  $h = \sum_{i=1}^r (2\rho^\vee, \omega_i) \alpha_i^\vee$ . Let  $f = \sum_{i=1}^r (2\rho^\vee, \omega_i) f_i$ . Then  $\{h, e, f\}$  spans the *principal  $\mathfrak{sl}_2$ -subalgebra* of  $\mathfrak{g}$ .

**Example 6.2.1.** Let  $\mathfrak{g} = \mathfrak{sl}_{n+1}$ . Then the restriction of the defining representation to the principal  $\mathfrak{sl}_2$  is  $L_n$ , the irreducible representation of  $\mathfrak{sl}_2$  of highest weight  $n$ .

**Remark.** Let  $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ , so that  $\mathfrak{g} = \sum \mathfrak{g}[2m]$  where  $m$  is the height of the corresponding root subspace (and  $2m$  is the weight with respect to  $h$ ). Note  $\mathfrak{g}[0] = \mathfrak{h}$  and  $\dim \mathfrak{g}[0] = r$ . Let  $r_m = \dim \mathfrak{g}[2m]$ .

**Definition 6.3.** We say that  $m$  is an *exponent* of  $\mathfrak{g}$  if  $r_m > r_{m+1}$ . The *multiplicity* of an exponent  $m$  is  $r_m - r_{m+1}$ .

**Remark.** We have  $r_0 = r$  and there are  $r$  exponents (counted with multiplicities)  $m_1 \leq m_2 \leq \dots \leq m_r$ . The roots of height 2 are given by  $\alpha_i + \alpha_j$  (where  $i, j$  are connected in the in the Dynkin diagram). So  $r_0 = r_1 = 1$  and  $r_2 = r - 1$ . Thus  $m_1 = 1$  and  $m_2 > 1$ . We have

$$m_r = (\rho^\vee, \theta) = h_{\mathfrak{g}} - 1,$$

where  $\theta$  is the highest root. We call  $h_{\mathfrak{g}}$  the *Coxeter number* of  $\mathfrak{g}$ . Note that  $\sum_{i=1}^r m_i = |R_+|$ .

**Proposition 6.3.** The restriction of  $\mathfrak{g}$  to its principal  $\mathfrak{sl}_2$ -subalgebra decomposes as  $\bigoplus_{i=1}^r L_{2m_i+1}$ .

**Example 6.3.1.** The exponents for  $\mathfrak{sl}_n$  are  $1, 2, \dots, n-1$ .

**Definition 6.4.** The *Coxeter number* of  $\mathfrak{g}$  is  $h_{\mathfrak{g}} = \langle \theta, \rho^\vee \rangle + 1 = m_r + 1$ , and the *dual Coxeter number* is

$$h_{\mathfrak{g}}^\vee = \langle \tilde{\theta}^\vee, \rho \rangle + 1,$$

where  $\tilde{\theta}^\vee = 2\theta/(\theta, \theta)$ . If we normalize  $(\theta, \theta) = 2$ , then  $h_{\mathfrak{g}}^\vee = \frac{1}{2}(\theta, \theta + 2\rho)$ , which is the eigenvalue of  $\frac{1}{2}C$  (where  $C$  is the Casimir operator).

### 6.4 Complex, Real, and Quaternionic Types

**Definition 6.5.** Let  $G$  be a Lie group. An irreducible representation  $V$  of  $G$  or  $\mathfrak{g}$  is of *complex type* if  $V \not\cong V^*$ , *real type* if there exists a symmetric isomorphism  $V \rightarrow V^*$  (i.e. a symmetric inner product for  $V$ ), and *quaternionic (or symplectic) type* if the isomorphism is given through an anti-symmetric inner product.

**Exercise 6.1.** Let  $V$  be an irreducible representation of a finite group  $G$ . Show that  $\text{End}_{\mathbb{R}G}(V)$  (i.e.  $V \otimes V^*$ ) can only be one of three types:

- complex type if  $\text{End}_{\mathbb{R}G}(V) \cong \mathbb{C}$ ,
- real type if  $\text{End}_{\mathbb{R}G}(V) \cong \text{Mat}_{2 \times 2}(\mathbb{R})$ ,
- quaternionic type if  $\text{End}_{\mathbb{R}G}(V) \cong \mathbb{H}$ .

**Example 6.5.1.** Let  $L_n$  be an irreducible representation of  $\mathfrak{sl}_2$ . Then  $L_n$  is of real type for even  $n$  and of quaternionic type for odd  $n$ . Thus  $L_n = S^n V$  where  $V = L_1$  is 2-dimensional. The invariant form on  $S^n V$  is  $S^n B$ , where  $B$  is a skew-symmetric invariant form on  $V$ .

**Proposition 6.4.** *Assume  $\lambda = -w_0(\lambda)$ , so that the corresponding representation is of real or quaternionic type. Then  $L_\lambda$  is of real type if  $(2\rho^\vee, \lambda)$  is even and of quaternionic type if it is odd.*

*Proof.* The number  $n = (2\rho^\vee, \lambda)$  is the eigenvalue of  $h$  (from the principal  $\mathfrak{sl}_2$ -subalgebra) on the highest weight vector. Thus we have a decomposition

$$L_\lambda|_{\mathfrak{sl}_2} = L_n \oplus \bigoplus_{m < n} k_m L_m,$$

where  $L_n$  has multiplicity 1. One can determine the type based on  $L_n$ . □

## 6.5 Review of Compact Lie Groups

**Remark.** Let  $G$  be a real Lie group of dimension  $n$ . Then  $\xi \in \wedge^n \mathfrak{g}^*$  gives a generating  $n$ -form  $\omega$ , which is non-vanishing if  $\xi$  is non-vanishing. This gives rise to left- and right-invariant measures  $\mu_L$  and  $\mu_R$  on  $G$ , which are unique up to a constant. We say that  $G$  is *unimodular* if  $\mu_L = \mu_R$  (up to constants).

When does  $\mu_L = \mu_R$ ? For a 1-dimensional representation  $V$  of  $G$ , let  $|V|$  be the representation of  $G$  on the same space where  $\rho_{|V|}(g) = |\rho_V(g)|$  (where  $\rho_V : G \rightarrow \text{Aut}(V) = \mathbb{R}^\times$ ).

**Proposition 6.5.** *We have  $\mu_L = \mu_R$  if and only if  $|\wedge^n \mathfrak{g}^*|$  is a trivial representation of  $G$ .*

*Proof.* We have  $\mu_L = \mu_R$  if and only if the left-invariant form is right- or left-invariant up to a sign. This is equivalent to  $\xi \in \wedge^n \mathfrak{g}^*$  being invariant up to a sign under the action of  $\mathfrak{g}$ . □

**Proposition 6.6.** *A compact group is unimodular.*

*Proof.* For compact groups, the representation  $|\wedge^n \mathfrak{g}^*|$  gives a continuous homomorphism  $G \rightarrow \mathbb{R}^+$ , whose only compact subgroup is  $\{1\}$ . The result follows by Proposition 6.5. □

**Proposition 6.7.** *Let  $V$  be an irreducible representation of  $G$ . Then  $V$  admits a  $G$ -invariant unitary structure.*

*Proof.* Take any positive Hermitian form  $B$  on  $V$ , and define

$$B_{\text{av}}(v, w) = \int_G B(\rho_V(g)v, \rho_V(g)w) dg.$$

This is well-defined and invariant by construction. □

**Corollary 6.0.1** (Weyl unitary trick). *Any finite-dimensional representation is completely reducible.*

*Proof.* Write  $V = W \oplus W^\perp$ . If  $W$  is invariant, then so is  $W^\perp$ . □

# Lecture 7

## Feb. 4 — Compact Groups

### 7.1 More on Exponents

**Theorem 7.1** (Chevalley's restriction theorem). *There is an isomorphism  $\mathbb{C}[\mathfrak{g}]^G \xrightarrow{\cong} \mathbb{C}[\mathfrak{h}]^W$ .*

**Theorem 7.2** (Harish-Chandra theorem). *There is an isomorphism  $\mathbb{C}[\mathfrak{h}]^W \xrightarrow{\cong} \mathcal{Z}(U(\mathfrak{g}))$ .*

**Remark.** Pick an ordering  $s_{i_1}, \dots, s_{i_r}$  of the simple roots. Then  $c = s_{i_1} \cdots s_{i_r}$  is the *Coxeter element*, and  $c^h = 1$  where  $h$  is the Coxeter number. Then the eigenvalues of  $c$  are  $\zeta^{m_i+1}$  where  $\zeta = e^{2\pi i/h}$  and the  $m_i$  are the exponents. Also note that  $|W| = \prod_{i=1}^r (m_i + 1)$ .

If  $e, f, h$  is the principal  $\mathfrak{sl}_2$ -triple, then one can consider  $e + \mathfrak{g}^f$  where  $\mathfrak{g}^f = \ker \text{ad}_f$ .

### 7.2 Matrix Coefficients

**Remark.** For the rest of this lecture, let  $G$  be a real compact group and  $V$  a finite-dimensional continuous complex representation of  $G$ .

**Definition 7.1.** A *matrix coefficient* of  $\rho_V : G \rightarrow \text{GL}(V)$  is a function  $G \rightarrow \mathbb{C}$  of the form

$$g \longmapsto \langle f, \rho_V(g)v \rangle$$

for some  $v \in V$  and  $f \in V^*$ .

**Proposition 7.1.** *Matrix coefficients are smooth.*

*Proof.* Call  $v \in V$  a smooth vector if  $\langle f, \rho_V(g)v \rangle$  is smooth for all  $f \in V^*$ . It is obvious that such vectors form a subspace of  $V$ , call it  $V_{\text{sm}} \subseteq V$ . Fix  $v \in V$  and  $\phi : G \rightarrow \mathbb{C}$  smooth and with compact support. Let

$$w = w(\phi, v) = \int_G \phi(g) \rho_V(g)v \, dg.$$

We claim that  $w$  is smooth. We have

$$f(\rho(h)w) = f\left(\rho_V(h) \int_G \phi(g) \rho_V(g)v \, dg\right) = \int_G f(\phi(g) \rho_V(hg)v) \, dg = \int_G f(\phi(h^{-1}g) \rho_V(g)v) \, dg.$$

Differentiating under the integral sign and noting that  $\phi(h^{-1}g)$  is smooth in  $h$ , we see that the above expression is smooth in  $h$ . Now choose a delta-like sequence  $\phi_n$  with compact support around 1 so that

$$\int_G \phi_n(g) \, dg = 1.$$



Then  $w_n = w(\phi_n, v) \rightarrow v$  and each  $w_n$  is smooth, so  $v$  is smooth.  $\square$

**Remark.** Let  $V$  be an irreducible representation of  $G$ . Then:

1.  $V$  has an invariant positive-definite inner product which is unique up to scaling;
2. one can use an orthonormal basis  $v_1, \dots, v_n$  to define matrix coefficients:

$$\psi_{V,i,j}(g) = v_j^*(\rho_V(g)v_i) = (\rho_V(g)v_i, v_j)$$

(note that this definition is independent of normalization).

**Theorem 7.3** (Orthonormality of matrix coefficients). *Let  $V, W$  be irreducible representations of  $G$ .*

1. *If  $V, W$  are not isomorphic, then*

$$\int_G \psi_{V,i,j}(g) \bar{\psi}_{W,k,\ell}(g) dg = 0.$$

2. *For  $V = W$ , we have*

$$\int_G \psi_{V,i,j}(g) \bar{\psi}_{V,k,\ell}(g) dg = \frac{\delta_{i,k} \delta_{j,\ell}}{\dim V}.$$

*Proof.* Let  $\{v_i\}$  and  $\{w_k\}$  be orthonormal bases for  $V$  and  $W$ , respectively. We have

$$\int_G \psi_{V,i,j}(g) \bar{\psi}_{W,k,\ell}(g) dg = \int_G ((\rho_V(g) \otimes \rho_{\bar{W}}(g))(v_i \otimes w_k), v_j \otimes w_\ell) dg$$

Define the operator

$$P = \int_G (\rho_V \otimes \rho_{\bar{W}})(g) dg = \int_G \rho_{V \otimes \bar{W}}(g) dg.$$

Since  $\bar{W} \cong W^*$ , we have  $P : V \otimes W^* \rightarrow V \otimes W^*$ . Thus

$$\text{Im } P \subseteq (V \otimes W^*)^G,$$

which is 0 if  $V \not\cong W$ . On the other hand, if  $V \cong W$ , then the only invariant is

$$\vec{u} = \sum_k (v_k \otimes \bar{v}_k),$$

so  $P$  is the orthogonal projection onto  $\vec{u}$ . Thus

$$P\vec{x} = \frac{(\vec{x}, \vec{u})}{(\vec{u}, \vec{u})} \vec{u},$$

so we have  $(P(v_i \otimes w_k), v_j \otimes w_\ell) = \delta_{i,j} \delta_{k,\ell} / (\dim V)$ .  $\square$

## 7.3 Peter-Weyl Theorem

**Theorem 7.4** (Peter-Weyl theorem). *The matrix coefficients  $\psi_{V,i,j}$  form an orthogonal basis in  $L^2(G)$ .*

**Remark.** Let  $V$  be a finite-dimensional irrep of  $G$ . There is a natural inclusion

$$\begin{aligned} i_V : V^* &\hookrightarrow \text{Hom}_G(V, L^2(G)), \\ f &\longmapsto [v \mapsto (\rho_{V^*}(\cdot)f)(v)]. \end{aligned}$$

We claim that  $i_V$  is also surjective. To see this, let  $\phi \in \text{Hom}_G(V, L^2(G))$ , i.e. an  $L^2$  function left-invariant

under  $G$ . Thus we have that

$$\phi(x) = \rho_{V^*}(xg^{-1})\phi(g)$$

(after modifying  $\phi$  on a set of measure zero). Setting  $g = 1$ , we get  $\phi(x) = \rho_{V^*}(x)\phi(1)$ , so we have

$$\xi : \bigoplus_{V \in \text{Irr}(G)} V \otimes V^* \cong \bigoplus_{V \in \text{Irr}(G)} V \otimes \text{Hom}_G(V, L^2(G)) \hookrightarrow L^2(G),$$

an embedding of  $(G \times G)$ -modules. Call the left-hand side  $L^2_{\text{alg}}(G)$ .

**Theorem 7.5** (Peter-Weyl theorem, alternative).  $L^2_{\text{alg}}(G)$  is dense in  $L^2(G)$ , i.e.

$$L^2(G) = \widehat{\bigoplus_{V \in \text{Irr}(G)} V \otimes V^*}.$$

**Example 7.1.1.** Let  $G = S^1 = U(1)$ . The irreducible representations of  $G$  are  $\psi_n(\theta) = e^{in\theta}$ . The  $e^{in\theta}$  form a basis of  $L^2(G) = L^2(S^1)$ , where the norm is given by

$$\|f\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta.$$

This is the usual Fourier series on  $S^1$ . The Peter-Weyl theorem extends this to non-abelian groups.

**Exercise 7.1.** Let  $G$  be a compact group and  $H$  a closed subgroup of  $G$ .

1. Show that  $L^2(G/H) = \widehat{\bigoplus_{V \in \text{Irr}(G)} N_H(V)V}$ , where  $N_H(V) = \dim V^H$  (the space of  $H$ -invariants).
2. Let  $G = \text{SO}(3)$  and  $H = \text{SO}(2)$ . Then show that  $L^2(G/H) = L^2(S^2) = \widehat{\bigoplus_{m \geq 0} N_H(m)L_{2m}}$ , and that  $N_H(m) = 1$  for every  $m$ .

## 7.4 Introduction to Quantum Mechanics

**Remark.** Let  $\mathcal{H}$  be a Hilbert space and  $H$  a self-adjoint operator on  $\mathcal{H}$ . The spectrum of  $H$  gives the *energy levels* of the system. The elements  $\psi(x, y, z) \in L^2(\mathbb{R}^3)$  are called *wave functions*, and we assume that they are normalized so that  $\|\psi\|_{L^2} = 1$ . This is so that

$$|\psi(x, y, z)|^2 \Delta V$$

gives the probability of a quantum particle to be in the region  $\Delta V$ .

In general, there is also a time dependence in the wave function  $\psi$ , so we have  $\psi(x, y, z, t)$ . The time dependence is governed by the Schroedinger equation:

$$i\partial_t \psi = H\psi.$$

One can solve this equation via separation of variables, and we can write

$$\psi(x, y, z, t) = \sum_N e^{-iE_N t} \psi_N(x, y, z),$$

where the  $\psi_N$  are eigenvectors satisfying  $H\psi_N = E_N\psi_N$ .

**Example 7.1.2.** For the hydrogen atom, we have

$$H = -\frac{1}{2}\Delta - \frac{1}{r},$$

where  $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$  is the Laplacian and  $r = \sqrt{x^2 + y^2 + z^2}$ . The  $\Delta/2$  is called the *kinetic part* of  $H$ , and the  $1/r$  is called the *potential part* of  $H$ .

# Lecture 8

## Feb. 9 — Hydrogen Atom

### 8.1 Bound States of the Hydrogen Atom

**Remark.** We are looking for eigenvectors for  $H = -\frac{1}{2}\Delta - \frac{1}{r}$ , i.e.  $\psi_N \in L^2(\mathbb{R}^3)$  such that  $H\psi_N = E_N\psi_N$  with  $E_N < 0$ . We first write the Laplacian in spherical coordinates:

$$\begin{aligned}\Delta &= \Delta_r + \frac{1}{r}\Delta_{\text{sph}} \\ \Delta_r &= \partial_r^2 + \frac{2}{r}\partial_r \\ \Delta_{\text{sph}} &= \frac{1}{\sin^2\theta}\partial_\phi^2 + \frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta \cdot),\end{aligned}$$

where  $\phi$  is the angle in the  $xy$ -plane and  $\theta$  is the angle from the positive  $z$ -axis. Then we have

$$\partial_r^2\psi + \frac{2}{r}\partial_r\psi + \frac{1}{r^2}\Delta_{\text{sph}}\psi = -2E\psi,$$

which is solved by  $\psi(r, \vec{u}) = f(r)\xi(\vec{u})$  for  $\vec{u} \in S^2$  satisfying

$$\begin{aligned}\Delta_{\text{sph}}\xi + \lambda\xi &= 0 \\ f''(r) + \frac{2}{r}f'(r) + \left(\frac{2}{r} - \frac{\lambda}{r^2} + 2E\right)f(r) &= 0.\end{aligned}\tag{*}$$

Note that (\*) implies  $\Delta_{\text{sph}}$  is rotationally invariant. By the Peter-Weyl theorem, we have

$$L^2(S^2) = \widehat{\bigoplus_{\ell \geq 0} L_{2\ell}},$$

where  $S^2 = \text{SO}(3)/\text{SO}(2)$  and  $L_{2\ell}$  are the irreps of  $\text{SO}(3)$ .

Let  $Y_\ell^0 \subseteq L_{2\ell}$  be a vector of weight 0, which is invariant under  $\text{SO}(2)$ . Thus it depends only on  $\theta$ . So we can write  $Y_\ell^0(\theta) = P_\ell(\cos\theta)$ , where  $P$  is a polynomial of degree  $\ell$ . By orthogonality,

$$\int_{-1}^1 P_k(z)P_\ell(z) dz = 0, \quad k \neq \ell.$$

Thus we can write

$$-\lambda_e P_\ell(z) = \Delta_{\text{sph}} P_\ell(z) = \partial_z(1 - z^2)\partial_z P_\ell(z).$$

From looking at the leading term we must have  $\lambda_\ell = \ell(\ell + 1)$ .

Now take  $Y_\ell^m \in L_{2\ell}$  for  $-\ell \leq m \leq \ell$ . Write  $Y_\ell^m(\phi, \theta) = e^{im\phi} P_\ell^m(\cos \theta)$ . So we have

$$\frac{-m^2}{1-z^2} P_\ell^m + \partial_z(1-z^2) \partial_z P_\ell^m + \ell(\ell+1) P_\ell^m = 0, \quad -\ell \leq m \leq \ell.$$

This equation has a unique solution (up to scaling) on  $[-1, 1]$ , given by

$$P_\ell^m = (1-z^2)^{m/2} \partial_z^{\ell+m} (1-z^2)^\ell.$$

Now we return to the radial equation:

$$f''(r) + \frac{2}{r} f'(r) + \left( \frac{2}{r} - \frac{\ell(\ell+1)}{r^2} + 2E \right) f(r) = 0.$$

Write  $f(r) = r^\ell e^{-r/n} h(2r/n)$ , where  $n$  is to be chosen later and  $h$  satisfies

$$\rho h''(\rho) + (2\ell + 2 - \rho) h'(\rho) + \left( n - \ell - 1 + \frac{1}{4}(1 + 2En^2) \rho \right) h(\rho) = 0.$$

Now choose  $n = 1/\sqrt{-2E}$ , so that  $E = -1/2n^2$ . Then the above equation becomes

$$\rho h''(\rho) + (2\ell + 2 - \rho) h'(\rho) + (n - \ell - 1) h(\rho) = 0.$$

This equation is known as the *generalized Laguerre equation*. To get  $\|\psi\|_{L^2}^2 < \infty$ , we must have

$$\int_0^\infty \rho^{2\ell+2} e^{-\rho} |h(\rho)|^2 d\rho < \infty,$$

where the extra  $+2$  in  $\rho^{2\ell+2}$  comes from the Jacobian. Solutions around 0 behave like  $\rho^s(1 + o(1))$ , so

$$s(s + 2\ell + 1) = 0.$$

Thus either  $s = 0$  or  $s = -2\ell - 1$ .

First consider when  $\ell = 0$ . Then  $s = -1$  and we have  $\rho^{-1}(1 + o(1))$ , so  $\psi \sim 1/r$  as  $r \rightarrow 0$ . Then

$$H\psi = E\psi + C\delta_0,$$

where  $\delta_0$  is the delta function at 0, so we do not get an eigenvector in this case.

Thus  $s = -2\ell - 1$ . Expanding  $h(\rho)$  in a series and substituting, we get the recursive formula

$$h_n(\rho) = \sum_{k=0}^{\infty} \frac{(1 + \ell - n) \cdots (k + \ell - n)}{(2\ell + 2) \cdots (2\ell + 1 + k) k!} \rho^k.$$

This series converges, and we have

$$\lim_{\rho \rightarrow \infty} \frac{h_n(\rho)}{\rho} = 1$$

unless the series terminates. Thus  $n - \ell - 1 \in \mathbb{Z}_{\geq 0}$ , so we can write

$$h_n(\rho) = \sum_{k=0}^{n-\ell-1} \frac{(1 + \ell - n) \cdots (k + \ell - n)}{(2\ell + 2) \cdots (2\ell + 1 + k) k!} \rho^k = L_{n-\ell-1}^{2\ell+1}(\rho),$$

which is known as the *generalized Laguerre polynomial*:

$$L_N^\alpha(\rho) = \sum_{k=0}^N (-1)^k \frac{N \cdots (N - k + 1)}{(\alpha + 1) \cdots (\alpha + k) k!} \rho^k.$$

**Theorem 8.1.** *The bound states (i.e. solutions to  $H\psi = E\psi$  in  $L^2(\mathbb{R}^3)$ ) of the hydrogen atom are*

$$\psi_{n,\ell,m}(r, \phi, \theta) = r^\ell e^{-r/n} L_{n-\ell-1}^{2\ell+1}(2r/n) Y_\ell^m(\theta, \phi),$$

where  $n \in \mathbb{Z}_{>0}$ ,  $\ell$  is an integer from  $0, \dots, n-1$ ,  $E_n = -1/2n^2$ , and  $m$  is an integer between  $-\ell, \dots, \ell$ .

**Remark.** In Theorem 8.1,  $n$  is known as the *principal quantum number*, and  $\ell$  is known as the *azimuthal quantum number*. Note that if  $\vec{L}^2 = L_x^2 + L_y^2 + L_z^2 = -\Delta_{\text{sph}}$ , where  $iL_x, iL_y, iL_z$  are the generators of  $\mathfrak{so}(3)$  satisfying  $[L_x, L_y] = -iL_z$ , then  $\vec{L}^2 = C = \ell(\ell+1)$  is the Casimir operator.

**Corollary 8.1.1.** *The space  $W_n$  of states with principal number  $n$  has dimension  $n^2$ .*

*Proof.* This follows from  $\sum_{\ell=0}^{n-1} (2\ell+1) = n^2$ . □

**Remark.** Note that  $\widehat{\bigoplus_n W_n}$  forms a proper, closed subspace  $L_0^2(\mathbb{R}^3)$  of  $L^2(\mathbb{R}^3)$ . We need to find all  $\varphi$  with  $(H\varphi, \varphi) \geq 0$  to reconstruct all of  $L^2(\mathbb{R}^3)$ . This corresponds to the continuous spectrum of  $H$ .

## 8.2 Spin

**Remark.** *Spin* is a kind of intrinsic angular momentum. Instead of just  $L^2(\mathbb{R}^3)$ , we should consider

$$L^2(\mathbb{R}^3) \otimes \mathbb{C}^2 = L^2(\mathbb{R}^3) \otimes L_1$$

to be the space of states for the hydrogen atom. We have

$$V_n = (L_0 \oplus L_2 \oplus \dots \oplus L_{2n-2}) \otimes L_1 = 2L_1 \oplus 2L_3 \oplus \dots \oplus 2L_{2n-3} \oplus 2L_{2n-1},$$

so  $\dim V_n = 2n^2$ . We have an additional *spin operator* given by

$$S_z = \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix},$$

which acts on  $\mathbb{C}^2$  in the standard basis  $e_+, e_-$ . Then we have

$$\psi_{n,\ell,m,+} = \psi_{n,\ell,m} \otimes e_+ \quad \text{and} \quad \psi_{n,\ell,m,-} = \psi_{n,\ell,m} \otimes e_-.$$

The *total spin* is  $m + s$  (where  $s$  is the eigenvalue for  $S_z$ ), which is either  $m + 1/2$ , or  $m - 1/2$ .

## 8.3 Pauli Exclusion Principle

**Remark.** The space  $\wedge^k V_n$  corresponds to the space of states for  $k$  electrons at energy level  $n$ . Note that we must have  $k \leq 2n^2$  to have  $\wedge^k V_n \neq 0$ , which gives the *Pauli exclusion principle*.

In the periodic table, one has *orbitals*  $s, p, d, f$  corresponding to  $\ell = 0, 1, 2, 3$ , respectively, written with coefficient  $n$  and with exponent  $k$  corresponding to the number of electrons in the orbital. For example, the element Ruthenium has

$$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^7 5s^1.$$

The periodic table is organized as follows: from left to right ordered by how many *valent* electrons (i.e. the number of electrons in the outermost orbital), and from top to bottom ordered by how many energy levels. For Ruthenium, it is on column 8 and row 5. The number is 44, for 44 total electrons.

**Exercise 8.1.** Let  $\vec{r} = (x, y, z)$  and  $\vec{p} = (-\partial_x, -i\partial_y, -i\partial_z)$  be the *position* and *momentum* operators. Let  $\vec{L} = \vec{r} \times \vec{p}$  and  $H = \frac{1}{2}\vec{p}^2 + U(r)$ , where  $U$  is rotationally invariant. Show that:

1. The components  $i\vec{L}$  are generators of the rotations on  $\mathbb{R}^3$ , and  $[\vec{L}, \vec{p}^2] = 0$ .
2.  $\vec{A}_0 = \frac{1}{2}(\vec{p} \times \vec{L} - \vec{L} \times \vec{p})$  satisfies  $[\vec{A}_0, \vec{p}^2] = 0$ .
3. Let  $A = \vec{A}_0 + \phi(r)\vec{r}$ . There exists  $\phi$  such that  $[\vec{A}, H] = 0$  if and only if  $U$  is a *Columb potential* (i.e.  $U(r) = \frac{C}{r} + D$ ), and in this case  $\phi$  is completely determined.
4. (Hidden symmetry of the hydrogen atom) Use the commutation relations between  $\vec{A}$  and  $\vec{L}$  to define an action on  $\mathfrak{so}_4 = \mathfrak{so}_3 \oplus \mathfrak{so}_3$ , so that  $\vec{L}$  is the diagonal copy in this decomposition.
5.  $W_n = L_{n-1} \boxtimes L_{n-1}$  as representations of  $\mathfrak{so}_4 = \mathfrak{so}_3 \oplus \mathfrak{so}_3 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ .

# Lecture 9

## Feb. 11 — Real Forms

### 9.1 Automorphisms of Semisimple Lie Algebras

**Remark.** Recall that we can identify  $\text{Aut}(\mathfrak{g})$  with a Lie group with Lie algebra  $\mathfrak{g}$ . The connected component of the identity  $\text{Aut}^0(\mathfrak{g})$  (also known as the *adjoint group*  $G_{\text{ad}}$ ) acts transitively on the set of Cartan subalgebras. If  $\mathfrak{h} \subseteq \mathfrak{g}$  is a Cartan subalgebra, then there is a connected subgroup  $H \subseteq G_{\text{ad}}$  which acts as 1 on  $\mathfrak{h}$  and as  $e^{\alpha(x)}$  on  $g_\alpha$  for  $x \in \mathfrak{h}$ . Then we have

$$\mathfrak{h}/2\pi i P^\vee \cong H,$$

and  $H$  is called a *maximal torus*.

**Proposition 9.1.** *The normalizer  $N(H)$  of  $H$  in  $G_{\text{ad}}$  coincides with the stabilizer of  $\mathfrak{h}$  and contains  $H$  as a normal subgroup such that  $N(H)/H = W$  (the Weyl group).*

*Proof.* Note that  $\text{SL}_2(\mathbb{C})$  is simply connected, so for any simple root  $\alpha_i$  there is a homomorphism

$$\eta_i : \text{SL}_2(\mathbb{C}) \longrightarrow G_{\text{ad}} = \tilde{G}/\mathcal{Z}(\tilde{G}).$$

Define  $S_i = \eta_i \left( \begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix} \right)$ . Consider  $w = s_{i_1} \cdots s_{i_n}$  and  $\tilde{w} = S_{i_1} \cdots S_{i_n}$ . Note that  $\tilde{w}$  acts on  $\mathfrak{h}$  acts  $w$ , and if  $w = w_1 w_2$ , then  $\tilde{w} = \tilde{w}_1 \tilde{w}_2 h$  for some  $h \in H$ . To see the latter claim, note that  $h$  has the preserve the root decomposition, hence  $h|_{\mathfrak{g}_{\alpha_j}} = \exp(b_j)$ . Thus  $h = \exp(\sum_j b_j w_j^\vee) \in H$  (where  $\langle w_j^\vee, \alpha_i \rangle = \delta_{j,i}$ ).

So  $\tilde{w}$  and  $H$  generate a subgroup  $N \subseteq N(H)$  such that  $N/H = W$ . It remains to show that  $N(H) = N$ . Let  $x \in N(H)$  and consider simple roots  $\alpha'_i = x(\alpha_i)$ . Then there exists  $w \in W$  such that  $w(\alpha'_i) = \alpha_{p(i)}$  for some permutation  $p$ . Then  $\tilde{w}x(\alpha_i) = \alpha_{p(i)}$ , so this is a Dynkin diagram automorphism. Now  $\tilde{w}x$  is an element of a group, so the fundamental weights are fixed. Thus  $p = \text{id}$ .  $\square$

**Remark.** Although  $N(H)/H = W$ , in general  $N(H)$  is not a semidirect product of  $H$  and  $W$ .

**Proposition 9.2.** *The map  $\xi : \text{Aut}(D) \ltimes G_{\text{d}} \rightarrow \text{Aut}(\mathfrak{g})$  is an isomorphism.*

*Proof.* We have to show that  $\xi$  is surjective. Let  $a \in \text{Aut}(\mathfrak{g})$ . We can say that  $a$  preserves the Cartan subalgebra (if not, we can shift it by  $g \in G_{\text{ad}}$ ). Multiplying by  $\text{Aut}(D)N(H)$ , we can make it act trivially on  $\mathfrak{h}$  and  $\mathfrak{g}_{\alpha_i}$ . Then  $a = 1$ , so  $a \in \text{Im } \xi$ .  $\square$

### 9.2 Real Forms of Semisimple Lie Algebras

**Remark.** Recall the Serre presentation for  $\mathfrak{g}$ , i.e. generators  $\{h_i, f_i, e_i\}$  with certain relations. In this setting, everything was defined over  $\mathbb{Q}$ .

**Definition 9.1.** A semisimple Lie algebra is *split* if it admits a Chevalley-Serre basis over base field  $K$ .

**Remark.** Let  $L$  be a Galois extension of  $K$  ( $\text{char } K = 0$ ), and assume that  $\mathfrak{g}_L$  is a split semisimple Lie algebra. We want to find  $\mathfrak{g}$  over  $K$  such that  $\mathfrak{g} \otimes_K L = \mathfrak{g}_L$ . The problem is then to find a classification of all such  $\mathfrak{g}$ . Let  $\Gamma = \text{Gal}(L/K)$ . Define an action of  $\Gamma$  on  $\mathfrak{g}_L$  by

$$g(\lambda x) = g(\lambda)g(x), \quad x \in \mathfrak{g}_L, \lambda \in L, g \in \Gamma,$$

which is twisted linear. We can reconstruct  $\mathfrak{g}$  as the invariants  $\mathfrak{g}_L^\Gamma$ .

The simplest action of this kind is  $\rho_0(g)$ , which acts on scalars and preserves  $\{h_i, e_i, f_i\}$ . Any twisted linear action takes the form  $\rho(g) = \eta(g)\rho_0(g)$  for some  $\eta : \Gamma \rightarrow \text{Aut}(\mathfrak{g}_L)$ . As  $\rho$  is a homomorphism,

$$\eta(gh)\rho_0(gh) = \eta(g)\rho_0(g)\eta(h)\rho_0(h),$$

and upon rearranging, we have

$$\eta(gh) = \eta(g)g(\eta(h)),$$

where  $g(a) = \rho_0(g)a\rho_0(g)^{-1}$  for  $a \in \text{Aut}(\mathfrak{g}_L)$ . The above is called a *1-cocycle condition*.

Denote the Lie algebra associated to the cocycle  $\eta$  by  $\mathfrak{g}_\eta$ . When do we have  $\mathfrak{g}_{\eta_1} \cong \mathfrak{g}_{\eta_2}$ ? This is the case when  $\rho_1$  and  $\rho_2$  are isomorphic, i.e. there exists  $a \in \text{Aut}(\mathfrak{g}_L)$  such that  $\rho_1(g)a = a\rho_2(g)$ . Then

$$\eta_1(g)\rho_0(g)a = a\eta_2(g)\rho_0(g),$$

so  $\eta_1(g) = a\eta_2(g)g(a)^{-1}$ . Thus  $\eta_1$  and  $\eta_2$  are cohomologous cocycles.

**Proposition 9.3.** *The semisimple Lie algebras  $\mathfrak{g}$  over  $K$  which split over  $L$  (where  $L/K$  is Galois) are classified by  $H^1(\Gamma, \text{Aut}(\mathfrak{g}_L))$ , where  $\Gamma = \text{Gal}(L/K)$ .*

**Remark.** We will now specialize to  $K = \mathbb{R}$ ,  $L = \mathbb{C}$ , where  $\Gamma = \mathbb{Z}/2\mathbb{Z}$ , generated by complex conjugation. We have  $\text{Aut}(\mathfrak{g}_L) = \text{Aut}(D) \ltimes G_{\text{ad}}$ . Since  $\eta(1) = 1$ ,  $\eta$  is determined by  $\eta(-1)$ . The cocycle condition is

$$s\bar{s} = 1, \quad s = \eta(-1).$$

The corresponding Lie algebra (up to isomorphism) depends only on the cohomology class of  $s$ , where  $s \rightarrow as\bar{a}^{-1}$  for  $a \in \text{Aut}(D)$ .

**Theorem 9.1.** *The real semisimple Lie algebras whose complexification is  $\mathfrak{g}$  (i.e. the real forms of  $\mathfrak{g}$ ) are classified by  $s \in \text{Aut}(D) \ltimes G_{\text{ad}}$  such that  $s\bar{s} = 1$ , modulo the equivalence  $s \rightarrow as\bar{a}^{-1}$  for  $a \in \text{Aut}(D)$ .*

**Remark.** Note that complex conjugation acts trivially on  $\text{Aut}(D)$ .

**Remark.** Denote by  $\mathfrak{g}_{(s)} = \{x \in \mathfrak{g} : \bar{x} = s(x)\}$  the real form corresponding to  $s$ . Denote by  $\mathfrak{g}_{(1)}$  the split form consisting of real  $x \in \mathfrak{g}$  (so that  $x = \bar{x}$ ).

Alternatively, we can define an antilinear involution  $\sigma_s(x) = \overline{s(x)}$ . Then  $\mathfrak{g}_{(s)}$  is the fixed point set of  $\sigma_s$ .

**Remark.** Note that  $s$  defines  $s_0 \in \text{Aut}(D)$  with  $s_0^2 = 1$ .

**Corollary 9.1.1.** *The conjugacy class of  $s_0$  is invariant under equivalences.*

**Remark.** Since  $s_0$  permutes the connected components of the Dynkin diagram  $D$ , it preserves some and divides some into pairs. So every semisimple real Lie algebra is a direct sum of simple ones, and each simple one has either connected Dynkin diagram or consists of two identical components.



**Remark.** We now consider the case when  $D$  is connected and  $\mathfrak{g}$  is simple.

**Definition 9.2.** A real form  $\mathfrak{g}_{(s)}$  of a complex simple Lie algebra is said to be *inner to*  $\mathfrak{g}_{(s')}$  if  $s' = gs$  up to equivalence, where  $g \in G_{\text{ad}}$  (i.e.  $s, s'$  differ by an inner automorphism). The *inner class* of  $\mathfrak{g}_{(s)}$  is the collection of all real forms inner to  $\mathfrak{g}_{(s)}$ . An *inner form* is a form inner to a split form. We call  $\mathfrak{g}_{(s)}$  *quasi-split* if  $s = s_0 \in \text{Aut}(D)$ .

**Corollary 9.1.2.** *We have the following:*

1. *Any real form is inner to a unique quasi-split form.*
2. *A real form which is both inner and quasi-split is split.*

**Example 9.2.1.** Consider the *Cartan involution*  $\tau$  defined by

$$\tau(h_j) = -h_j, \quad \tau(e_j) = -f_j, \quad \tau(f_j) = -e_j.$$

Then  $\mathfrak{g}_{(\tau)} = \mathfrak{g}^c$  is called the *compact real form* of  $\mathfrak{g}$ .

# Lecture 10

## Feb. 16 — Real Forms, Part 2

### 10.1 Compact Real Forms

**Proposition 10.1.** *Let  $\tau$  be the Cartan involution (defined in Example 9.2.1) and  $\mathfrak{g}^c = \mathfrak{g}_{(\tau)}$ . Then the Killing form of  $\mathfrak{g}^c$  is negative definite.*

*Proof.* We can write  $\mathfrak{g}^c = (\mathfrak{h} \cap \mathfrak{g}^c) \oplus \bigoplus_{\alpha \in R_+} (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}) \cap \mathfrak{g}^c$ . The Killing form is negative definite on  $\mathfrak{h} \cap \mathfrak{g}^c$  since the inner product on a coroot lattice is positive definite. Thus it is negative definite on  $\mathfrak{g}^c$ , as  $\{i\alpha_j^\vee\}$  is a basis for  $\mathfrak{g}^c \cap \mathfrak{h}$ . Now we need to show that it is negative definite on  $(\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}) \cap \mathfrak{g}^c$ . Note that for  $\mathfrak{g} = \mathfrak{sl}_2$ , we have a basis for  $\mathfrak{g}^c$  given by  $ih, e - f, i(e + f)$ , so  $\mathfrak{g}^c = \mathfrak{su}(2)$ . So the statement holds there. For general  $\mathfrak{g}$ , we have that  $S_i$  preserves  $\mathfrak{g}^c$ , since

$$S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SU}(2).$$

So we have  $\mathrm{Lie}(\mathrm{SU}(2)_i) \subseteq \mathfrak{g}^c$ . For any  $w \in W$ , the lift  $\tilde{w}$  preserves  $\mathfrak{g}^c$ , so the restriction of the Killing form of  $\mathfrak{g}^c$  to  $\mathfrak{g}^c \cap (\mathfrak{sl}_2)_\alpha$  is negative definite.  $\square$

**Remark.** Consider  $\mathrm{Aut}(\mathfrak{g}^c)$ . Since the Killing form is negative definite,  $\mathrm{Aut}(\mathfrak{g}^c)$  is a closed subgroup of  $\mathrm{O}(\mathfrak{g}^c)$ , so it is compact. Moreover, this is a Lie group with Lie algebra  $\mathfrak{g}^c$ .

**Corollary 10.0.1.**  $G_{\mathrm{ad}}^c = \mathrm{Aut}(\mathfrak{g}^c)^\circ$  is a connected, compact Lie group with Lie algebra  $\mathfrak{g}^c$ .

**Example 10.0.1.** We have the following:

1. For  $\mathfrak{g} = \mathfrak{sl}_n$ , we have  $G_{\mathrm{ad}}^c = \mathrm{PSU}(n) = \mathrm{SU}(n)/\mu_n$ , where  $\mu_n$  is the  $n$ th roots of unity.
2. For  $\mathfrak{g} = \mathfrak{so}_n$ , we have  $G_{\mathrm{ad}}^c = \mathrm{SO}(n)$  for odd  $n$  and  $G_{\mathrm{ad}}^c = \mathrm{SO}(n)/\{\pm 1\}$  for even  $n$ .
3. For  $\mathfrak{g} = \mathfrak{sp}_{2n}$ , we have  $G_{\mathrm{ad}}^c = \mathrm{U}(n, \mathbb{H})/\{\pm 1\}$ . The group

$$\mathrm{U}(n, \mathbb{H}) = \mathrm{Sp}_{2n}(\mathbb{C}) \cap \mathrm{U}(2n)$$

is called the *quaternionic unitary group*.

**Example 10.0.2.** We have the following:

1. Consider  $A_{n-1}$ , which corresponds to the split form  $\mathfrak{sl}(n, \mathbb{R})$  and compact form  $\mathfrak{su}(n)$ . For  $n > 2$ , we have a quasi-split real form as follows: Let  $s(A) = -JA^T J^{-1}$  where  $J_{i,j} = (-1)^i \delta_{i,n+1-j}$ . Then

$$e_i, f_i, h_i, \mapsto e_{n+1-i}, f_{n+1-i}, h_{n+1-i}.$$

Note that  $J$  is a Hermitian or skew-Hermitian form of signature  $(p, p)$  with  $n = 2p$  or of signature  $(p + 1, p), (p, p + 1)$ , which are isomorphic when  $n = 2p + 1$ .

For  $n = 2$ , we have  $\mathfrak{su}(1, 1) = \mathfrak{sl}(2, \mathbb{R})$  (and  $\mathrm{PSU}(1, 1) = \mathrm{PSL}(2, \mathbb{R})$ ).<sup>1</sup>

2. For type  $B_n$ , we have compact form  $\mathfrak{so}(2n + 1)$  and split form  $\mathfrak{so}(n + 1, n)$ . There are no nontrivial automorphisms, so there are no non-split quasi-split forms.

Particular cases of interest are  $\mathrm{SO}(3) \cong \mathrm{SU}(2)$  and  $\mathrm{SO}^+(2, 1) = \mathrm{PSU}(1, 1) = \mathrm{PSL}(2, \mathbb{R})$ .

3. For type  $C_n$ , we have the split form  $\mathfrak{sp}(2n, \mathbb{R})$  and compact form  $\mathfrak{u}(n, \mathbb{H})$ . There are no non-split quasi-split real forms, as there are no nontrivial automorphisms of the Dynkin diagram.

Note that  $B_2 = C_2$ , so we have  $\mathfrak{so}(3, 2) = \mathfrak{sp}_4(\mathbb{R})$  and  $\mathfrak{so}(5) = \mathfrak{u}(2, \mathbb{H})$ .

4. For type  $D_n$ , we have split form  $\mathfrak{so}(n, n)$  and compact form  $\mathfrak{so}(2n)$ . For  $n > 4$ , there is a unique nontrivial involution, while for  $n = 4$ , we have  $\mathrm{Aut}(D) = S_3$ . However, there is still a unique non-split quasi-split form as there is only one nontrivial involution up to conjugation. Recall

$$A = -JA^T J^{-1}, \quad J_{i,j} = \delta_{i, 2n+1-j}.$$

Then the quasi-split form is given by  $J \mapsto J' = gJ$ , where  $g$  permutes  $e_n, e_{n+1}$  (which corresponds to  $\alpha_{n-1}, \alpha_n$ ). The signature defined by  $J'$  is  $(n + 1, n - 1)$ , so the quasi-split form is  $\mathfrak{so}(n + 1, n - 1)$ .

Note that  $D_2 = A_1 \oplus A_1$ , so we have the following isomorphisms:

$$\begin{aligned} \mathfrak{so}(4) &= \mathfrak{su}(2) \oplus \mathfrak{su}(2), \\ \mathfrak{so}(2, 2) &= \mathfrak{su}(1, 1) \oplus \mathfrak{su}(1, 1), \\ \mathfrak{so}(3, 1) &= \mathfrak{sl}_2(\mathbb{C}), \end{aligned}$$

where in the last isomorphism we view  $\mathfrak{sl}_2(\mathbb{C})$  as a real Lie algebra.

We also have  $D_3 = A_3$ , which gives the following isomorphisms:

$$\begin{aligned} \mathfrak{so}(6) &= \mathfrak{su}(4), \\ \mathfrak{so}(3, 3) &= \mathfrak{sl}_4(\mathbb{R}), \\ \mathfrak{so}(4, 2) &= \mathfrak{su}(2, 2). \end{aligned}$$

## 10.2 Classification of Real Forms

**Remark.** Write  $\mathfrak{g} = \mathfrak{g}^c \otimes_{\mathbb{R}} \mathbb{C}$  and  $\omega = \sigma_{\tau}$  the Cartan antilinear involution (so that  $\mathfrak{g}^c$  is the fixed points of  $\sigma_{\tau}$ ). Another real structure on  $\mathfrak{g}$  is given by  $\sigma = \omega \circ g$  for  $g \in \mathrm{Aut}(\mathfrak{g})$ , as

$$\sigma^2 = \omega \circ g \circ \omega \circ g = 1$$

(note that  $\omega \circ \omega = \mathrm{id}$ , and  $\omega(g)g = 1$ ). Define

$$(X, Y) = \mathrm{tr}(\mathrm{ad}_X \mathrm{ad}_{\omega(Y)}),$$

which is the Hermitian extension of the Killing form from  $\mathfrak{g}^c$  to  $\mathfrak{g}$ . Note that  $\omega(g) = (g^{\dagger})^{-1}$ , where  $g^{\dagger}$  is the adjoint of  $g$ . Thus we see that  $g$  is self-adjoint.

<sup>1</sup>Note that  $\mathrm{PSU}(1, 1)$  is the group of automorphisms of the unit disk, and  $\mathrm{PSL}(2, \mathbb{R})$  is the group of automorphisms of the upper half-plane. The isomorphism comes from the Cayley transform from the unit disk to the upper half-plane.

In particular,  $g$  is diagonalizable with real eigenvalues. So we can write

$$\mathfrak{g} = \bigoplus_{\gamma \in \mathbb{R}} \mathfrak{g}(\gamma),$$

where  $\mathfrak{g}(\gamma)$  is the eigenspace of  $\mathfrak{g}$  corresponding to eigenvalue  $\gamma$ . Note that

$$[\mathfrak{g}(\beta), \mathfrak{g}(\gamma)] = \mathfrak{g}(\beta\gamma).$$

Consider the operator  $|g|^t$  for  $t \in \mathbb{R}$ , which acts on  $\mathfrak{g}(\gamma)$  as  $|\gamma|^t$ . We can rewrite

$$|g|^t = \exp(t \log |g|) \in G_{\text{ad}},$$

which is a 1-parameter subgroup of  $G_{\text{ad}}$ . Then we can define  $\theta := g|g|^{-1}$ , and we have

$$\begin{cases} \theta \circ \omega = \omega \circ \theta, \\ \theta^2 = 1, \end{cases}$$

where the first identity follows from  $(\theta^\dagger)^{-1} = \theta$ . We have

$$\theta = |g|^{-1/2} g \omega(|g|^{1/2}).$$

We can assume  $g = \theta$  with  $\theta \circ \omega = \omega \circ \theta$ , or equivalently, that  $\theta \in \text{Aut}(\mathfrak{g}^c)$  and  $\theta^2 = 1$ . Thus  $\theta$  has  $\pm 1$  eigenspaces. Note that  $\theta'$  defines the same real form if and only if

$$\theta' = x\theta(\omega(x))^{-1}$$

for some  $x \in \text{Aut}(\mathfrak{g})$ . Then we have  $x\theta(\omega(x))^{-1} = \omega(x)\theta x^{-1}$  (since  $\theta'^2 = 1$ , so  $\theta'^{-1} = \theta'$ ). Let

$$z = (\omega(x))^{-1}x,$$

so that  $\omega(z) = z^{-1}$ . Then  $\theta z = z^{-1}\theta$ . Now note that  $z = x^\dagger x$  is positive definite, so if  $y = xz^{-1/2}$ ,

$$\omega(y) = \omega(x)z^{1/2} = (x^\dagger)^{-1}z^{1/2} = xz^{-1/2} = y.$$

Thus  $y \in \text{Aut}(\mathfrak{g}^c)$ , and we also have that

$$\theta' = x\theta\omega(x)^{-1} = x\theta z x^{-1} = xz^{-1/2}\theta z^{1/2}x^{-1} = y\theta y^{-1}.$$

**Theorem 10.1.** *The real forms of  $\mathfrak{g}$  are in one-to-one correspondence with the conjugacy classes of involutions  $\theta \in \text{Aut}(\mathfrak{g}^c)$ , where  $\theta \mapsto \omega_\theta = \theta \circ \omega = \omega \circ \theta$ .*

**Remark.** For  $\theta : \mathfrak{g} \rightarrow \mathfrak{g}$ , denote by  $\mathfrak{g}_\theta$  the corresponding real form. For example,

$$\mathfrak{g}_1 = \mathfrak{g}^c = \mathfrak{g}_{(\tau)},$$

where the latter is our old notation using the split forms.

**Remark.** We now have a canonical (up to automorphisms of  $\mathfrak{g}^c$ ) decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p},$$

where  $\theta = 1$  on  $\mathfrak{k}$  and  $\theta = -1$  on  $\mathfrak{p}$ . Here  $\mathfrak{k}$  is a Lie subalgebra and  $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$ . For  $\mathfrak{g}^c$  itself, we have

$$\mathfrak{g}^c = \mathfrak{k}^c \oplus \mathfrak{p}^c.$$

Moreover, we have  $\mathfrak{g}_\theta = \mathfrak{k}^c \oplus \mathfrak{p}_\theta$ , where  $\mathfrak{p}_\theta = i\mathfrak{p}^c$  (a fixed point of  $\sigma = \omega \circ \theta$  has to have an extra  $-$  sign, which we can achieve by multiplying by  $i$ ).

**Exercise 10.1.** Show that  $\mathfrak{k}$  is reductive but not necessarily semisimple.

# Lecture 11

## Feb. 18 — Real Forms, Part 3

### 11.1 Classification of Real Forms, Continued

**Proposition 11.1.** *There exists a Cartan subalgebra  $\mathfrak{h}$  in  $\mathfrak{g}$  which is invariant under  $\theta$  and such that  $\mathfrak{h} \cap \mathfrak{k}$  is a Cartan subalgebra in  $\mathfrak{k}$ .*

*Proof.* Consider a generic element  $t \in \mathfrak{k}^c$ . It is regular and semisimple. Consider  $\mathfrak{h}_+^c$ , the centralizer of  $t$  in  $\mathfrak{k}^c$ . Then necessarily  $\mathfrak{h}_+ := \mathfrak{h}_+^c \otimes_{\mathbb{R}} \mathbb{C}$  is a Cartan subalgebra in  $\mathfrak{k}$ . Let  $\mathfrak{h}_-^c$  be a maximal subspace of  $\mathfrak{p}^c$  so that  $\mathfrak{h}^c = \mathfrak{h}_+^c \oplus \mathfrak{h}_-^c$  is a commutative subalgebra of  $\mathfrak{g}^c$ . Then we claim that  $\mathfrak{h} = \mathfrak{h}^c \otimes_{\mathbb{R}} \mathbb{C}$  is a Cartan subalgebra of  $\mathfrak{g}$ . Note that  $\mathfrak{h}$  consists of the semisimple elements, and all elements in  $\mathfrak{g}^c$  are anti-self-adjoint operators. If  $z \in \mathfrak{g}$  commutes with  $\mathfrak{h}$ , then

$$z = z_+ + z_-, \quad z_+ \in \mathfrak{k}, z_- \in \mathfrak{p},$$

where  $z_{\pm}$  commute with  $\mathfrak{h}$ . Then  $z_+ \in \mathfrak{h}_+$ , and

$$z_- = x + iy, \quad x, y \in \mathfrak{p}^c,$$

where  $x, y$  commute with  $\mathfrak{h}$ . Then  $x, y \in \mathfrak{h}_-^c$  (by the definition of  $\mathfrak{h}_-^c$ ), so  $z \in \mathfrak{h}$ . □

**Corollary 11.0.1.** *We have  $\mathfrak{h} = \mathfrak{h}_+ \oplus \mathfrak{h}_-$ , where  $\theta = 1$  on  $\mathfrak{h}_+$  and  $\theta = -1$  on  $\mathfrak{h}_-$ .*

**Lemma 11.1.** *There are no coroots of  $\mathfrak{g}$  in  $\mathfrak{h}_-$ .*

*Proof.* Suppose otherwise that  $\alpha^\vee \in \mathfrak{h}_-$ . Then  $\theta(\alpha^\vee) = -\alpha^\vee$ , so

$$\theta(e_\alpha) = e_{-\alpha} \quad \text{and} \quad \theta(e_{-\alpha}) = e_\alpha$$

for some  $e_{\pm\alpha} \in \mathfrak{g}_{\pm\alpha}$ . Then  $x = e_\alpha + e_{-\alpha}$  satisfies  $\theta(x) = x$ , so  $x \in \mathfrak{k}$ . But  $x \notin \mathfrak{h}_+$  (since  $x \perp \mathfrak{h}_+$ ). Thus  $[\mathfrak{h}_+, x] = 0$  since  $\alpha$  vanishes on  $\mathfrak{h}_+$ , a contradiction as  $\mathfrak{h}_+$  is a maximal commutative subalgebra of  $\mathfrak{k}$ . □

**Remark.** Pick a generic element  $t \in \mathfrak{h}_+$ , which is regular in  $\mathfrak{g}$ . Choose  $t$  so that

$$\operatorname{Re}(t, \alpha^\vee) \neq 0$$

for all  $\alpha^\vee$  of  $\mathfrak{g}$ . Then we can define a *polarization* on  $R$  by

$$R_+ = \{\alpha \in R : \operatorname{Re}(t, \alpha^\vee) > 0\}$$

which satisfies  $\theta(R_+) = R_+$ . Now  $\{\theta(i) : i \in D\}$  gives the action of  $\theta$ : If  $\theta = i$ , then

$$\theta(e_i) = \pm e_i, \quad \theta(h_i) = h_i, \quad \theta(f_i) = \pm f_i.$$

Otherwise, if  $\theta(i) \neq i$ , then we can choose generators  $h_i, e_i, e_{\theta(i)}, f_i, f_{\theta(i)}, h_{\theta(i)}$  such that

$$\theta(x_i) = x_{\theta(i)}, \quad x = e, f, h.$$

We can then construct *markings* on the Dynkin diagram as follows:

- Connect vertices  $i$  and  $\theta(i)$  if  $\theta(i) \neq i$ .
- Mark a vertex  $i$  as white if  $\theta(e_i) = e_i$ .
- Mark a vertex  $i$  as black if  $\theta(e_i) = -e_i$ .

This is the *Vogan diagram* associated to the Dynkin diagram. Note that  $e_i \in P$  is a non-compact root.

**Exercise 11.1.** Showing the following:

1. The signature of the Killing form of  $g_\theta$  is  $(\dim \mathfrak{p}, \dim \mathfrak{k})$ . Moreover, the Killing form is negative definite if and only if  $\theta = 1$ , i.e.  $\mathfrak{g} = \mathfrak{g}^c$ .
2. For a split real form,  $\dim \mathfrak{k} = |R_+|$ .
3. Show that for any real form in a compact inner class,  $\text{rank}(\mathfrak{k}) = \text{rank}(\mathfrak{g})$ .

## 11.2 Real Forms of Classical Lie Algebras

**Example 11.0.1.** We have the following real forms of the classical Lie algebras:

1. Type  $A_{n-1}$ , compact inner class.

Let  $\theta$  be the inner automorphism element of  $\text{PSU}(n)$  of order 2. Let  $g \in \text{U}(n)$  such that  $g^2 = 1$ . Then  $\theta(x) = gxg^{-1}$ , so  $g = \text{id}_p \otimes (-\text{id}_q)$  with  $p + q = n$ . Thus

$$\mathfrak{g}_\theta = \mathfrak{su}(p, q) \quad \text{and} \quad \mathfrak{k} = \mathfrak{gl}_p \oplus \mathfrak{sl}_q.$$

For  $n = 2$ , we have  $\mathfrak{su}(2)$  and  $\mathfrak{su}(1, 1)$ , with  $\mathfrak{k} = \mathfrak{gl}_1$ .

2. Type  $A_{n-1}$ , split inner class.

If  $n$  is odd (so all vertices are divided into connected pairs), then

$$\mathfrak{g}_\theta = \mathfrak{sl}_n(\mathbb{R}).$$

If  $n$  is even, then there is 1 stable vertex (which is either black or white). In these cases we either have  $\mathfrak{k} = \mathfrak{sp}_{2k}$  (in which case  $\mathfrak{g}_\theta = \mathfrak{sl}(k, \mathbb{H})$ ) or  $\mathfrak{k} = \mathfrak{so}_{2k}$  (which is just the split form  $\mathfrak{sl}_n(\mathbb{R})$ ).

3. Type  $B_n$  (i.e.  $\mathfrak{so}_{2n+1}$ ).

Let  $\theta$  be the inner automorphism of order  $\leq 2$ . We can write

$$\theta = \text{id}_{2p+1} \oplus (-\text{id}_{2q})$$

where  $p + q = n$ . The real forms are  $\mathfrak{so}(2p + 1, 2q)$ , with  $\mathfrak{k} = \mathfrak{so}_{2p+1} \oplus \mathfrak{so}_{2q}$ .

4. Type  $C_n$ .

We can have  $g \in \text{Sp}_{2n}(\mathbb{C})$  such that  $g^2 = 1$  or  $g^2 = -1$ . The adjoint compact group is

$$(\text{Sp}(2n) \cap \text{U}(n))/\pm 1.$$

If  $g^2 = 1$ , then the eigenspace with eigenvalue 1 has dimension  $2p$ , and the eigenspace for  $-1$  has dimension  $2q$  (where  $p + q = n$ ). Assume that  $p \geq q$  (otherwise take  $g \mapsto -g$ ). Then we have

$$\mathfrak{sp}(2p, 2q) = \mathfrak{sp}_{2n} \cap \mathfrak{u}(p, q) = \mathfrak{u}(p, q, \mathbb{H}).$$

In this case, we have  $\mathfrak{k} = \mathfrak{sp}_{2p} \oplus \mathfrak{sp}_{2q}$ .

If  $g^2 = -1$ , then  $\mathbb{C}^{2n} = V(i) \oplus V(-i)$ . In this case,  $\mathfrak{k} = \mathfrak{gl}_n(\mathbb{C})$ , as for any  $(w, w) = w \cdot \bar{w}$  in  $\mathbb{C}^n$ ,

$$\mathrm{Im}(w, w) = i \mathrm{Im}(w \wedge \bar{w})$$

defines a symplectic form on  $\mathbb{R}^{2n}$ . Thus we can view  $U(n) \subseteq \mathrm{Sp}_{2n}(\mathbb{R})$ .

5. Type  $D_n$ , compact inner class.

In this case,  $\theta$  is given by  $g \in \mathrm{SO}(2n)$  with  $g^2 = \pm 1$ .

If  $g^2 = 1$ , then  $\mathbb{C}^{2n} = V(1) \oplus V(-1)$  (where  $\dim V(1) = 2p$  and  $\dim V(-1) = 2q$  with  $p + q = n$ ). Note that  $\det(g) = 1$ , so the eigenspaces are even-dimensional. Then

$$\mathfrak{g}_\theta = \mathfrak{so}(2p, 2q) \quad \text{and} \quad \mathfrak{k} = \mathfrak{so}(2p) \oplus \mathfrak{so}(2q).$$

If  $g^2 = -1$ , then  $\mathbb{C}^{2n} = V(i) \oplus V(-i)$ , so  $\mathfrak{k} = \mathfrak{gl}_n(\mathbb{C})$ . In this case, we have

$$\mathfrak{g}_\theta = \mathfrak{so}^*(2n),$$

which is known as the *quaternionic orthogonal Lie algebra*.

6. Type  $D_n$ , the other inner class.

In this case,  $\theta$  is given by  $g \in \mathrm{O}(2n)$  with  $\det(g) = -1$  and  $g^2 = \pm 1$ .

Note that if  $g^2 = -1$ , then  $\det(g) = 1$ , which is a contradiction. So we can only have  $g^2 = 1$ . Then

$$\mathbb{C}^{2n} = V(1) \oplus V(-1),$$

where  $\dim V(1) = 2p + 1$  and  $\dim V(-1) = 2q + 1$ . Here  $\mathfrak{k} = \mathfrak{so}(2p + 1) \oplus \mathfrak{so}(2q + 1)$ .

## 11.3 More on Compact Groups

**Exercise 11.2.** Show that if  $K^c$  is a compact Lie group, then  $\mathfrak{k} = \mathrm{Lie}_{\mathbb{C}}(K^c)$  is a reductive Lie algebra.

**Example 11.0.2.** Let  $G_{\mathrm{ad}} = \mathrm{Aut}(\mathfrak{g})^0$  for a semisimple Lie algebra  $\mathfrak{g}$ , and let  $G_{\mathrm{ad}}^c$  be its compact form. Consider the following product:

$$(S^1)^r \times G_{\mathrm{ad}}^c.$$

We will see that any Lie algebra of a compact group is isomorphic to a Lie algebra of such a product. We can also consider covering spaces, i.e. what is  $\pi_1(G_{\mathrm{ad}}^c)$ ?

# Lecture 12

## Feb. 23 — Classifications of Lie Groups

### 12.1 Classification of Compact Lie Groups

**Remark.** Let  $\mathfrak{g}$  be semisimple and  $G$  the corresponding simply connected Lie group (which is the universal cover for  $G_{\text{ad}} = \text{Aut}(\mathfrak{g})^0$ ). Define

$$Z = \ker(G \rightarrow G_{\text{ad}}) = \pi_1(G_{\text{ad}}).$$

Recall that the finite-dimensional representations of  $G$  are in one-to-one correspondence with the finite-dimensional representations of  $\mathfrak{g}$ , which (the irreducible ones) are given by  $\{L_\lambda\}$  for  $\lambda \in P_+$ . Then  $Z$  acts as  $\chi_\lambda : Z \rightarrow \mathbb{C}^\times$  on every  $L_\lambda$ , and  $\chi_\lambda \chi_\mu = \chi_{\lambda+\mu}$ . Note that

$$\chi : P \longrightarrow \text{Hom}(Z, \mathbb{C}^\times),$$

and  $\chi_\theta = 1$  (where  $\theta$  is the longest root) as  $Z$  acts trivially on the adjoint representation.

**Exercise 12.1.** If  $\lambda(h_i)$  is sufficiently large, then show that for all  $\alpha \in \mathfrak{g}$  we have

$$L_{\lambda+\alpha} \subseteq L_\lambda \otimes \mathfrak{g},$$

so in particular,  $\chi_{\lambda+\alpha} = \chi_\lambda$ , i.e.  $\chi_\alpha = 1$ .

**Remark.** We can define maps  $P/Q \rightarrow \text{Hom}(Z, \mathbb{C}^\times)$  and  $Z \rightarrow P^\vee/Q^\vee$  (and similarly for  $G_{\text{ad}}^c, G^c, Z^c$ ).

**Proposition 12.1.** *A representation  $L_\lambda$  of  $\mathfrak{g}$  of highest weight  $\lambda \in P_+$  lifts to a representation of  $G_{\text{ad}}$  (equivalently, of  $G_{\text{ad}}^c$ ) if and only if  $\lambda \in P_+ \cap Q$ .*

*Proof.* We have shown if  $\lambda \in P_+ \cap Q$ , then  $L_\lambda$  lifts. The converse follows from Proposition 12.2.  $\square$

**Proposition 12.2.** *If  $V$  is a faithful finite-dimensional representation of a compact Lie group, then any irrep  $Y$  is contained in  $V^{\otimes n} \otimes (V^*)^{\otimes m}$  for some  $n, m$ .*

**Lemma 12.1.** *If  $X$  is a compact manifold, then  $\pi_1(X)$  is finitely generated.*

*Proof.* Cover  $X$  with balls around each point, by the compactness of  $X$  there is a finite subcover. Let  $x_1, \dots, x_n$  be the centers of the finitely many balls, and create a graph  $G$  by connecting  $x_i$ . Then there is a surjection  $\pi_1(G) \rightarrow \pi_1(X)$ , and  $\pi_1(G)$  is finitely generated.  $\square$

**Theorem 12.1.** *Let  $\mathfrak{g}$  be a semisimple complex Lie algebra and  $G_{\text{ad}}^c$  the corresponding adjoint compact group. Then  $\pi_1(G_{\text{ad}}^c) = P^\vee/Q^\vee$ . In particular, the universal cover of  $G_{\text{ad}}^c$  is a compact Lie group.*



*Proof.* Let  $G_*^c$  be a finite cover of  $G_{\text{ad}}^c$  and define

$$Z_{G_{\text{ad}}^c} = \ker(G_*^c \rightarrow G_{\text{ad}}^c) \subseteq G_*^c.$$

A finite-dimensional irrep is classified by  $P_+(G_*^c) \subseteq P_+$  with  $P_+ \cap Q \subseteq P_+(G_*^c)$ . Let  $P(G_*^c) \subseteq P$  be the lattice generated by  $P_+(G_*^c)$ , and consider the character  $\chi_\lambda$  for the action of  $Z_{G_{\text{ad}}^c}$  on  $L_\lambda$  (an irrep of  $G_*^c$ ). Then  $\chi$  gives a map

$$\xi : P(G_*^c)/Q \longrightarrow Z_{G_{\text{ad}}^c}^\vee = \text{Hom}(Z_{G_{\text{ad}}^c}, \mathbb{C}^\times).$$

Since  $G_*^c$  is compact, by the Peter-Weyl theorem  $\xi$  is surjective. It just remains to show that  $\pi_1(G_{\text{ad}}^c)$  is finite. Let  $G_*^c = G^c$ , the universal cover. Then  $P(G_*^c) = P$ , so  $P/Q \cong Z^\vee$ , and thus

$$Z = \pi_1(G_{\text{ad}}^c) \cong P^\vee/Q^\vee.$$

By Lemma 12.1,  $\pi_1(G_{\text{ad}}^c)$  is finitely generated, and it is also abelian. Take a subgroup of finite index  $N$  and take  $G_*^c$  to be the corresponding cover. Then we have

$$N = |Z_{G_*^c}| \leq |P(G_*^c)/Q| \leq |P/Q|,$$

which for abelian groups implies that the group is finite.  $\square$

**Corollary 12.1.1.** *We have the following:*

1. *If  $\mathfrak{g}$  is a semisimple complex Lie algebra, then the simply connected Lie group  $G^c$  corresponding to the Lie algebra  $\mathfrak{g}^c$  is compact and its center is  $P^\vee/Q^\vee$ , which is the same as  $\pi_1(G_{\text{ad}}^c)$ .*
2. *Let  $\Gamma \subseteq P^\vee/Q^\vee$ . Then the irreps of  $G/\Gamma$  are the  $L_\lambda$  such that  $\lambda$  defines the trivial character of  $\Gamma$ .*
3. *Let  $G_i^c$  be compact Lie groups corresponding to simple Lie algebras  $\mathfrak{g}_i$ , and let  $\mathfrak{g} = \bigoplus_{i=1}^n \mathfrak{g}_i$ . Then any connected Lie group with Lie algebra  $\mathfrak{g}^c$  is compact and of the form*

$$\left( \prod_{i=1}^n G_i^c \right) / Z,$$

*where  $Z = \pi_1(G^c)$  is a subgroup of  $\prod_i Z_i$  for  $Z_i = P_i^\vee/Q_i^\vee$  (the center of  $G_i^c$ ). Moreover, every semisimple connected compact Lie group has this form.*

**Example 12.0.1.** Let  $G_*^c = \text{SO}(4, \mathbb{R})$ . Then  $G_{\text{ad}}^c = \text{SO}(3, \mathbb{R}) \times \text{SO}(3, \mathbb{R})$  and  $G^c = \text{SU}(2) \times \text{SU}(2)$ , and

$$\text{SO}(4, \mathbb{R}) = (\text{SU}(2) \times \text{SU}(2)) / \{\pm(1, 1)\}.$$

**Example 12.0.2.** Let  $G_*^c = \text{SO}(4, \mathbb{C})$ . Then  $G = \text{SL}(2, \mathbb{C}) \times \text{SL}(2, \mathbb{C}) = \text{Spin}(4)$ , and

$$\text{SO}(4, \mathbb{C}) = (\text{SL}(2, \mathbb{C}) \times \text{SL}(2, \mathbb{C})) / \{\pm(1, 1)\}.$$

In this case,  $G_{\text{ad}} = \text{PSL}(2, \mathbb{C}) \times \text{PSL}(2, \mathbb{C})$ .

**Corollary 12.1.2.** *Any connected compact Lie group with Lie algebra of the form  $\mathfrak{g}^c \oplus \mathfrak{a}$  with  $\mathfrak{a}$  abelian is a quotient of  $T \times C$  by a finite central subgroup, where  $T = (S^1)^m$  and  $C$  is compact, semisimple, and simply connected.*

## 12.2 Polar Decomposition

**Remark.** Consider  $G_{\text{ad}, \theta} \subseteq G_{\text{ad}}$  corresponding to  $\mathfrak{g}_\theta \subseteq \mathfrak{g}$ . Note that  $G_{\text{ad}, \theta}$  is a closed subgroup, but it may be disconnected (e.g. if  $\mathfrak{g}_\theta = \mathfrak{sl}(2, \mathbb{R}) \subseteq \mathfrak{sl}(2, \mathbb{C})$ , then  $G_{\text{ad}} = \text{PGL}_2(\mathbb{C})$  and  $G_{\text{ad}, \theta} = \text{PGL}_2(\mathbb{R})$ , which is disconnected as  $\det : \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$  and  $\mathbb{R} \setminus \{0\}$  is disconnected; but  $\mathbb{C} \setminus \{0\}$  is connected).

Now let  $K^c \subseteq G_{\text{ad},\theta}$  be the subgroup of elements acting on  $\mathfrak{g}$  by unitary operators, i.e.  $K^c$  is the fixed points of  $\omega_\theta$ . Note that  $K^c$  is a closed but possibly disconnected subgroup. Let  $\text{Lie}(K^c) = \mathfrak{k}^c$ , and note that  $K^c$  is compact. Let  $P_\theta = \exp(\mathfrak{p}_\theta) \subseteq G_{\text{ad},\theta}$ , where  $\mathfrak{p}_\theta = i\mathfrak{p}^c$  (note that  $P_\theta$  need not be a subgroup). Now  $P_\theta$  acts on  $\mathfrak{g}$  by Hermitian operators, so we get a diffeomorphism

$$\exp : \mathfrak{p}_\theta \rightarrow P_\theta,$$

thus  $P_\theta$  is a closed embedded submanifold in  $G_{\text{ad},\theta}$ .

**Theorem 12.2** (Polar decomposition). *The multiplication  $K^c \times P_\theta \rightarrow G_{\text{ad},\theta}$  is a diffeomorphism, hence*

$$G_{\text{ad},\theta} \cong K^c \times \mathbb{R}^{\dim \mathfrak{p}}$$

*as manifolds. In particular,  $G_{\text{ad},\theta}$  is homotopy equivalent to  $K^c$ .*

*Proof.* By the polar decomposition for matrices, any invertible matrix  $A$  can be written as  $A = U_A R_A$ , where  $U_A$  is unitary and  $R_A$  is positive Hermitian. Explicitly, we have

$$R_A = (A^\dagger A)^{1/2} \quad \text{and} \quad U_A = A(A^\dagger A)^{-1/2}.$$

Let  $g \in G_{\text{ad},\theta} \subseteq \text{Aut}(\mathfrak{g}) \subseteq \text{GL}(\mathfrak{g})$ . Note that  $g^\dagger g$  is an automorphism of  $\mathfrak{g}$  with positive eigenvalues, so  $(g^\dagger g)^{1/2} = R_g \in P_\theta$ , the positive self-adjoint elements. Since  $U_g$  is unitary, it has to belong to  $K^c$ . Thus we have constructed an inverse map to  $\mu$ .  $\square$

**Corollary 12.2.1.** *We have  $G_{\text{ad}} \cong G^{\text{ad}} \times \mathbb{P}$ , where  $\mathbb{P}$  is the set of elements acting on  $\mathfrak{g}$  by positive Hermitian operators. In particular,  $\pi_1(G_{\text{ad}}) = \pi_1(G_{\text{ad}}^c) = P^\vee / Q^\vee$ .*

**Corollary 12.2.2.** *If  $G$  is a semisimple complex Lie group, then the center  $Z$  of  $G$  is contained in  $G^c$ , hence  $Z$  coincides with the center  $Z^c$  of  $G^c$ . Thus the restriction of finite-dimensional representations from  $G$  to  $G^c$  defines an equivalence of categories.*

**Remark.** By considering coverings, the polar decomposition also applies to the real form  $G_\theta \subseteq G$  of any connected complex semisimple Lie group  $G$ . However, note that  $G_\theta$  need not be simply connected even if  $G$  is (e.g. for  $G = \text{SL}_2(\mathbb{C})$  we have  $G_\theta = \text{SL}_2(\mathbb{R}) \cong \text{SO}(2) \cong S^1$ , so  $\pi_1(G_\theta) \cong \mathbb{Z}$ ).

**Example 12.0.3.** We have the following:

1. If  $G = \text{SL}_n(\mathbb{C})$ , then  $K^c = \text{SU}(n)$  and  $P$  is the set of positive Hermitian matrices of determinant 1.
2. If  $G_\theta = \text{SL}_n(\mathbb{R})$ , then  $K^c = \text{SO}(n)$  and  $P_\theta$  is the positive symmetric matrices of determinant 1.

# Lecture 13

## Feb. 25 — Maximal Tori

### 13.1 Classification of Reductive Lie Groups

**Definition 13.1.** A connected complex Lie group  $G$  is *reductive* if it has the form

$$((\mathbb{C}^\times)^r \times G_{\text{ss}})/Z,$$

where  $G_{\text{ss}}$  is semisimple and  $Z$  is a finite central subgroup. A complex Lie group  $G$  is called *reductive* if  $G^0$  is reductive and  $G/G_0$  is finite.

**Example 13.1.1.** Recall that  $\text{GL}_n(\mathbb{C}) = (\mathbb{C}^\times \times \text{SL}_n(\mathbb{C}))/\mu_n$ , hence  $\text{GL}_n(\mathbb{C})$  is reductive.

**Remark.** With this definition, a simply connected complex Lie group with reductive Lie algebras is not necessarily reductive (e.g.  $\mathbb{C}$ ).

**Remark.** Given  $G = ((\mathbb{C}^\times)^r \times G_{\text{ss}})/Z$ , we have  $Z \subseteq (S^1)^r \times G_{\text{ss}} \subseteq (\mathbb{C}^\times)^r \times G_{\text{ss}}$ . Then

$$G^c = ((S^1)^r \times G_{\text{ss}}^c)/Z \subseteq G.$$

Recall that restriction from  $G$  to  $G^c$  defines an equivalence of categories between their representations, so the representations of  $G$  are completely reducible and the irreps are characterized by

$$(n_1, \dots, n_r, \lambda), \quad \lambda \in P_+(G_{\text{ss}}), \quad n_i \in \mathbb{Z}.$$

Thus we have the characterize the trivial character of  $Z$ .

**Definition 13.2.** A connected Lie group  $G$  is called *linear* if it can be realized as a subgroup of  $\text{GL}_n(\mathbb{C})$  (or  $\text{GL}_n(\mathbb{R})$ ) for some  $n$ .

**Remark.** Any complex semisimple Lie group can be realized as a linear one, but this is not true for real Lie groups (e.g. the universal cover of  $\text{SL}_2(\mathbb{R})$ ).

**Proposition 13.1.** Suppose  $\mathfrak{g}_\theta$  is a real form of a semisimple Lie algebra  $\mathfrak{g}$ , and let  $G$  be the connected complex Lie group corresponding to  $\mathfrak{g}$ . Define  $G_\theta = G^{\omega_\theta}$ . Then  $G_\theta$  and  $G_\theta^0$  are linear Lie groups, and every connected real semisimple Lie group is of the form  $G_\theta^0$  for some  $\mathfrak{g}_\theta$ .

### 13.2 Maximal Tori

**Remark.** Let  $\mathfrak{g}$  be a complex semisimple Lie algebra and  $\mathfrak{g}^c$  its compact form. Let  $G$  be the corresponding connected Lie group and  $G^c$  its compact part. Let  $\mathfrak{h}^c \subseteq \mathfrak{g}^c$  be a maximal commutative Lie subalgebra. Note that  $\mathfrak{h}^c \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{h} \subseteq \mathfrak{g}$ , where  $\mathfrak{h}$  is a Cartan subalgebra in  $\mathfrak{g}$ .

**Remark.** Recall that we have  $\{(\mathfrak{h}, \Pi)\}$  (here  $\mathfrak{h}$  is a Cartan subalgebra and  $\Pi$  the set of simple roots) are all conjugate: For any  $(\mathfrak{h}, \Pi)$  and  $(\mathfrak{h}', \Pi')$ , there exists  $g \in G$  such that

$$\text{Ad}_g(\mathfrak{h}, \Pi) = (\mathfrak{h}', \Pi').$$

The same happens for  $\mathfrak{g}^c$ .

**Lemma 13.1.** *Any two Cartan subalgebras in  $\mathfrak{g}^c$  equipped with systems of simple roots are conjugate under some  $g \in G^c$ .*

*Proof.* Fix  $(\mathfrak{h}^c, \Pi)$  and  $((\mathfrak{h}^c)', \Pi')$ . Then there exists  $g \in G$  such that

$$\text{Ad}_g(\mathfrak{h}^c, \Pi) = ((\mathfrak{h}^c)', \Pi'),$$

and for  $\bar{g} = \omega(g)$ , we have  $\text{Ad}_{\bar{g}}(\mathfrak{h}^c, \Pi) = ((\mathfrak{h}^c)', \Pi')$ . Then  $\bar{g}^{-1}g$  commutes with  $\mathfrak{h}^c$ , so it preserves  $\Pi$ . Now  $\bar{g}h = g$  for some  $h \in H = \exp(\mathfrak{h}_{\mathbb{C}}^c)$ . Write  $g = kp$  for some  $k \in G^c$  and  $p \in P$  by the polar decomposition. Then  $\bar{g} = kp^{-1}$ , so  $kp^{-1}h = kp$ , so  $h = p^2$ . So  $p = h^{1/2}$  commutes with  $\mathfrak{h}^c$  and preserves  $\Pi$ . Thus we can take  $k \in G^c$  to be the desired element.  $\square$

**Definition 13.3.** Given a Cartan subalgebra  $\mathfrak{h}^c \subseteq \mathfrak{g}^c$ , the corresponding Lie group  $H^c = \exp(\mathfrak{h}^c) \subseteq G^c$  is called a *maximal torus*.

**Corollary 13.0.1.** *Any two maximal tori in  $G$  or  $G^c$  equipped with systems of simple roots are conjugate.*

**Theorem 13.1.** *Every element of a connected compact Lie group  $K$  is contained in a maximal torus, and all maximal tori are conjugate.*

*Proof.* We can assume  $K$  is semisimple and  $K = G^c$ . Let  $K' \subseteq K$  be the set of elements contained in some maximal torus. Fix a maximal torus  $T \subseteq K$ , and define

$$\begin{aligned} f : K \times T &\longrightarrow K \\ k, t &\longmapsto ktk^{-1}. \end{aligned}$$

Clearly  $\text{Im } f = K'$ . Note that  $K'$  is compact, so  $K'$  is closed and hence  $K \setminus K'$  is open. Note that a generic  $x \in \mathfrak{g}^c$  is a regular element, i.e. its centralizer  $\mathfrak{z}_x$  has  $\dim \mathfrak{z}_x = \text{rank } \mathfrak{g}$ . Then every regular element  $g \in K$  is contained in the maximal torus  $\exp(\mathfrak{z}_x)$ , so the elements of  $K' \setminus K$  are non-regular. Since the non-regular elements are defined by a set of polynomial equations,  $K' \setminus K$  is also closed. Hence  $K' \setminus K$  must be empty by connectedness.  $\square$

**Corollary 13.1.1.** *On compact groups, the exponential map  $\exp : \mathfrak{g}^c \rightarrow G^c$  is surjective.*

## 13.3 Semisimple and Unipotent Elements

**Definition 13.4.** Let  $G$  be a connected complex reductive Lie group. An element  $g \in G$  is *semisimple* (resp. *unipotent*) if it acts in every finite-dimensional representation of  $G$  by semisimple (resp. unipotent, i.e. all eigenvalues are 1) operators.

**Exercise 13.1.** Recall that for any compact  $G$ , there is a faithful finite-dimensional representation  $Y$ . Show that  $g \in G$  is semisimple (resp. unipotent) if and only if it acts on  $Y$  by a semisimple (resp. unipotent) operator.

**Exercise 13.2.** Show that if  $G$  is semisimple, then the exponential map defines a homeomorphism between the set of nilpotent elements in  $\mathfrak{g} = \text{Lie}(G)$  and the set of unipotent elements in  $G$ .

**Exercise 13.3.** Let  $Z$  be the center of a connected complex reductive Lie group  $G$ .

1. Show that the homomorphism  $\pi : G \rightarrow G/Z$  defines a bijection between the unipotent elements of  $G$  and  $G/Z$ .
2. Show that the set of semisimple elements of  $G$  is the preimage under  $\pi$  of the set of semisimple elements in  $G/Z$ .

**Proposition 13.2** (Jordan decomposition). *Any  $g \in G$  has a unique factorization*

$$g = g_s g_u,$$

where  $g_s \in G$  is semisimple,  $g_u \in G$  is unipotent, and  $g_s g_u = g_u g_s$ .

*Proof.* We can reduce to the adjoint group. Then we can decompose

$$\text{Ad}_g = su$$

uniquely on the level of matrices, where  $s$  is semisimple and  $u$  is unipotent. It then suffices to show that  $s, u$  are  $\text{Ad}_{g_s}, \text{Ad}_{g_u}$  for some group elements. Fill in the details as an exercise.  $\square$

## 13.4 Cartan Decomposition

**Remark.** Let  $G$  be a connected complex semisimple Lie group. Let  $G_\theta \subseteq G$  be a real form, and write  $G_\theta = K^c P_\theta$ . Then  $\mathfrak{g}_\theta = \mathfrak{k}^c \oplus \mathfrak{p}_\theta$  and  $\mathfrak{g}^c = \mathfrak{k}^c \oplus \mathfrak{p}^c$  where  $\mathfrak{p}_\theta = i\mathfrak{p}^c$ .

**Proposition 13.3.** *We have the following:*

1. Let  $\mathfrak{a}$  be a maximal abelian subspace in  $\mathfrak{p}_\theta$ . Then the centralizer  $\mathfrak{z}$  of  $\mathfrak{a}$  in  $\mathfrak{g}^c$  has the form  $\mathfrak{m} \oplus \mathfrak{a}$ , where  $\mathfrak{m}$  is a reductive Lie algebra in  $\mathfrak{k}^c$ . Moreover,  $\mathfrak{t}$  is a Cartan subalgebra in  $\mathfrak{m}$ ,  $\mathfrak{t} \oplus i\mathfrak{a}$  is a Cartan subalgebra in  $\mathfrak{t} \oplus \mathfrak{a}$  is a Cartan subalgebra in  $\mathfrak{g}_\theta$ .
2. If  $a \in \mathfrak{a}$  is sufficiently generic in  $\mathfrak{p}_\theta$ , then the centralizer of  $a$  in  $\mathfrak{p}_\theta$  is  $\mathfrak{a}$ .
3. For all  $p \in \mathfrak{p}_\theta$ , there exists  $k \in K^c$  such that  $\text{Ad}_k(p) \in \mathfrak{a}$ .
4. All maximal subspaces of  $\mathfrak{p}_\theta$  are conjugated by  $K^c$ .

**Theorem 13.2** (Cartan decomposition). *Let  $\mathfrak{a} \subseteq \mathfrak{p}_\theta$  be a maximal abelian subspace. Let  $A = \exp(\mathfrak{a}) \subseteq P_\theta \subseteq G_\theta$ , which is isomorphic to  $\mathbb{R}^n$  where  $n = \dim \mathfrak{a}$ . Then  $G_\theta = K^c A K^c$ .*

**Remark.** The decomposition in Theorem 13.2 is not unique.

**Remark.** For  $G_\theta = \text{GL}_n(\mathbb{C})$ , the Cartan decomposition is  $U_1 D U_2$  where  $U_1, U_2$  are unitary and  $D$  is diagonal with positive entries, a classical theorem of linear algebra.

**Theorem 13.3** (E. Cartan). *Let  $G_\theta$  be a real form of a connected semisimple complex Lie group. Then any compact subgroup  $L$  of  $G_\theta$  is conjugated to a subgroup of  $K^c$  by an element of  $P_\theta$ . Also, every compact subgroup of  $G_\theta$  is contained in a maximal one. Therefore, all maximal compact subgroups are conjugate to each other.*

# Lecture 14

## Mar. 2 — Verma Modules

### 14.1 Integral Form of Weyl Character Formula

**Theorem 14.1.** *If  $f$  is a conjugation invariant continuous function on a compact connected Lie group  $K$  with maximal torus  $T \subseteq K$  and Haar probability measures  $dk$  and  $dt$ , then*

$$\int_K f(k) dk = \frac{1}{|W|} \int_T f(t) |\delta(t)|^2 dt,$$

where  $\delta(t) = \rho(t)^{-1} \prod_{\alpha \in R_+} (\alpha(t) - 1)$ .<sup>1</sup>

*Proof.* First (exercise) use the Peter-Weyl theorem to show that the characters form a dense set in the space of conjugation invariant functions. So it suffices to show the formula for  $f$  a character. Then

$$\int_K \chi_\lambda(k) dk = \delta_{\lambda,0}$$

is the left-hand side by the orthogonality of characters. Now write

$$\chi_\lambda(t) = \frac{\sum_{w \in W} \varepsilon(w) t^{w(\lambda+\rho)}}{\Delta}, \quad \Delta = \sum_{w \in W} \varepsilon(w) t^{w(\rho)} = t^{-\rho} \prod_{\alpha \in R_+} (t^\alpha - 1) = \rho(t)^{-1} \prod_{\alpha \in R_+} (\alpha(t) - 1),$$

so  $\Delta = \delta(t)$ . Here  $\varepsilon(w) = (-1)^{\ell(w)}$  is the sign of  $w$ . Then we get

$$\frac{1}{|W|} \int_T \chi_\lambda(t) \overline{\chi_\mu(t)} |\delta(t)|^2 dt = \frac{1}{|W|} \int_T \sum_{w \in W} \varepsilon(w) t^{w(\lambda+\rho)} \sum_{w' \in W} \varepsilon(w') t^{-w'(\mu+\rho)} dt$$

This is equal to  $\delta_{\lambda,\mu}$  by the orthogonality of characters, so we get the formula.  $\square$

**Remark.** If  $T \cong \mathbb{R}^r / \Lambda$  where  $\Lambda \cong \mathbb{Z}^r$  and  $\mathcal{F}$  is a fundamental domain for  $T$ , then  $H = \sum_{i=1}^r x_i e_i$  and

$$dt = \frac{dx_1 \dots dx_r}{\text{vol}(\mathcal{F})}.$$

For  $e^{i\theta_j}$  on  $U(n)$ , this is given by  $(d\theta_1 \dots d\theta_n) / (2\pi)^n$ .

**Exercise 14.1.** Let  $f$  be a conjugation invariant function on  $U(n)$ . Show that

$$\int_{U(n)} f(k) dk = \frac{1}{(2\pi)^n n!} \int_{|z_1|=\dots=|z_n|=1} f(\text{diag}(z_1, \dots, z_n)) \prod_{m < j} |z_m - z_j|^2 d\theta_1 \dots d\theta_n, \quad z_j = e^{i\theta_j}.$$

<sup>1</sup>Recall that if  $\alpha \in R_+$  and  $e^H = t$  for  $H \in \mathfrak{t} = \text{Lie } T$ , then we denote  $e^{\alpha(H)} = t^\alpha = \alpha(t)$ .

## 14.2 More on Verma Modules

**Remark.** Let  $\mathfrak{g}$  be a complex semisimple Lie algebra and  $\mathfrak{h} \subseteq \mathfrak{g}$  a Cartan subalgebra. Let  $\lambda \in \mathfrak{h}^*$  and  $M_\lambda$  the Verma module of highest weight  $\lambda$ , i.e. it is generated by one vector  $v_\lambda$  with

$$e_i v_\lambda = 0 \quad \text{and} \quad h(v_\lambda) = \lambda(h)v_\lambda$$

for  $1 \leq i \leq r$ . Note that  $M_\lambda = \text{Ind}_{\mathfrak{h} \oplus \mathfrak{n}^+} \mathbb{C}_\lambda$ , where  $\mathbb{C}_\lambda$  is the one-dimensional representation for  $\mathfrak{h} \oplus \mathfrak{n}^+$  defined by the above actions. Note that  $M_\lambda$  is irreducible for generic  $\lambda$ . In these cases,

$$L_\lambda = M_\lambda / I_\lambda$$

is irreducible, where  $I_\lambda$  is the maximal proper submodule of  $M_\lambda$ . Note that exchanging  $e_i$  and  $f_i$  in the above definitions gives the lowest weight Verma modules.

There is a unique symmetric bilinear form on  $M_\lambda$  satisfying the contravariance property:

$$\langle v_\lambda, v_\lambda \rangle = 1 \quad \text{and} \quad \langle e_i u, w \rangle = \langle u, f_i w \rangle$$

for  $u, w \in M_\lambda$ . This form is known as the *Shapovalov form*, and its kernel is precisely  $I_\lambda$ . Note that if  $\omega$  is the *Chevalley involution* given by  $\omega(e_i) = -f_i$ ,  $\omega(f_i) = -e_i$ , and  $\omega(h_i) = -h_i$ , then

$$\langle xu, v \rangle = -\langle u, \omega(x)v \rangle$$

is the contravariance property from above.

Let  $V = \bigoplus V^\mu$  be the weight decomposition (note that the weight spaces are finite-dimensional). Then the  $V^\mu$  and  $V^\nu$  are orthogonal with respect to the Shapovalov form for  $\mu \neq \nu$ . Let

$$V^* = \bigoplus_{\lambda} (V^\lambda)^*$$

be the *restricted dual* of  $V$ , and let  $\text{Hom}_{\mathbb{C}}(V_1, V_2)$  be the space of all linear maps.

**Proposition 14.1.** *If  $M_\lambda$  is irreducible, then its restricted dual is the lowest weight Verma module with lowest weight  $-\lambda$ .*

**Definition 14.1.** Define  $V^c$  to be the restricted dual of  $V$  with action of  $\mathfrak{g}$  twisted by  $\omega$ :

$$\langle xu^c, v \rangle = -\langle u^c, \omega(x)v \rangle, \quad u^c \in V^c, v \in V, x \in \mathfrak{g}.$$

Naturally,  $V^c$  has the same weight subspaces as  $V$ . We call  $M_\lambda^c$  the *contragredient Verma module*.

**Remark.** There is a unique  $M_\lambda \rightarrow M_\lambda^c$  with image  $L_\lambda$ , and combining it with the pairing above gives

$$M_\lambda^c \otimes M_\lambda \longrightarrow \mathbb{C},$$

which is exactly the Shapovalov form.

**Theorem 14.2.** *The Verma module is irreducible if and only if the following condition is satisfied: For any  $\alpha \in R_+$ , we have  $\langle \lambda + \rho, \alpha^\vee \rangle \notin \mathbb{N}$ .*

**Definition 14.2.** We will call  $\lambda$  *generic* if for all  $\alpha \in R_+$ , we have  $\langle \lambda, \alpha^\vee \rangle \notin \mathbb{Z}$ .

**Theorem 14.3.** *We have the following:*

1. The module  $L_\lambda$  is irreducible if and only if  $\lambda \in P^+$ .
2. The representations  $L_\lambda$  for  $\lambda \in P^+$  are pairwise non-isomorphic, and any finite-dimensional irrep is isomorphic to one of them.
3. If  $\lambda = \sum n_i \omega_i$  with  $n_i \in \mathbb{Z}$ , then  $L_\lambda$  is generated by  $v_\lambda$  and the action of  $\mathfrak{g}$  on  $v_\lambda$  with the constraints  $f_i^{n_i+1} v_\lambda = 0$  for  $i = 1, \dots, r$ . Moreover, we have  $L_\lambda^* \cong L_{-w_0(\lambda)}$ .

**Proposition 14.2.** The category  $C(\mathfrak{g})$  of finite-dimensional representations is semisimple, i.e. every module  $V$  is isomorphic to a direct sum of simple modules  $L_\lambda$  for  $\lambda \in P^+$ . Moreover, the map

$$\Phi \longmapsto \Phi(v_\lambda)$$

identifies the multiplicity space  $\text{Hom}_{\mathfrak{g}}(L_\lambda, V)$  for  $\lambda \in P^+$  with the space

$$(V^\lambda)^{\mathfrak{n}^+} = \{v \in V^\lambda : e_i v = 0\}$$

of singular vectors of weight  $\lambda$ .

### 14.3 Maximal Roots and the Casimir Operator

**Remark.** Recall that the *maximal root*  $\theta$  is the highest weight in the adjoint representation.

**Proposition 14.3.** The maximal root is uniquely determined by the following conditions: For all  $\alpha \in R$ , we have  $\theta - \alpha \in Q^+$ .

**Remark.** We normalize the pairing so that  $(\theta, \theta) = 2$ . Let

- $\theta = \sum_i h_i \alpha_i$ ,
- $h = 1 + \sum_i h_i$ ,
- $h_i^\vee = h_i(\alpha_i, \alpha_i)/2$  (these are the coefficients of the expansion of  $\tilde{\theta}^\vee$ ),
- $h^\vee = 1 + \sum_i h_i^\vee = (\rho, \theta) + 1$  (this is the dual Coxeter number).

For root systems of type A, D, E, we have  $h = h^\vee$ .

**Remark.** Recall that  $C = \sum x_i x^i$  is the *Casimir operator*, where  $\{x^i\}$  is the dual basis for  $x_i$ . Then

$$\Omega = \sum (x_i \otimes x^i)$$

does not depend on the choice of basis, and the value of  $C$  on any irreducible highest weight module  $L_\lambda$  is given by  $(\lambda, \lambda + 2\rho)$  (and for the lowest weight  $L_{-\mu}$  by  $(\mu, \mu + 2\rho)$ ). The value of  $C$  in the adjoint representation is given by  $2h^\vee$ .

**Exercise 14.2.** Show that in a tensor product of two representations, the action of  $\mathfrak{g}$  commutes with  $\Omega$ .

**Proposition 14.4.** We have

$$\Omega|_{V_\lambda \otimes V_\mu \otimes V_\nu} = \frac{(\lambda, \lambda + 2\rho) - (\mu, \mu + 2\rho) - (\nu, \nu + 2\rho)}{2}.$$

*Proof.* The proof relies on the identity

$$\Omega = \frac{1}{2}(\Delta(C) - c \otimes 1 - 1 \otimes c),$$

where  $\Delta : U(\mathfrak{g}) \rightarrow U(\mathfrak{g}) \otimes U(\mathfrak{g})$  is given by  $x \mapsto x \otimes 1 + 1 \otimes x$ . Note that  $(\Delta \otimes I)\Delta = (I \otimes \Delta)\Delta$ .  $\square$



## 14.4 Affine Lie Algebras

**Remark.** Consider a manifold  $M$  and a 1-form  $J$  satisfying  $d(*J) = 0$ . The 1-form  $J$  takes values in a Lie algebra, and we have

$$[J^a(x), J^b(y)] = f^{a,b,c} J^c(x) \delta(x - y),$$

where the  $f^{a,b,c}$  are the structure constants of the Lie algebra. These are called *current algebras*, see the book by J. Mickelson for more details (e.g. for maps  $S^3 \rightarrow \mathfrak{g}$ ).

We will consider the case where  $M$  is 2-dimensional. Then the conservation property becomes

$$\partial_{\bar{z}} J_z + \partial_z J_{\bar{z}} = 0.$$

In some cases, this can be separated, so that  $\partial_{\bar{z}} J_z = 0$ , i.e.  $J_z$  is holomorphic. Then we can write

$$J_z = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}.$$

The  $J_n$  in this expansion are called *modes* (one can think of this as some kind of wave expansion). Then the commutation relations become (with some correction term)

$$[J_n^a, J_m^b] = f^{a,b,c} J_{n+m}^c.$$

**Remark.** One can think of the modes above as loops. The *loop algebra* is

$$L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[t, t^{-1}].$$

One can also think of them as maps  $S^1 \rightarrow \mathfrak{g}$ . The commutation relations for  $L\mathfrak{g}$  are

$$[x \otimes P, y \otimes Q] = [x, y] \otimes PQ,$$

so that  $x \otimes t^n = x[n]$  has commutation relations

$$[x[n], y[m]] = [x, y][m + n].$$

This algebra has a unique nontrivial central extension, i.e. a short exact sequence of Lie algebras

$$0 \longrightarrow \mathfrak{g} \longrightarrow \widehat{\mathfrak{g}} \longrightarrow \mathbb{C}c \longrightarrow 0$$

Here  $\mathfrak{g}$  is a simple Lie algebra. Then on  $\widehat{\mathfrak{g}}$ , we have  $[c, x \otimes P] = 0$  for  $P \in \mathbb{C}[t, t^{-1}]$  and

$$[x \otimes P, y \otimes Q] = [x, y] \otimes PQ + c \frac{(x, y)}{2\pi i} \oint_{|t|=1} Q dP,$$

which gives the commutation relation

$$[x[n], y[m]] = [x, y][m + n] + n\delta_{n, -m}(x, y)c.$$

# Lecture 15

## Mar. 4 — Affine Lie Algebras

### 15.1 Affine Lie Algebras, Continued

**Remark.** We can extend  $\widehat{\mathfrak{g}}$  again by the grading operator  $d = t \frac{d}{dt}$  to get  $\widetilde{\mathfrak{g}} = \widehat{\mathfrak{g}} \oplus \mathbb{C}d$ , where

$$[d, x[n]] = nx[n] \quad \text{and} \quad [d, x \otimes P] = x \otimes t \frac{d}{dt} P.$$

We can also define a nondegenerate form on  $\widehat{\mathfrak{g}}$  by

$$(x \otimes P, y \otimes Q) = \frac{(x, y)}{2\pi i} \oint_{|t|=1} PQ \frac{dt}{t},$$

and on  $\widetilde{\mathfrak{g}}$  by  $(c, d) = 1$  and  $(c, c) = (d, d) = (c, x \otimes P) = (d, x \otimes P) = 0$ .

**Definition 15.1.** Define  $\widehat{\mathfrak{h}} \subseteq \widehat{\mathfrak{g}}$  by  $\widehat{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}c$  and  $\widetilde{\mathfrak{h}} \subseteq \widetilde{\mathfrak{g}}$  by  $\widetilde{\mathfrak{h}} = \widehat{\mathfrak{h}} \oplus \mathbb{C}d$ , these are Cartan subalgebras. Their duals are

$$\widehat{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \quad \text{and} \quad \widetilde{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta,$$

where  $\Lambda_0(c) = 1$ ,  $\Lambda_0(d) = \Lambda_0(h) = 0$  for  $h \in \mathfrak{h}$ , and  $\delta(d) = 1$ ,  $\delta(c) = \delta(h) = 0$  for  $h \in \mathfrak{h}$ .

**Remark.** We can write

$$\widetilde{\mathfrak{g}} = \widetilde{\mathfrak{h}} \oplus \bigoplus_{\gamma \in \widehat{R}} \widehat{\mathfrak{g}}^\gamma,$$

where  $\widehat{\mathfrak{g}}^\gamma = \{x \in \widetilde{\mathfrak{g}} : [h, x] = \gamma(h)x \text{ for } h \in \widetilde{\mathfrak{h}}\}$  and  $\widehat{R} = \{\alpha + n\delta : \alpha \in R \sqcup \{0\}, n \in \mathbb{Z} \text{ not both } 0\}$ . Then

$$\widetilde{\mathfrak{g}}^{\alpha+n\delta} = \mathfrak{g}^\alpha \otimes t^n \quad \text{and} \quad \widetilde{\mathfrak{g}}^{n\delta} = \mathfrak{h} \otimes t^n,$$

with  $\alpha \in R$ ,  $n \in \mathbb{Z}$ , and  $n \in \mathbb{Z} \setminus \{0\}$ , respectively. We can choose a polarization

$$\widehat{R} = \widehat{R}_+ \sqcup \widehat{R}_-, \quad \widehat{R}_+ = \{\alpha + n\delta : n > 0 \text{ or } n = 0, \alpha \in R_+\},$$

where as usual  $\widehat{R}_- = -\widehat{R}_+$ . The simple roots in  $\widehat{R}_+$  are the ones which cannot be represented as a nontrivial sum of positive roots. Let  $\alpha_1, \dots, \alpha_r$  be the simple roots of  $\mathfrak{g}$  and

$$\alpha_0 = \delta - \theta,$$

where  $\theta$  is the highest root. Then we can define  $\widehat{\mathfrak{n}}^\pm = \mathfrak{n}^\pm \oplus (\mathfrak{g} \otimes t^{\pm 1} \mathbb{C}[t^{\pm 1}])$  and write

$$\widetilde{\mathfrak{g}} = \widehat{\mathfrak{n}}^+ \oplus \widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{n}}^-.$$

The root lattices  $\widehat{Q} = \bigoplus_{i=0}^r \mathbb{Z}\alpha_i$  and  $\widehat{Q}^+ = \sum_{i=0}^r \mathbb{Z}_{\geq 0}\alpha_i$ . We can define the coroot

$$\alpha_0^\vee = \frac{2h_{\alpha_0}}{(\alpha_0, \alpha_0)} = h_{\alpha_0} = c - \theta^\vee.$$

Then we can define the weight lattice by  $\widehat{P} = \langle \Lambda \in \widetilde{\mathfrak{h}}^* : \Lambda(\alpha_i^\vee) \in \mathbb{Z} \rangle = P \oplus \mathbb{Z}\Lambda_0 \oplus \mathbb{C}\delta$ , and

$$\widehat{P}^+ = \{ \Lambda \in \widetilde{\mathfrak{h}}^* : \Lambda(\alpha_i^\vee) \in \mathbb{Z}_+ \} = \{ \Lambda = \lambda + k\Lambda_0 - \Delta\delta : \lambda \in P^+, k \in \mathbb{Z}_+, \Delta \in \mathbb{C}, \langle \lambda, \theta^\vee \rangle \leq k \}.$$

From this we get a Cartan matrix  $A = \{a_{i,j}\}_{i,j=0}^r$ .

## 15.2 Generalized Cartan Matrices

**Remark.** A *generalized Cartan matrix* is a matrix  $A = \{a_{i,j}\}_{i,j=0}^r$  satisfying

1.  $a_{i,i} = 2$  for all  $i$ ,
2.  $a_{i,j} \leq 0$  for  $i \neq j$ ,
3.  $a_{i,j} = 0$  if and only if  $a_{j,i} = 0$ .

**Definition 15.2.** Let  $A$  be a generalized Cartan matrix. The *Kac-Moody algebra*  $\mathfrak{g}(A)$  is the Lie algebra generated by elements  $\{e_i, h_i, f_i\}$  together with the standard Serre relations.

**Theorem 15.1.** The Lie algebra  $\widehat{\mathfrak{g}}$  is the Kac-Moody algebra associated with the Cartan matrix which we constructed above.

**Remark.** We have  $e_0 \in \mathfrak{g}^{-\theta} \otimes t$ ,  $f_0 \in \mathfrak{g}^{\theta} \otimes t^{-1}$ ,  $(e_0, f_0) = 1$ , and  $h_0 = [e_0, f_0] = c - \theta^\vee$ .

**Remark.** Let  $A$  be a generalized Cartan matrix and assume that it is symmetrizable, i.e. there exist  $d_i$  such that  $d_i a_{i,j} = d_j a_{j,i}$ . Then we can define an inner product

$$(h_i, h_j) := \frac{a_{i,j}}{d_i}$$

on the Cartan subalgebra. Let  $Z$  be the kernel of this inner product. Let

$$\mathfrak{g}_e(A) = \mathfrak{g}(A) \oplus Z^*$$

with commutation relations  $[z_1, z_2] = [z_1, h] = 0$ ,  $[z, e_j] = z(\alpha_j)e_j$ , and  $[z, f_j] = -z(\alpha_j)f_j$ . Then

$$\langle h_i, h_j \rangle = \frac{a_{i,j}}{d_j}, \quad \langle h, z \rangle = z(h), \quad \langle z_1, z_2 \rangle = 0$$

defines a nondegenerate pairing on  $\mathfrak{g}_e(A)$ .

## 15.3 Verma Modules and Weyl Modules

**Definition 15.3.** Let  $\Lambda \in \widetilde{\mathfrak{h}}^*$ . The *Verma module* corresponding to  $\Lambda$  is

$$M_\Lambda = \text{Ind}_{\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{n}}^+}^{\widehat{\mathfrak{g}}} \mathbb{C}_\Lambda$$

where  $\mathbb{C}_\Lambda$  is the 1-dimensional representation of  $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{n}}^+$ . Let  $v_\Lambda$  be the basis vector, with  $h v_\Lambda = \Lambda(h)v_\Lambda$ .

**Remark.** We have a weight decomposition given by

$$M_\Lambda = \bigoplus_{m \in \Lambda - \hat{Q}^+} M_\Lambda^m,$$

where  $\Lambda = \lambda + k\Lambda_0 - \Delta\delta$  for some  $\lambda \in \mathfrak{h}^*$ . From the  $\hat{\mathfrak{g}}$  perspective, we have

$$\Delta = \Delta(\lambda) = \frac{(\lambda, \lambda + 2\rho)}{2(k + h^\vee)}$$

(here  $h^\vee$  is the dual Coxeter number). We call  $k$  the *central charge* (or *level*) of the representation. In this case, we say that a  $\hat{\mathfrak{g}}$ -module  $V$  is of level  $k$  or  $V$  is a module with central charge  $k$ . We denote the corresponding Verma module  $M_\Lambda$  by  $M_{\lambda,k}$ .

The case  $k = -h^\vee$  is called the *critical level*, though this generally does not happen.

**Remark.** The Verma module  $M_{\lambda,k}$  has a natural  $\mathbb{Z}$ -grading given by

$$M_{\lambda,k} = \bigoplus_{n \geq 0} M_{\lambda,k}[-n],$$

where  $M_{\lambda,k}[-n]$  is naturally a  $\mathfrak{g}$ -module and an eigenspace of  $d$  with eigenvalue  $-n - \Delta(\lambda)$ .

**Example 15.3.1.** We have  $M_{\lambda,k}[0] = M_\lambda$ , the Verma module for  $\mathfrak{g}$ .

**Remark.** Let  $L_\Lambda = M_\Lambda / I_\Lambda$ , where  $I_\Lambda$  is the maximal proper submodule of  $M_\Lambda$ . Then  $L_{\lambda,k} := L_\Lambda$  is an irreducible module. The *Chevalley involution*  $\tilde{\omega}$  is given by

$$\tilde{\omega}(d) = -d$$

and on  $L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[t, t^{-1}]$  by  $t \mapsto t^{-1}$ . Then  $\tilde{\omega}$  gives a contravariant *Shapovalov form* with kernel  $I_\Lambda$ .

**Remark.** We can construct a class of representations of  $\tilde{\mathfrak{g}}$  induced from  $\mathfrak{g}$ -modules. Let

$$\tilde{\mathfrak{g}}^+ = \mathfrak{g} \otimes \mathbb{C}[t] \oplus \mathbb{C}c \oplus \mathbb{C}d \subseteq \tilde{\mathfrak{g}}.$$

If  $V$  is a module for  $\mathfrak{g}$ , then automatically  $V$  is also a module for  $\tilde{\mathfrak{g}}^+$  by letting

1.  $\mathfrak{g} \otimes t\mathbb{C}[t]$  act by 0,
2.  $c, d$  act by  $k, -\Delta$ , respectively.

Then one can define  $\text{Ind}_{\tilde{\mathfrak{g}}^+}^{\tilde{\mathfrak{g}}} V$ , which is a module for  $\tilde{\mathfrak{g}}$ .

**Example 15.3.2.** Verma modules belong to this class, as we can write

$$M_{\lambda,k} = \text{Ind}_{\tilde{\mathfrak{g}}^+}^{\tilde{\mathfrak{g}}} M_\lambda$$

where  $d$  acts on  $M_\lambda$  by  $-\Delta(\lambda)$ .

**Example 15.3.3.** Let  $V_{\lambda,k} = \text{Ind}_{\tilde{\mathfrak{g}}^+}^{\tilde{\mathfrak{g}}} L_\lambda$ , where  $d$  acts as  $-\Delta(\lambda)$ . If  $\lambda \in P_+$ , we call  $V_{\lambda,k}$  a *Weyl module*.

**Remark.** We get a contragredient Verma module by replacing  $\omega$  with  $\tilde{\omega}$ . This gives maps  $M_{\lambda,k} \rightarrow M_{\lambda,k}^c$  and  $V_{\lambda,k} \rightarrow V_{\lambda,k}^c$ , which are isomorphisms if  $M_{\lambda,k}, V_{\lambda,k}$  are irreducible.

## 15.4 Integrable Representations

**Theorem 15.2.** *The Verma module  $M_\Lambda$  is irreducible if the following condition is satisfied: For every  $\hat{\alpha} \in \hat{R}_+$  and  $N \in \{1, 2, \dots\}$ , we have  $(\Lambda + \hat{\rho}, \hat{\alpha}) \neq N(\hat{\alpha}, \hat{\alpha})/2$ , where  $\hat{\rho} = \rho + h^\vee \Lambda_0$ .*

**Remark.** For  $(\hat{\alpha}, \hat{\alpha}) \neq 0$ , the above condition becomes  $(\Lambda + \hat{\rho}, \hat{\alpha}^\vee) \neq N$ , which is the analogue of the simple Lie algebra case.

**Remark.** We call weights satisfying  $(\Lambda + \hat{\rho}, \hat{\alpha}) \neq N(\hat{\alpha}, \hat{\alpha})/2$  *generic*. For generic weights,  $L_\Lambda = M_\Lambda$ .

**Theorem 15.3.** *Let  $\lambda \in \mathfrak{h}^*$  and  $k \in \mathbb{C}$  be such that  $k \notin \mathbb{Q}\langle \alpha^\vee, \lambda \rangle + \mathbb{Q}$  for all  $\alpha \in R_+$ . Then the induced module  $V_{\lambda, k}$  is irreducible. If this holds, we say that  $k$  is generic with respect to  $\lambda$ .*

**Corollary 15.3.1.** *If  $\lambda \in P^+$  and  $k \notin \mathbb{Q}$ , then the Weyl module  $V_{\lambda, k}$  is irreducible.*

# Lecture 16

Mar. 9 —

# Lecture 17

Mar. 11 —

# Lecture 18

## Mar. 16 — Correlation Functions

### 18.1 Correlation Functions

**Remark.** Recall that if we write  $J_x(z) = J_x^+(z) - J_x^-(z)$ , then

$$[J_x^\pm(z), \Phi_n^{(g)}(w)] = \frac{1}{z-w} \Phi_{xn}^g(w).$$

Then  $\widehat{\Phi}^g(z) = z^{-\Delta(\mu)} \widetilde{\Phi}^g(z) = \sum_{n \in \mathbb{Z}} \Phi[n] z^{-n-\Delta}$ , where  $\Delta = \Delta(\lambda_1) - \Delta(\lambda_0) + \Delta(\mu)$  and

$$\Phi^g(z) : V_{\lambda_1, k} \longrightarrow V_{\lambda_0, k} \widehat{\otimes} V_\mu(z).$$

**Theorem 18.1** (FR). *We have*

$$(k + h^\vee) \frac{d}{dz} \widehat{\Phi}^g(z) = \sum_{a \in B} :J_a(z) \widehat{\Phi}_{an}^g(z):.$$

**Remark.** Let  $\widehat{\Phi}^{g_i}(z_i) : V_{\lambda_i, k} \widehat{\otimes} z^{-\Delta} V_{\mu_i}[z_i^{\pm 1}]$  and

$$\psi(z_1, \dots, z_N) = (\widehat{\Phi}^{g_1}(z_1) \otimes \dots \otimes 1) \dots (\widehat{\Phi}^{g_{N-1}}(z_{N-1}) \otimes 1) \widehat{\Phi}^{g_N}(z_N),$$

where  $\psi : V_{\lambda_N, k} \rightarrow V_{\lambda_0, k} \widehat{\otimes} V_{\mu_1}(z_1) \widehat{\otimes} \dots \widehat{\otimes} V_{\mu_N}(z_N)$ . Here  $V_{\mu_i}$  are the lowest weight modules with lowest weight  $-\mu_i$ . Let  $u_0 \in (V_{\lambda_0, k}[0])^* = L_{\lambda_0}^*$  and  $u_{N+1} \in V_{\lambda_N, k}[0] = L_{\lambda_N}$ . Then

$$\langle u_0, \psi(z_1, \dots, z_N) u_{N+1} \rangle \in V_{\mu_1} \otimes \dots \otimes V_{\mu_N}.$$

We can also treat this as lying in  $z_1^{-\Delta_1} \dots z_N^{-\Delta_N} \mathbb{C}[[z_2/z_1, \dots, z_N/z_{N-1}]]$ . Fix  $u_0$  and define a  $V$ -valued function (called the *correlation function*)

$$\psi(z_1, \dots, z_N) = \langle u_0, \psi(z_1, \dots, z_N) \cdot \rangle,$$

where  $V = V_{\mu_1} \otimes \dots \otimes V_{\mu_N} \otimes L_{\lambda_N}^*$ . For  $u_i \in V_{\mu_i}^*$ , we have

$$\psi_{u_1, \dots, u_{N+1}}(z_1, \dots, z_N) = \langle u_0, \Phi_{u_1}^{g_1} \dots \Phi_{u_N}^{g_N} u_{N+1} \rangle \in \mathbb{C}.$$

**Theorem 18.2.** *The correlation function  $\psi(z_1, \dots, z_N)$  satisfies the following system of differential equations:*

$$(k + h^\vee) \partial_{z_i} \psi = \left( \sum_{j=1, j \neq i} \frac{\Omega_{i,j}}{z_i - z_j} + \frac{\Omega_{i,N+1}}{z_i} \right) \psi,$$

where  $\Omega = \sum x_i \otimes x^i$  and  $A_{i,j} = (x)_i (y)_j$  for  $A = x \otimes y \in \mathfrak{g} \otimes \mathfrak{g}$ .



*Proof.* We can write

$$\begin{aligned}
& (k + h^\vee) \frac{\partial}{\partial z_i} \psi_{u_1, \dots, u_{N+1}}(z_1, \dots, z_N) \\
&= \sum_{a \in B} \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, (J_a^+(z_i) \widehat{\Phi}_{au_i}(z_i) - \widehat{\Phi}_{au_i}(z_i) J_a^-(z_i)), \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle \\
&= \sum_{j=1}^{i-1} \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, [\widehat{\Phi}_{u_j}(z_j), J_a^+(z_i)], \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle \tag{*}
\end{aligned}$$

$$+ \sum_{j=i+1}^N \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, [\widehat{\Phi}_{u_j}(z_j), J_a^-(z_i)], \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle \tag{**}$$

$$\begin{aligned}
& + \langle u_0, J_a^+(z_i) \widehat{\Phi}_{u_1}(z_1), \dots, \widehat{\Phi}_{au_i}(z_i), \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle \\
& - \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, \widehat{\Phi}_{au_i}(z_i), \dots, \widehat{\Phi}_{u_N}(z_N) J_a^-(z_i) u_{N+1} \rangle. \tag{***}
\end{aligned}$$

Now we have  $J_a^-(z) u_{N+1} = (-1/z) a u_{N+1}$  and  $\langle u_0, J_a^+(z) w \rangle = 0$  since  $w \in V_{\lambda, k}$ . Then

$$(*) = \sum_{j=1}^{i-1} \sum_{a \in B} \frac{1}{z_i - z_j} \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, \widehat{\Phi}_{au_j}(z_j), \dots, \widehat{\Phi}_{au_i}(z_i), \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle,$$

and similarly,

$$(**) = \sum_{j=i+1}^N \sum_{a \in B} \frac{1}{z_i - z_j} \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, \widehat{\Phi}_{au_i}(z_i), \dots, \widehat{\Phi}_{au_j}(z_j), \dots, \widehat{\Phi}_{u_N}(z_N) u_{N+1} \rangle.$$

We also get

$$(***) = \sum_{a \in B} \frac{1}{z_i} \langle u_0, \widehat{\Phi}_{u_1}(z_1), \dots, \widehat{\Phi}_{au_i}(z_i), \dots, \widehat{\Phi}_{u_N}(z_N) a u_{N+1} \rangle.$$

This is the action of the Casimir element  $\Omega$ . □

**Remark.** For  $j > i$ , we can write

$$\frac{1}{z_i - z_j} = z_i^{-1} \sum_{n=0}^{\infty} (z_j/z_i)^n.$$

**Remark.** The usual form of the KZ equations (call this the *symmetric* form) is written as

$$(k + h^\vee) \frac{\partial}{\partial z_i} \psi = \left( \sum_{j=1, j \neq i}^{N+1} \frac{\Omega_{i,j}}{z_i - z_j} \right) \psi$$

for  $i = 1, \dots, N+1$ . These systems are equivalent: If  $\tilde{\psi}(z_1, \dots, z_{N+1})$  is a solution to the above system, then  $\psi(z_1, \dots, z_N, 0)$  is a solution to the system in Theorem 18.2 (call this the *non-symmetric* form). Similarly, if  $\psi(z_1, \dots, z_N)$  is a solution to the system in Theorem 18.2, then  $\tilde{\psi}(z_1, \dots, z_{N+1}) = \psi(z_1 - z_{N+1}, \dots, z_N - z_{N+1})$  is a solution to the above system.

**Remark.** If  $\psi(z_1, \dots, z_{N+1})$  is a solution to the KZ equations, then so is  $\psi(z_1 + c, \dots, z_{N+1} + c)$  for constant  $c$ . Also notice that for  $\lambda_N = 0$ , the non-symmetric system turns into the symmetric one.

**Remark.** If  $u_0 \in (V_{\lambda_0, k}[0])^*$  is the lowest weight vector with respect to  $\mathfrak{g}$ , then

$$\psi(z_1, \dots, z_N) = \langle u_0, \psi(z_1, \dots, z_N) \cdot \rangle$$

(which a priori has values in  $V = V_{\mu_1} \otimes \dots \otimes V_{\mu_N} \otimes L_{\lambda_N}^*$ ) takes values in  $V^{n-}$ . If  $\lambda_0 = 0$ , then this is  $\mathfrak{g}$ -invariant, and  $\psi$  takes values in  $V^{\mathfrak{g}}$ .

**Theorem 18.3.** *The KZ system is consistent, i.e. for all  $p, q = 1, \dots, N+1$ ,*

$$\left[ (k + h^\vee) \partial_{z_p} - \sum_{j=1, j \neq p}^{N+1} \frac{\Omega_{p,j}}{z_p - z_j}, (k + h^\vee) \partial_{z_q} - \sum_{j=1, j \neq q}^{N+1} \frac{\Omega_{q,j}}{z_q - z_j} \right] = 0.$$

*Proof.* We trivially have  $\Omega_{p,q} = \Omega_{q,p}$ , so

$$[\Omega_{p,q}, \Omega_{p,r} + \Omega_{q,r}] = \left[ \sum_a a_p \otimes a_q, \sum_b b_p \otimes b_r + \sum_b b_q \otimes b_r \right] = 0.$$

Expanding this out, we get that

$$0 = \sum_{a, b \in B} (a_q [a_p, b_p] b_r + a_p [a_q, b_q] b_r) = \sum_{b \in B} [\Omega_{p,q}, b_p + b_q] b_r,$$

so that  $[\Omega, b \otimes 1 + 1 \otimes b] = 0$ . Since  $\Omega = \sum_a a \otimes a$ , this proves consistency.  $\square$

**Remark.** The above results imply that the KZ equations define a flat connection on a trivial vector bundle  $V$  over

$$X_{N+1} = \{(z_1, \dots, z_{N+1}) \in \mathbb{C}^N : z_i \neq z_j\}.$$

We will see later that for any open disk  $D \subseteq X_{N+1}$ , the space of solutions  $\Gamma(D, V_{\text{KZ}})$  is a space of flat sections of  $V_{\text{KZ}}$  over  $D$ . If  $D$  is simply connected, then  $\Gamma(D, V_{\text{KZ}}) \cong V$  by  $\psi \mapsto \psi(p)$ . In other words,  $V_{\text{KZ}}$  is a direct sum of finite dimensional vector bundles (i.e. *local systems*).

**Theorem 18.4.** *The KZ system is  $\mathfrak{g}$ -invariant, meaning that if  $x \in \mathfrak{g}$  and  $\psi$  is a solution, then  $x\psi$  is also a solution.*

*Proof.* Check as an exercise that  $[\sum_{j=1, j \neq p}^{N+1} \Omega_{p,j} / (z_p - z_j), x] = 0$ , where  $x = \sum_{i=1}^{N+1} x_i$ .  $\square$

## 18.2 Analyticity of Correlation Functions

**Remark.** We change variables from  $z_1, \dots, z_N$  to  $\zeta_i = z_{i+1}/z_i$  for  $i = 1, \dots, N-1$ , and  $\zeta_N = z_N$ . Then the KZ equations become the following system:

$$\zeta_i \frac{\partial}{\partial \zeta_i} \psi = A_i(\zeta_1, \dots, \zeta_{N-1}) \psi.$$

To see this, one can take logarithmic derivatives and then take differences. Here the  $A_i$  are holomorphic functions in the region

$$\{|\zeta_j| < 1 : 1 \leq j \leq N\},$$

and we have  $[A_i(0), A_j(0)] = 0$  for all  $i, j$ .

**Proposition 18.1.** *Let  $A_i(\zeta_1, \dots, \zeta_m)$  for  $1 \leq i \leq m$  be  $n \times n$  matrix-valued functions holomorphic in the region  $R = \{|\zeta_j| < 1 : 1 \leq j \leq m\}$  and such that the differential operators  $\zeta_i \partial_{\zeta_i} - A_i$  commute with each other. Assume also that for all  $i$ , the eigenvalues of  $A_i(0)$  do not differ by a nonzero integer. Then the differential equation*

$$\zeta_i \frac{\partial}{\partial \zeta_i} F = A_i(\zeta_1, \dots, \zeta_m) F, \quad i = 1, \dots, m$$

*has a unique  $n \times n$  matrix solution of the form*

$$F(\zeta_1, \dots, \zeta_m) = F_0(\zeta_1, \dots, \zeta_m) \zeta_1^{A_1(0)} \cdots \zeta_m^{A_m(0)},$$

*where  $F_0$  is an  $n \times n$  matrix valued function holomorphic in the region  $R$  and  $F_0(0) = \text{id}$ .*

**Corollary 18.4.1.** *In the notation of Proposition 18.1, let  $f$  be a vector-valued power series*

$$f(\zeta_1, \dots, \zeta_m) = \zeta_1^{\delta_1} \cdots \zeta_m^{\delta_m} f_0(\zeta_1, \dots, \zeta_m),$$

*where  $f_0 \in \mathbb{C}^n[[\zeta_1, \dots, \zeta_m]]$  and  $\delta_i \in \mathbb{C}$ . Then there exists such a vector  $v \in \mathbb{C}^n$  such that  $f(\zeta_1, \dots, \zeta_m) = F(\zeta_1, \dots, \zeta_m)v$ , where  $F$  is the fundamental solution from Proposition 18.1, and  $f$  converges to an analytic solution in the region  $R$ .*

*Proof.* Let  $v(\zeta_1, \dots, \zeta_m) = F^{-1}f$ , then differentiating gives  $\partial_{\zeta_i} v = 0$ . □

**Theorem 18.5.** *The correlation function is absolutely convergent in the region  $|z_1| > |z_2| > \cdots > |z_N| > 0$  and defines a regular, multi-valued function in this region.*

**Corollary 18.5.1.** *For  $|z_1| > \cdots > |z_N| > 0$ , the product  $\widehat{\Phi}_{u_1}(z_1) \cdots \widehat{\Phi}_{u_N}(z_N)$  is a well-defined operator*

$$V_{\lambda_N, k} \longrightarrow \widehat{V}_{\lambda_0, k}$$

*(i.e. all matrix elements are converging series).*

**Corollary 18.5.2.** *In the limit  $|z_1| \gg \cdots \gg |z_N|$ , the correlation function has the following asymptotics:*

$$\psi(z_1, \dots, z_N) \sim z_1^{-\Delta_1} \cdots z_N^{-\Delta_N} v,$$

*where  $v = \langle u_0, g_1 \cdots g_N \rangle \in V$ .*

**Remark.** By Corollary 18.5.2, we can write  $\psi = z_1^{-\Delta_1} \cdots z_N^{-\Delta_N} \psi_0$ , where

$$\psi_0 = \psi_0(\zeta_1, \dots, \zeta_{N-1}) = \psi_0(z_2/z_1, \dots, z_N/z_{N-1})$$

is analytic in a neighborhood of  $\{\zeta_i = 0\}$  and  $\psi_0(0) = v$ .

# Lecture 19

## Mar. 18 — KZ Equations, Part 2

### 19.1 Correlation Functions, Continued

**Proposition 19.1.** *The correlation functions span the space of solutions of the KZ equations.*

**Remark.** Let  $\xi = (\lambda_0, \dots, \lambda_N; g_1, \dots, g_N; u_0)$ ,  $\lambda_i \in \mathfrak{h}^*$ ,  $g_i \in \text{Hom}_{\mathfrak{g}}(L_{\lambda_i}, L_{\lambda_{i-1}} \otimes V_{\mu_i})$  where  $u_0 \in L_{\lambda_0}^*$ . Let

$$\Xi = \bigoplus_{\lambda_0, \dots, \lambda_{N-1}} \left( L_{\lambda_0}^* \otimes \bigotimes_{i=1}^N \text{Hom}_{\mathfrak{g}}(L_{\lambda_i}, L_{\lambda_{i-1}} \otimes V_{\mu_i}) \right).$$

The only nonzero terms satisfy  $\lambda_i + \mu_i - \lambda_{i-1} \in Q$ , and this space has a natural  $\mathfrak{g}$ -module structure, and the  $\text{Hom}_{\mathfrak{g}}(L_{\lambda_i}, L_{\lambda_{i-1}} \otimes V_{\mu_i})$  are trivial  $\mathfrak{g}$ -modules. Each  $\xi \in \Xi$  defines a unique correlation function

$$\psi^\xi(z_1, \dots, z_N) \in V = V_{\mu_1} \otimes \dots \otimes V_{\mu_N} \otimes L_{\lambda_N}^*.$$

**Theorem 19.1.** *Fix  $\lambda_N$  and  $\mu_i$  for  $i = 1, \dots, N$ . Let  $V_{\mu_i}$  be the lowest weight modules satisfying either of the two following conditions:*

1. Generic case: *The weights  $\lambda_N, \lambda_N + \mu_N, \dots, \lambda_N + \mu_N + \dots + \mu_1$  are generic and  $k$  is generic with respect to each;*
2. Finite-dimensional case:  *$\lambda_N, \mu_i \in P^+$ ,  $k \notin \mathbb{Q}$ , and the  $V_{\mu_i}$  are irreducible finite-dimensional modules with lowest weight  $-\mu_i$ .*

*Then the correlation functions span the space of  $V$ -valued solutions of the KZ equations in a region  $D$ . Moreover, the following map*

$$\begin{aligned} \Xi &\longrightarrow \Gamma(D, V_{\text{KZ}}) \\ \xi &\longmapsto \psi^\xi(z_1, \dots, z_N) \end{aligned}$$

*is an isomorphism of  $\mathfrak{g}$ -modules.*

**Remark.** In Case 1 of Theorem 19.1, the  $\lambda_i$  are generic, and in Case 2, the  $\lambda_i \in P^+$ .

**Lemma 19.1.** *Under assumptions (1) or (2) of Theorem 19.1, the map*

$$(\lambda_0, \dots, \lambda_{N-1}; g_1, \dots, g_N; u_0) \longmapsto \langle u_0, g_1 \cdots g_N \cdot \rangle$$

*is an isomorphism of  $\mathfrak{g}$ -modules.*

*Proof.* Note that  $\text{Hom}_{\mathfrak{g}}(L_{\lambda_i}, L_{\lambda_{i-1}} \otimes V_{\mu_i}) \cong \text{Hom}_{\mathfrak{g}}(L_{\lambda_{i-1}}^*, V_{\mu_i} \otimes L_{\lambda_i}^*)$ . So it suffices to show that

$$\bigoplus_v \text{Hom}_{\mathfrak{g}}(L_v^*, V_{\mu} \otimes L_{\lambda}^*) \otimes L_v^* \longrightarrow V_{\mu} \otimes L_{\lambda}^*$$

is an isomorphism. This is true if we are in the finite-dimensional case. In general, if  $\lambda$  and  $\lambda + \mu$  are generic, the  $L_v^*$  being irreducible implies that the above map is injective. Since  $v$  is generic,  $L_v^* \cong M_v^*$  and  $L_{\lambda}^* \cong M_{\lambda}^*$ , so  $\text{Hom}_{\mathfrak{g}}(L_v^*, V_{\mu} \otimes L_{\lambda}^*) \cong V_{\mu}^{(\lambda-v)}$  (the weight subspace).  $\square$

**Corollary 19.1.1.** *If  $u_0$  is the lowest weight vector, then the correlation functions span solutions with values in  $V^{\mathfrak{n}^-}$ . Moreover, if  $u_0$  is  $\mathfrak{g}$ -invariant, then the correlation functions take values in  $V^{\mathfrak{g}}$ .*

**Remark.** We had an identification  $\Gamma(D, V_{\text{KZ}}) \cong V$  via  $\psi \mapsto \psi(p)$  (i.e. evaluation at  $p$ ). Now we have

$$\Gamma(D, V_{\text{KZ}}) \cong \Xi \cong V,$$

which is a more sophisticated identification that is not attached to a particular point.

## 19.2 Trigonometric KZ Equations

**Remark.** Let  $\phi(z_1, \dots, z_N) = \langle u_0, \psi(z_1, \dots, z_N) u_{N+1} \rangle \in V_{\mu_1} \otimes \dots \otimes V_{\mu_N}$ . Then

$$\phi \in (V_{\mu_1} \otimes \dots \otimes V_{\mu_N})^{\lambda_N - \lambda_0},$$

and  $\phi$  satisfies the following differential equation

$$(k + h^{\vee}) \frac{\partial}{\partial z_i} \phi = \left( \sum_{j=1}^N \frac{\Omega_{ij}}{z_i - z_j} \right) \phi - \frac{1}{z_i} \sum_e (a_e)_i \langle u_0, \psi(z_1, \dots, z_N) a^{\ell} u_{N+1} \rangle,$$

where  $a^e, a_e$  are dual basis elements in  $\mathfrak{g}$ . For  $\alpha \in R_+$ , let  $e_{\alpha} \in \mathfrak{g}^{\alpha}$ ,  $f_{\alpha} \in \mathfrak{g}^{-\alpha}$  such that  $\langle e_{\alpha}, f_{\alpha} \rangle = 1$ . Then

$$\Omega = \sum_{\alpha \in R_+} (e_{\alpha} \otimes f_{\alpha} + f_{\alpha} \otimes e_{\alpha}) + \sum_p x_p \otimes x_p,$$

where  $x_p$  is the dual basis in  $\mathfrak{h}$ , and  $C = \sum_{\alpha \in R_+} (e_{\alpha} f_{\alpha} + f_{\alpha} e_{\alpha}) + \sum_p x_p^2$ . We can rewrite

$$\begin{aligned} \sum_e (a_e)_i \langle u_0, \psi a^{\ell} u_{N+1} \rangle &= \sum_{\alpha \in R_+} (e_{\alpha})_i \langle u_0, \psi f_{\alpha} u_{N+1} \rangle + \sum_p (x_p)_i \langle u_0, \psi \langle x_p, \lambda_N \rangle u_{N+1} \rangle \\ &= \sum_{\alpha \in R_+} (e_{\alpha})_i \sum_{j=1}^N (f_{\alpha})_j \phi + (h_{\lambda_N})_i \phi. \end{aligned}$$

Note that  $u_0$  is  $\mathfrak{n}_-$ -invariant. Now  $\phi$  has weight  $\lambda_N - \lambda_0$ , so

$$\sum_p \sum_{j=1}^N (x_p)_i (x_p)_j \phi = (h_{\lambda_N - \lambda_0})_i \phi.$$

Then we can rewrite

$$\begin{aligned}
\sum_e (a_e)_i \langle u_0, \psi a^\ell u_{N+1} \rangle &= \sum_{j=1}^N \left( \sum_{\alpha \in R_+} (e_\alpha)_i (f_\alpha)_j + \frac{1}{2} \sum_p (x_p)_i (x_p)_j \right) \phi + \frac{1}{2} (h_{\lambda_N} + \lambda_0)_i \phi \\
&= \sum_{j=1}^N \Omega_{i,j}^+ \phi + \frac{1}{2} (h_{\lambda_N + \lambda_0})_i \phi + \left( \sum_{\alpha \in R_+} e_\alpha f_\alpha + \frac{1}{2} \sum_p x_p^2 \right)_i \phi,
\end{aligned}$$

where  $\Omega^+ = \sum_{\alpha \in R_+} e_\alpha \otimes f_\alpha + \frac{1}{2} \sum_p x_p \otimes x_p$

$$\Omega^- = \Omega - \Omega^+ = \sum_{\alpha \in R_+} f_\alpha \otimes e_\alpha + \frac{1}{2} \sum_p x_p \otimes x_p.$$

Check as an exercise that  $[e_\alpha, f_\alpha] = h_\alpha$  and that

$$\sum e_\alpha f_\alpha + \frac{1}{2} \sum_p x_p^2 = \frac{1}{2} C + h_\rho.$$

Then we have that

$$\begin{aligned}
\sum_\ell (a_\ell)_i \langle u_0, \psi a^\ell u_{N+1} \rangle &= \sum_{\substack{j=1 \\ j \neq i}}^N \Omega_{i,j}^+ \phi + \frac{1}{2} (h_{\lambda_N + \lambda_0 + 2\rho})_i \phi + \frac{1}{2} (C)_i \phi \\
&= \left( \sum_{j=1}^N \Omega_{i,j}^+ + \frac{1}{2} (h_{\lambda_N + \lambda_0 + 2\rho})_i + \frac{1}{2} \langle \mu_i, \mu_i + 2\rho \rangle \right) \phi.
\end{aligned}$$

Thus we can rewrite the KZ equations as

$$(k + h^\vee) z_i \partial_{z_i} \phi = \left( \sum_{\substack{j=1 \\ j \neq i}}^N \frac{z_i \Omega_{i,j}^- + z_j \Omega_{i,j}^+}{z_i - z_j} - \frac{1}{2} (h_{\lambda_N + \lambda_0 + 2\rho})_i - \frac{1}{2} \langle \mu_i, \mu_i + 2\rho \rangle \right) \phi$$

**Theorem 19.2.** *Let  $z_i = e^{z_i}$ . Then*

$$(k + h^\vee) \partial_{x_i} \phi = \left( \sum_{\substack{j=1 \\ j \neq i}}^N r_{i,j} (x_i - x_j) - \frac{1}{2} (h_{\lambda_N + \lambda_0 + 2\rho})_i - \langle \mu_i, \mu_i + 2\rho \rangle \right) \phi,$$

where  $r(x) = (e^x \Omega^- + \Omega^+) / (e^x - 1)$ .

## 19.3 Factorizable Systems

**Remark.** The equations we are studying take the form

$$\frac{\partial}{\partial z_i} \phi(z_1, \dots, z_N) = A_i(z_1, \dots, z_N) \phi(z_1, \dots, z_N), \quad \phi \in V, A_i \in \text{End}(V),$$

where  $\partial_i A_j \partial_j A_i + [A_i, A_j] = 0$  and the  $A_i$  are holomorphic in  $D \subseteq \mathbb{C}^N$ .

**Definition 19.1.** We say that such a system is *factorizable* if

1.  $V = V_1 \otimes \cdots \otimes V_N$ , where the  $\{V_i\}$  are  $\mathfrak{g}$ -modules,
2. there exists a function  $r(z) \in \mathfrak{g} \otimes \mathfrak{g}$  of one complex variable meromorphic in a neighborhood of the origin, and  $\kappa \in \mathbb{C}$  such that

$$A_i(z_1, \dots, z_N) = \frac{1}{\kappa} \left( \sum_{j>i} r_{i,j}(z_i - z_j) - \sum_{j<i} r_{j,i}(z_j - z_i) + s_i \right)$$

for  $s \in \mathfrak{g}$  such that  $[s \otimes 1 + 1 \otimes s, r(z)] = 0$ .

**Proposition 19.2.** *A factorizable system is consistent if*

$$[r_{i,j}(z_i - z_j), r_{j,k}(z_j - z_k)] + [r_{i,k}(z_i - z_k), r_{j,k}(z_j - z_k)] + [r_{i,j}(z_i - z_j), r_{i,k}(z_i - z_k)] = 0.$$

*These are called the classical Yang-Baxter equations.*

**Remark.** There are also *quantum Yang-Baxter equations*, which is of the form

$$R_{i,j}(z_i - z_j) = 1 + \hbar r(z_i - z_j) + O(\hbar^2),$$

and one can recover the classical equation from it.

**Remark.** We can naturally take  $V_i \cong U(\mathfrak{g})$  to make a factorizable system, and in this case the condition in Proposition 19.2 is satisfied in  $U(\mathfrak{g})^{\otimes 3}$ .

**Definition 19.2.** *Classical  $r$ -matrices* are solutions of the classical Yang-Baxter equations which are meromorphic in a neighborhood of the origin with at least one point  $z$  where  $r(z)$  is nondegenerate.

**Definition 19.3.** Two classical  $r$ -matrices  $r(z)$ ,  $\tilde{r}(z)$  are *equivalent* if there is  $\phi(x) \in \text{Aut}(\mathfrak{g})$  such that

$$\hat{r}(z_1 - z_2) = (\phi(z_1) \otimes \phi(z_2))r(z_1 - z_2).$$

**Theorem 19.3.** *A classical  $r$ -matrix  $r(z)$  extends to a meromorphic function on the complex plane satisfying  $r_{1,2}(-z) = -r_{2,1}(z)$ . Moreover, they fall into one of the following classes:*

1. Rational:  $r(z) = c\Omega/z + P(z)$  for some  $P \in \mathfrak{g} \otimes \mathfrak{g} \otimes \mathbb{C}[z]$  and  $c \in \mathbb{C}^\times$ .
2. Hyperbolic (or trigonometric):  $r = f(e^{az})$ , where

$$r(z) = \frac{P(e^{az})}{e^{a(n+1)z} - 1} + e^{-anz}Q(e^{az})$$

where  $P, Q$  are polynomials and  $r(z + 2\pi i/a) = r(z)$ .

3. Elliptic:  $r(z + n\omega_1) = r(z + n\omega_2) = r(z)$ . This case exists only for Type A Lie algebras.

# Lecture 20

## Mar. 30 — KZ Equations for $\mathfrak{sl}_2$

### 20.1 KZ Equations for $\mathfrak{sl}_2$

**Remark.** For this lecture, let  $\mathfrak{g} = \mathfrak{sl}_2$ . Then we have (letting  $\kappa = k + h^\vee$ )

$$\kappa \frac{\partial}{\partial z_i} \psi = \left( \sum_{j=1}^N \frac{\Omega_{i,j}}{z_i - z_j} \right) \psi, \quad i = 1, \dots, N.$$

Note that  $\psi$  takes values in  $V = V_{\mu_1} \otimes \cdots \otimes V_{\mu_N}$ , where  $V_{\mu_i}$  are the lowest weight modules (with height  $-\mu_i$ ). Let  $(V^{n-})^\lambda$  be the space of singular vectors, where  $\lambda$  is a weight. Write  $\lambda = -\sum_{i=1}^N \mu_i + \mu$ , where  $\mu = \sum_{i=1}^r n_i \alpha_i$ . Call  $|\mu| = \sum_{i=1}^r n_i$  the *level* of the solution. Let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and  $\Omega = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h$ . Let  $(x, y) = \text{tr}(xy)$ , so that

$$(\beta, \gamma) = (\beta\omega, \gamma\omega) = \frac{\beta\gamma}{2}.$$

With this normalization, we have  $h \rightarrow 1$ ,  $\alpha \rightarrow 2$ ,  $\rho = 1$ ,  $k + h^\vee \rightarrow k + 2$ , and  $h = h^\vee = 2$ .

Now we find the dimension of  $W$ . Let  $p_N(m)$  be the number of ordered partitions of  $m$  into  $N$  parts, i.e.

$$p_N(m) = \# \left\{ \vec{m} = (m_1, \dots, m_N) \in \mathbb{Z}_+^N : \sum m_i = m \right\} = \binom{N+m-1}{m} = \frac{(N+m-1)!}{m!(N-1)!}.$$

Note that  $p_N$  has generating function  $1/(1-q)^N$ . Let  $v_i$  be the lowest weight vector in  $V_{\mu_i}$ . Consider

$$e^{m_1} v_1 \otimes \cdots \otimes e^{m_N} v_N,$$

then we see that  $\dim V^\lambda = p_N(m)$  where  $\lambda = -\sum \mu_i + 2m$ .

**Proposition 20.1.** *Let  $W$  be the space of singular vectors of weight  $\lambda = -\sum \mu_i + 2m$  in  $V$ . Then for generic  $\mu_i$ , we have*

$$\dim W = p_{N-1}(m) = \binom{N+m-2}{m}$$

**Remark.** In level 0, we have  $W^0 = \mathbb{C}v$ , where  $v = v_1 \otimes \cdots \otimes v_N$ . Then

$$\Omega_{i,j} v = \frac{1}{2} (v_1 \otimes \cdots \otimes h v_i \otimes \cdots \otimes h v_j \otimes \cdots \otimes v_N) = \frac{\mu_i \mu_j}{2} v.$$

Define the function



$$\psi_0(z_1, \dots, z_N) = \prod_{i < j} (z_i - z_j)^{\mu_i \mu_j / 2\kappa}.$$

Then  $\psi_0 = \psi_0(z_1, \dots, z_n)v$  is a solution to the KZ equations.

We will now consider  $N = 3$ , which gives the simplest level 1 solution.

**Proposition 20.2.** *Any solution  $\psi(z_1, z_2, z_3)$  of the KZ equations can be rewritten as*

$$\psi(z_1, z_2, z_3) = (z_1 - z_3)^{(\Omega_{1,2} + \Omega_{1,3} + \Omega_{2,3})/\kappa} f\left(\frac{z_1 - z_2}{z_1 - z_3}\right).$$

where  $f$  satisfies the equation

$$\kappa \frac{\partial}{\partial z} f(z) = \left( \frac{\Omega_{1,2}}{z} + \frac{\Omega_{2,3}}{z-1} \right) f(z). \quad (*)$$

**Remark.** Consider  $L_0, L_{\pm 1}$  with  $-d/dz$ ,  $-zd/dz$ , and  $-z^2d/dz$ . Let

$$x = \frac{z_1 - z_2}{z_1 - z_3}, \quad y = z_1 - z_3, \quad t = z_1 + z_2 + z_3.$$

Then we have that

$$\kappa \frac{\partial \psi}{\partial x} = \left( \frac{\Omega_{1,2}}{x} + \frac{\Omega_{2,3}}{x-1} \right) \psi, \quad \kappa \frac{\partial \psi}{\partial y} = \frac{(\Omega_{1,2} + \Omega_{1,3} + \Omega_{2,3})}{y} \psi, \quad \kappa \frac{\partial \psi}{\partial t} = 0.$$

Then we see that  $f = y^{-(\Omega_{1,2} + \Omega_{1,3} + \Omega_{2,3})/\kappa} \psi(x, y)$  does not depend on  $y$ .

**Remark.** Note that  $W$  is 2-dimensional on level 1: Assume  $\mu_i \neq 0$ , then

$$\begin{aligned} w_1 &= \mu_1 ev_1 \otimes v_2 \otimes v_3 - \mu_2 v_1 \otimes ev_2 \otimes v_3 \\ w_2 &= \mu_3 v_1 \otimes ev_2 \otimes v_3 - \mu_2 v_1 \otimes v_2 \otimes ev_3 \end{aligned}$$

**Theorem 20.1.** *Any solution  $f(z)$  of the equation  $(*)$  can be written as follows:*

$$f(z) = z^{(\mu_1 \mu_2 - 2\mu_1 - 2\mu_2)/2\kappa} (1-z)^{(\mu_2 \mu_3)/2\kappa} \left( F(z) w_1 + z \frac{\kappa}{\mu_3} F'(z) w_2 \right),$$

where  $F(z)$  is a solution of the Gauss hypergeometric equation:

$$z(1-z) \frac{d^2}{dz^2} F = [c - (a+b+1)z] \frac{dF}{dz} - abF = 0,$$

where  $a = \mu_3/\kappa$ ,  $b = -\mu_1/\kappa$ , and  $c = 1 - (\mu_1 + \mu_2)/\kappa$ .

**Remark.** The unique solution of the Gauss hypergeometric equation with  $F(0) = 1$  is given by

$${}_2F_1(a, b, c, z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{n! (c)_n} z^n$$

in  $|z| < 1$ , where  $(a)_n = a(a+1) \cdots (a+n-1)$ .

**Proposition 20.3.** *For generic values of parameters,*

$$F^{(1)} = {}_2F_1(a, b, c, z) \quad \text{and} \quad F^{(2)} = z^{1-c} {}_2F_1(a-c+1, b-c+1, 2-c, z)$$

form a basis for the space of solutions.

## 20.2 Integral Formulas for Hypergeometric Equation

**Remark.** Let  $\vec{z} = (z_1, \dots, z_N)$ , and define  $\psi_1(\vec{z}, t) = \prod_{j=1}^N (t - z_j)^{-\mu_j/\kappa}$ . Then we claim that

$$\psi_C(\vec{z}) = \psi_0(\vec{z}) \sum_{r=1}^N \int_C \psi_1(\vec{z}, t) \frac{dt}{t - z_r} e_r v$$

is a well-defined integral, where  $v = v_1 \otimes \dots \otimes v_N$  and  $e_r = 1 \otimes \dots \otimes e \otimes \dots \otimes 1$  ( $r$ -th position).

**Remark.** Fix two distinct points  $z_1, z_2 \in \mathbb{C}$ . Consider a closed contour around  $z_1, z_2$ , which wraps around  $z_1$  once in the negative direction, then  $z_1$  once in the positive direction, then  $z_2$  once in the positive direction, and finally  $z_1$  once in the negative direction. This is called the *Pochhammer contour*. Using this, one can show that  $\psi_C$  is well-defined.

**Proposition 20.4.** *The function  $\psi_C$  as defined above is a  $W$ -valued solution of the KZ equations.*

*Proof.* This is a direct calculation. Define

$$T_i(\vec{z}, t) = \psi_1(\vec{z}, t) \frac{e_i}{t - z_i}, \quad Y(\vec{z}, t) = \sum_i T_i(\vec{z}, t), \quad H(\vec{z}, t) = \sum_i \frac{h_i + \mu_i}{t - z_i}.$$

We use the following lemma, which is also a direct calculation using commutators:

**Lemma 20.1.** *We have*

$$\kappa \frac{\partial}{\partial z_i} Y - \left[ \sum_{j=1}^N \frac{\Omega_{i,j}}{z_i - z_j}, Y \right] = Y \frac{h_i + \mu_i}{t - z_i} - T_i H - \kappa \frac{\partial}{\partial t} T_i$$

Now we check that  $\psi_C$  is a solution. We can write

$$\psi_C(\vec{z}) = \psi_0(\vec{z}) \int_C Y(\vec{z}, t) v dt.$$

Since  $\psi_0(\vec{z})v$  is a solution, it is enough to see that

$$\int_C \left[ \kappa \frac{\partial}{\partial t_i} - \sum_{j=1}^N \frac{\Omega_{i,j}}{z_i - z_j}, Y \right] v dt = \int_C \kappa \frac{\partial}{\partial t} T_i v dt = 0.$$

To check that  $\psi_C$  is  $W$ -valued, we want to show that  $f\psi_C = 0$ . Note that

$$[f, Y(\vec{z}, t)] = \psi_1(\vec{z}, t) \sum_i \frac{-h_i}{t - z_i},$$

so we get that

$$\begin{aligned} f\psi_C(\vec{z}, t) &= \psi_0(\vec{z}) \int_C \psi_1(\vec{z}, t) \sum_i \frac{-\mu_i}{t - z_i} v dt \\ &= \psi_0(\vec{z}) \int_C \psi_1(\vec{z}, t) \sum_i \frac{\mu_i}{t - z_i} v dt = -\psi_0(\vec{z})v \int_C \kappa \frac{\partial}{\partial t} \psi_1(\vec{z}, t) dt = 0. \end{aligned}$$

This proves the desired result. □

**Example 20.0.1.** Let  $N = 3$ . Then  $z_1 = 0$ ,  $z_2 = z$ , and  $z_3 = 1$ . We have

$$\psi_C(0, z, 1) = z^{\mu_1\mu_2/2\kappa}(1-z)^{\mu_2\mu_3/2\kappa} \int_C t^{-\mu_1/\kappa}(t-z)^{-\mu_2/\kappa}(t-1)^{-\mu_3/\kappa} \left( \frac{ev_1 \otimes v_2 \otimes v_3}{t} + \frac{v_1 \otimes ev_2 \otimes v_3}{t-z} + \frac{v \otimes v_2 \otimes ev_3}{t-1} \right) dt.$$

Note that the function

$$F_C(z) = z^{(\mu_1+\mu_2)/\kappa} \int_C t^{-\mu_1/\kappa}(t-z)^{-\mu_2/\kappa}(1-t)^{-\mu_3/\kappa} \frac{dt}{t}$$

is a solution of the hypergeometric equation. Let  $t = xz$ ,  $z^{-1} \in [0, 1]$ , and  $C$  be the Pochhammer contour around 0, 1. Then we have that

$$\begin{aligned} F_C(z) &= \int_C x^{-1-\mu_1/\kappa}(1-x)^{-\mu_2/\kappa}(1-zx)^{-\mu_3/\kappa} dx \\ &= (1 - e^{2\pi i\mu_1/\kappa})(1 - e^{2\pi i\mu_2/\kappa}) \int_0^1 x^{-1-\mu_1/\kappa}(1-x)^{-\mu_2/\kappa}(1-zx)^{-\mu_3/\kappa} dx. \end{aligned}$$

For  $\operatorname{Re}(\mu_i/\kappa) < 0$ ,  $\operatorname{Re}(\mu_2/\kappa) < 1$ , and  $\mu_i/\kappa \notin \mathbb{Z}$ , we have that

$$F(z) = \int_0^1 x^{-1-\mu_1/\kappa}(1-x)^{-\mu_2/\kappa}(1-zx)^{-\mu_3/\kappa} dx$$

is a solution of the hypergeometric equation.

**Proposition 20.5.** *We have*

$${}_2F_2(1)(a, b, c, z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(a)} = \int_0^1 x^{b-1}(1-x)^{c-b-1}(1-zx)^{-a} dx.$$

## 20.3 Solutions in Higher Levels

**Remark.** Let  $m = \mu/2 \in \mathbb{Z}_+$  be the level of the solution. Let

$$\vec{z} = (z_1, \dots, z_N), \quad \vec{t} = (t_1, \dots, t_m), \quad d\vec{t} = dt_1 \wedge \dots \wedge dt_m.$$

Define the multi-valued differential form

$$\omega_{\vec{z}, m} := \prod_{p < n} (t_p - t_n)^{2/\kappa} Y(\vec{z}, t_1) \cdots Y(\vec{z}, t_m) d\vec{t} \in Y_{\vec{z}, m} := \{t \in \mathbb{C}^m : t_i \neq t_j, t_i \neq z_j\} \subseteq \mathbb{C}^m$$

**Theorem 20.2.** *The function*

$$\psi_C(\vec{z}) = \psi_0(\vec{z}) \int_C \omega_{\vec{z}, m} v$$

*is a solution of the KZ equations if  $C$  is a “legitimate” cycle.*

**Remark.** Define the function

$$\psi_m(\vec{z}, t) = \prod_{i < j} (z_i - z_j)^{\mu_i\mu_j/2\kappa} \prod_{p, j} (t_p - z_j)^{-\mu_j/\kappa} \prod_{p < n} (t_p - t_n)^{2/\kappa},$$

and let (here  $\Sigma_m$  is the set of maps  $s : \{1, \dots, m\} \rightarrow \{1, \dots, N\}$  such that  $s^{-1}(j) = m_j$  for  $1 \leq j \leq N$ )

$$\rho_{\vec{m}}(\vec{z}, t) = \sum_{s \in \Sigma_m} \prod_{n=1}^n (t_n - z_{s(n)})^{-1}.$$

Then we have that

$$\psi_C(\vec{t}) = \left( \sum_{\vec{m}} \int_C \psi_m(\vec{z}, t) \rho_{\vec{m}}(\vec{z}, t) dt_1 \wedge \cdots \wedge dt_m \right) (e^{m_1} v_1 \otimes \cdots \otimes e^{m_N} v_N).$$